

https://github.com/pvermees/Thermo2025

isoplotr.es.ucl.ac.uk x pvermees/Thermo2025 +

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pvermees Added info about random vs systematic uncertainty 09b7e8f · yesterday 5 Commits

R	Fixed pop 4 FT age plot	2 days ago
handout	Added info about random vs systematic uncertainty	yesterday
.gitignore	Created README file, updated helper.R	last month
LICENSE	Initial commit	last month
README.md	Created README file, updated helper.R	last month

README GPL-3.0 license

A short course in statistical thermochronology

This repository contains a collection of teaching materials for the short course in statistical thermochronology (with a focus on detrital thermochronology), held on 20/09/2025 at the [Thermo2025](#) conference in Kanazawa, Japan. It comprises the following materials:

- A self-contained [handout](#) with instructions, and the associated [source code](#).
- [Data](#) and [code](#) to run the exercises in the notes.
- [Slides](#) in an Open Document Presentation format.

To run the exercises in the notes, you must have [R](#) installed on your computer, as well as the [IsoplotR](#) and [provenance](#) packages. Installation instructions for these dependencies are provided in the [notes](#).

About

Short course in statistical thermochronology

Readme

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Activity

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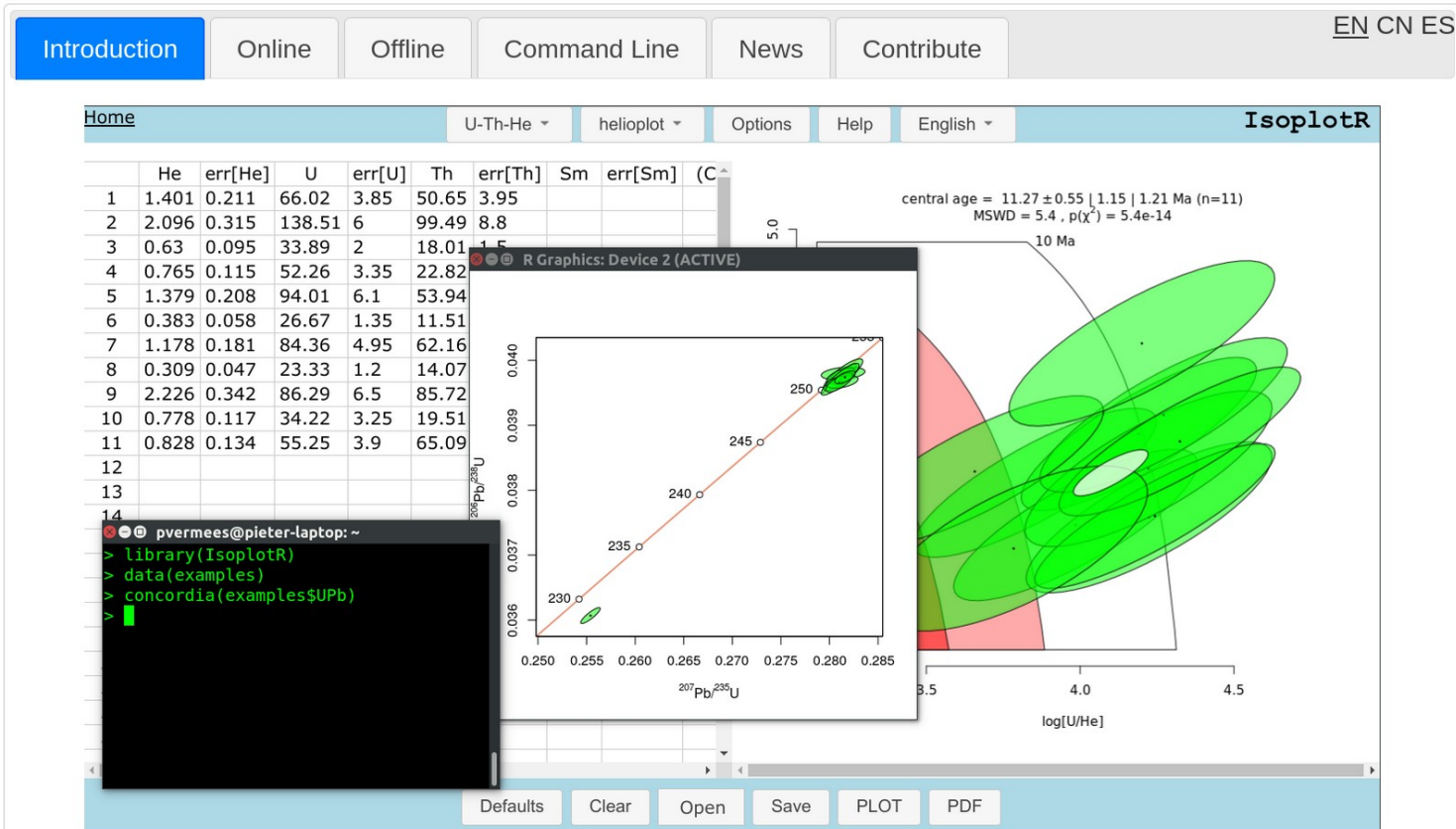
Languages

R 100.0%

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<https://isoplotr.es.ucl.ac.uk>



Install R from:

<https://cran.r-project.org/bin/>

OR

<https://posit.co/download/rstudio-desktop/>

 Parent Directory		-
 linux/	2020-07-07 11:45	-
 macos/	2005-04-19 09:45	-
 macosx/	2025-06-18 22:17	-
 windows/	2024-03-05 14:36	-

[EN](#) [CN](#) [ES](#)

Introduction

Online

Offline

Command Line

News

Contribute

To run the offline version of the program, install *R* from <http://r-project.org> and enter the following command at the prompt:

```
install.packages('IsoplotRgui')
```

Once the package has been installed, you can run the browser-based graphical user interface by typing:

```
library(IsoplotRgui)  
IsoplotR()
```

This will open a new tab in your internet browser. To stop the GUI, enter the following command at the prompt:

```
stopIsoplotR()
```


Ages vs. dates

true values	measured data
$\rho_s = 1 \times 10^6 \text{ cm}^{-2}$	$N_s = 15$
$\rho_i = 2 \times 10^6 \text{ cm}^{-2}$	$N_i = 42$
$\rho_d = 2.5 \times 10^6 \text{ cm}^{-2}$	$\hat{\rho}_d = 2.55 \pm 0.05 \times 10^6 \text{ cm}^{-2}$
$\zeta = 350 \text{ yr}^{-1} \text{ cm}^2$	$\hat{\zeta} = 355 \pm 10 \text{ yr}^{-1} \text{ cm}^2$

$$t = \frac{1}{\lambda_{238}} \ln \left(1 + \frac{1}{2} \lambda_{238} \zeta \rho_d \frac{\rho_s}{\rho_i} \right) = 215 \text{ Ma}$$

$$\hat{t} = \frac{1}{\lambda_{238}} \ln \left(1 + \frac{1}{2} \lambda_{238} \hat{\zeta} \hat{\rho}_d \frac{N_s}{N_i} \right) = 160 \text{ Ma}$$

$$s[\hat{t}] = \hat{t} \sqrt{\underbrace{\frac{1}{N_s} + \frac{1}{N_i}}_{\text{random}} + \underbrace{\left(\frac{s[\hat{\zeta}]}{\hat{\zeta}} \right)^2 + \left(\frac{s[\hat{\rho}_d]}{\hat{\rho}_d} \right)^2}_{\text{systematic}}} = 48 \text{ Ma}$$

$$s[\hat{t}] = \hat{t} \sqrt{\underbrace{\frac{1}{N_s} + \frac{1}{N_i}}_{\text{random}} + \underbrace{\left(\frac{s[\hat{\zeta}]}{\hat{\zeta}}\right)^2 + \left(\frac{s[\hat{\rho}_d]}{\hat{\rho}_d}\right)^2}_{\text{systematic}}}$$

#	1	2	3	4	5	6	7	8	9	10	Σ
N_s	38	48	114	13	25	49	39	18	26	24	394
N_i	132	187	565	51	122	249	159	50	80	79	1674
$\hat{t}$	125	111	88	111	89	86	106	156	141	132	102.1
$s[\hat{t}] = \sqrt{1/N_s + 1/N_i}$	23	18	9	34	20	13	19	43	32	31	5.7
$s[\bar{t}] = \sqrt{1/\sum N_s + 1/\sum N_i + (s[\hat{\zeta}]/\hat{\zeta})^2 + (s[\hat{\rho}_d]/\hat{\rho}_d)^2}$											6.4

Pooled date

$$[\text{He}] = a(t) [\text{U}] + b(t) [\text{Th}]$$

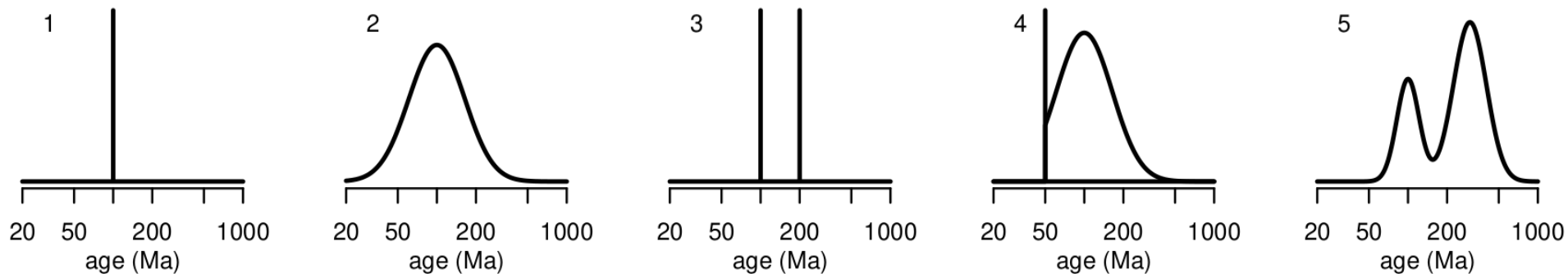
$$\text{where } a(t) = 8 \frac{137.818}{138.818} (e^{\lambda_{238} t} - 1) + 7 \frac{1}{138.818} (e^{\lambda_{235} t} - 1)$$

$$\text{and } b(t) = 6(e^{\lambda_{232} t} - 1)$$

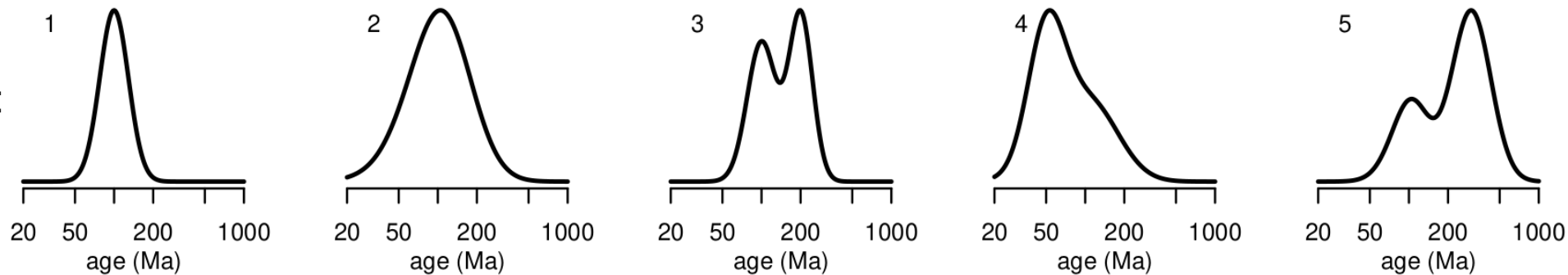
$$s[\hat{t}] = \frac{\sqrt{a(\hat{t})^2 s[\text{U}]^2 + 2 a(\hat{t}) b(\hat{t}) s[\text{U}, \text{Th}] + b(\hat{t})^2 s[\text{Th}]^2 + s[\text{He}]^2}}{8 \frac{137.818}{138.818} \lambda_{238} e^{\lambda_{238} \hat{t}} + 7 \frac{\lambda_{235}}{138.818} e^{\lambda_{235} \hat{t}}}$$

Dates are always more scattered than ages

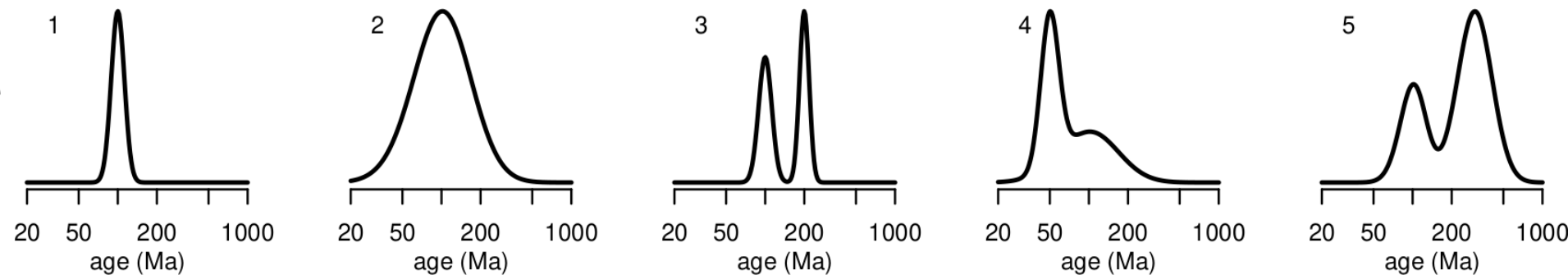
Ages:



FT dates:

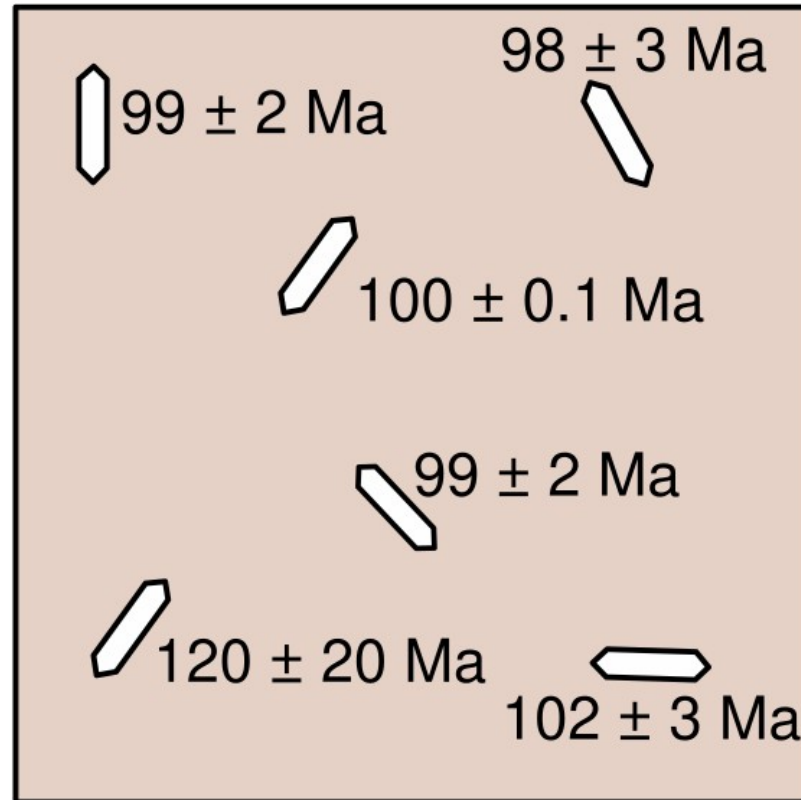


U-Th-He dates:



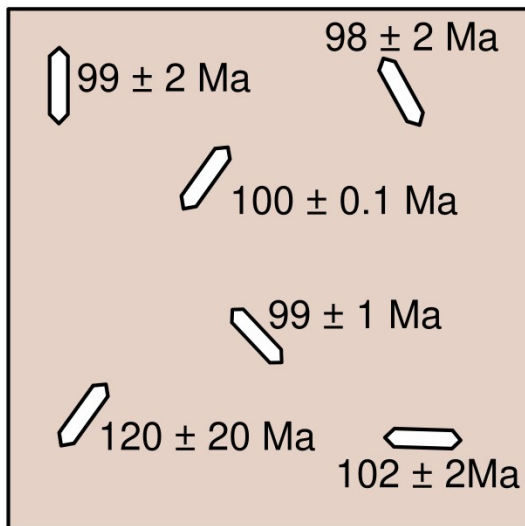
true age=100 Ma

measured dates:



true age=100 Ma

measured dates:



mean:

$$\bar{t} = \frac{99 + 98 + 100 + 99 + 120 + 102}{6} = 103$$

standard
deviation:

$$s[t] = \sqrt{\frac{(99-103)^2 + (98-103)^2 + (100-103)^2 + (99-103)^2 + (120-103)^2 + (102-103)^2}{5}} = 8.4$$

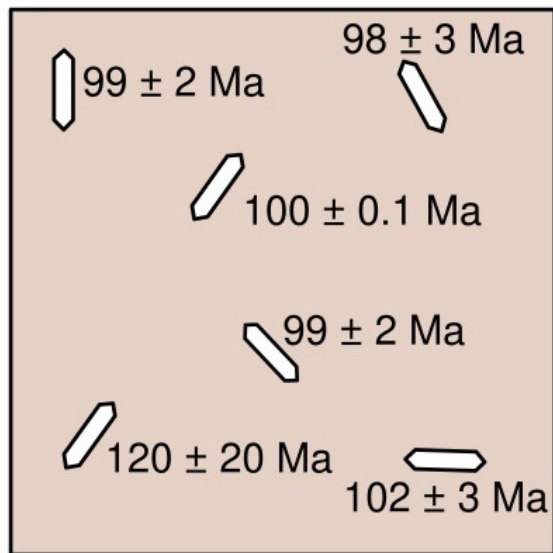
standard
error:

$$s[\bar{t}] = \sqrt{\frac{(99-103)^2 + (98-103)^2 + (100-103)^2 + (99-103)^2 + (120-103)^2 + (102-103)^2}{5 \cdot 6}} = 3.4$$

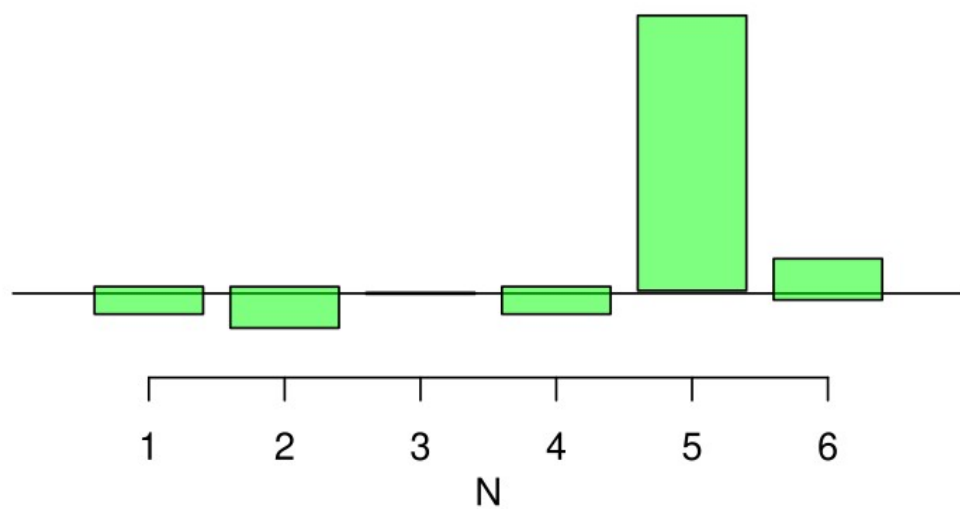
$$\bar{t} = \frac{\sum_{i=1}^n t_i / s[t_i]^2}{\sum_{i=1}^n 1 / s[t_i]^2} \quad s[\bar{t}] = \frac{1}{\sqrt{\sum_{i=1}^n 1 / s[t_i]^2}}$$

true age=100 Ma

measured dates:



weighted mean = 99.996 ± 0.098

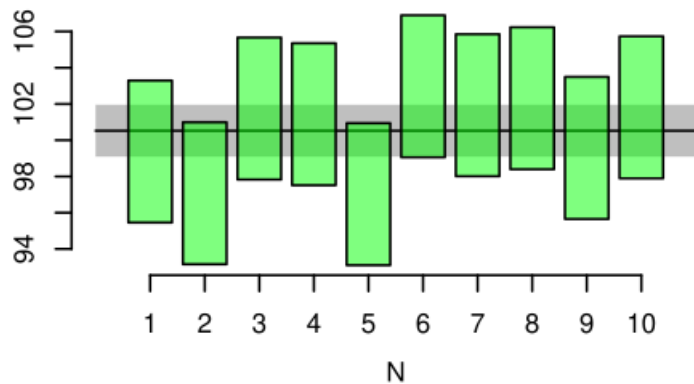


$$\chi^2 = \sum_{i=1}^n \frac{(t_i - \bar{t})^2}{s[t_i]^2} \quad MSWD = \frac{\chi^2}{n-1}$$

MSWD ≈ 1

A

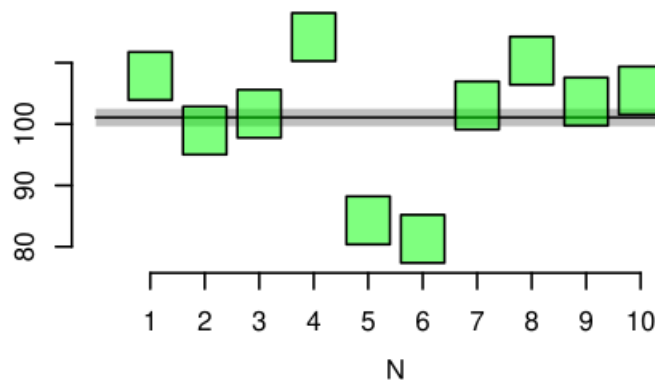
mean = 100.53 ± 1.43
MSWD = 1.16, $p(\chi^2) = 0.32$



MSWD $\gg 1$ (overdispersion)

B

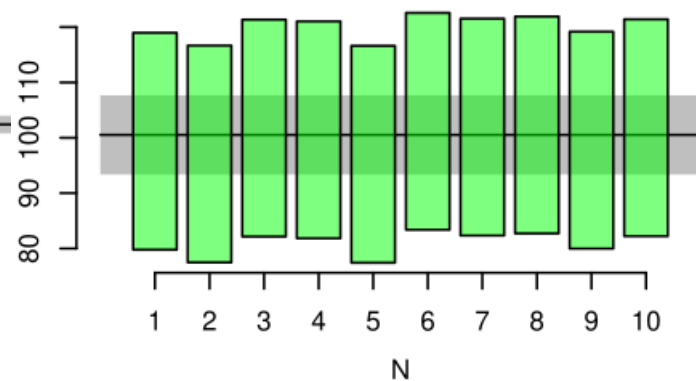
mean = $101.07 \pm 1.43 \mid 7.58$
MSWD = 28.1, $p(\chi^2) = 0$

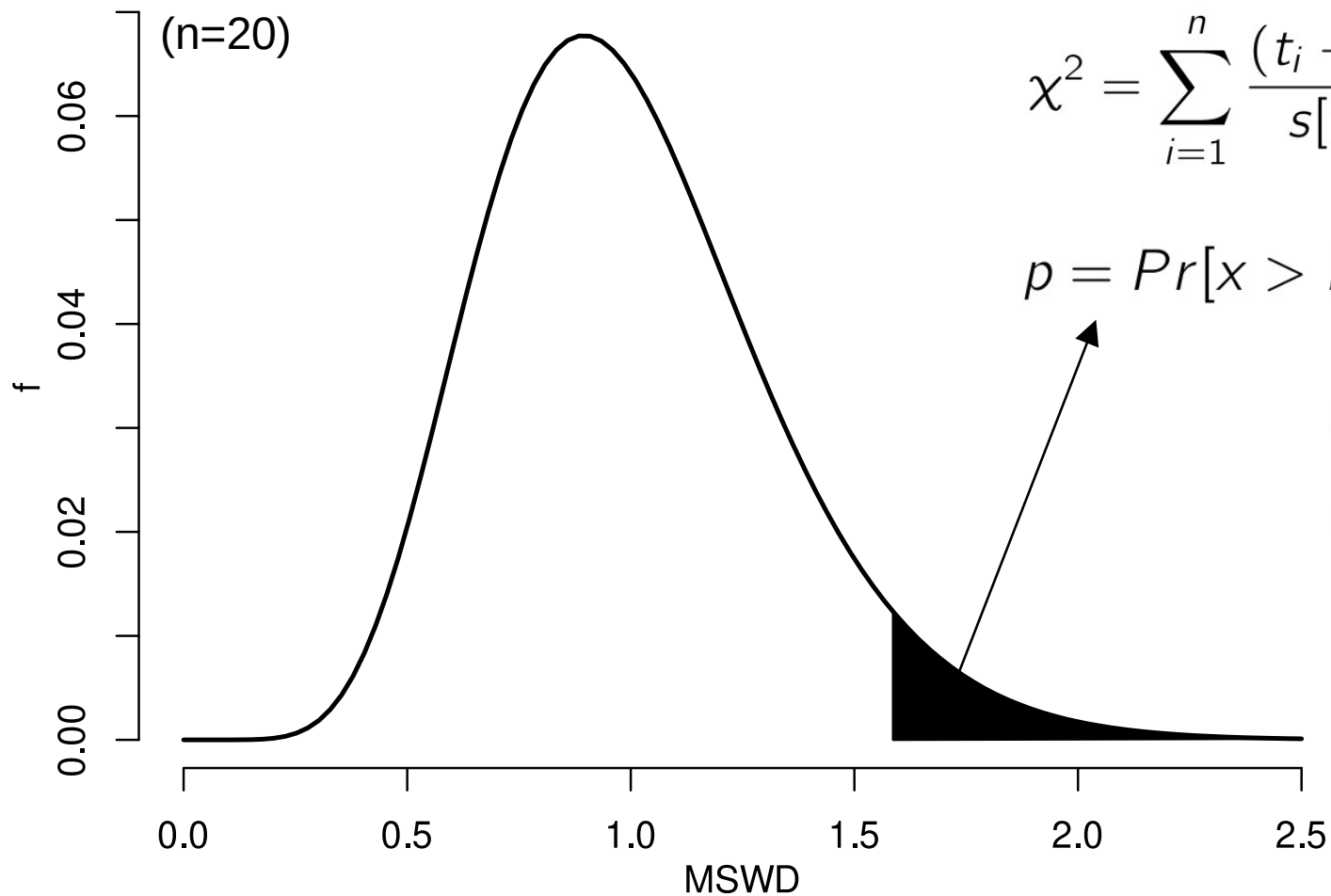


MSWD ≈ 0 (underdispersion)

C

mean = 100.53 ± 7.15
MSWD = 0.046, $p(\chi^2) = 1.00$





$$\chi^2 = \sum_{i=1}^n \frac{(t_i - \bar{t})^2}{s[t_i]^2} \quad MSWD = \frac{\chi^2}{n-1}$$

$$p = \Pr[x > MSWD]$$

$$MSWD \gg 1 \Leftrightarrow p \approx 0$$

$$MSWD \approx 1 \Leftrightarrow 0 < p < 1$$

$$MSWD \approx 0 \Leftrightarrow p \approx 1$$

$p < 0.05 \Rightarrow \text{overdispersed}$

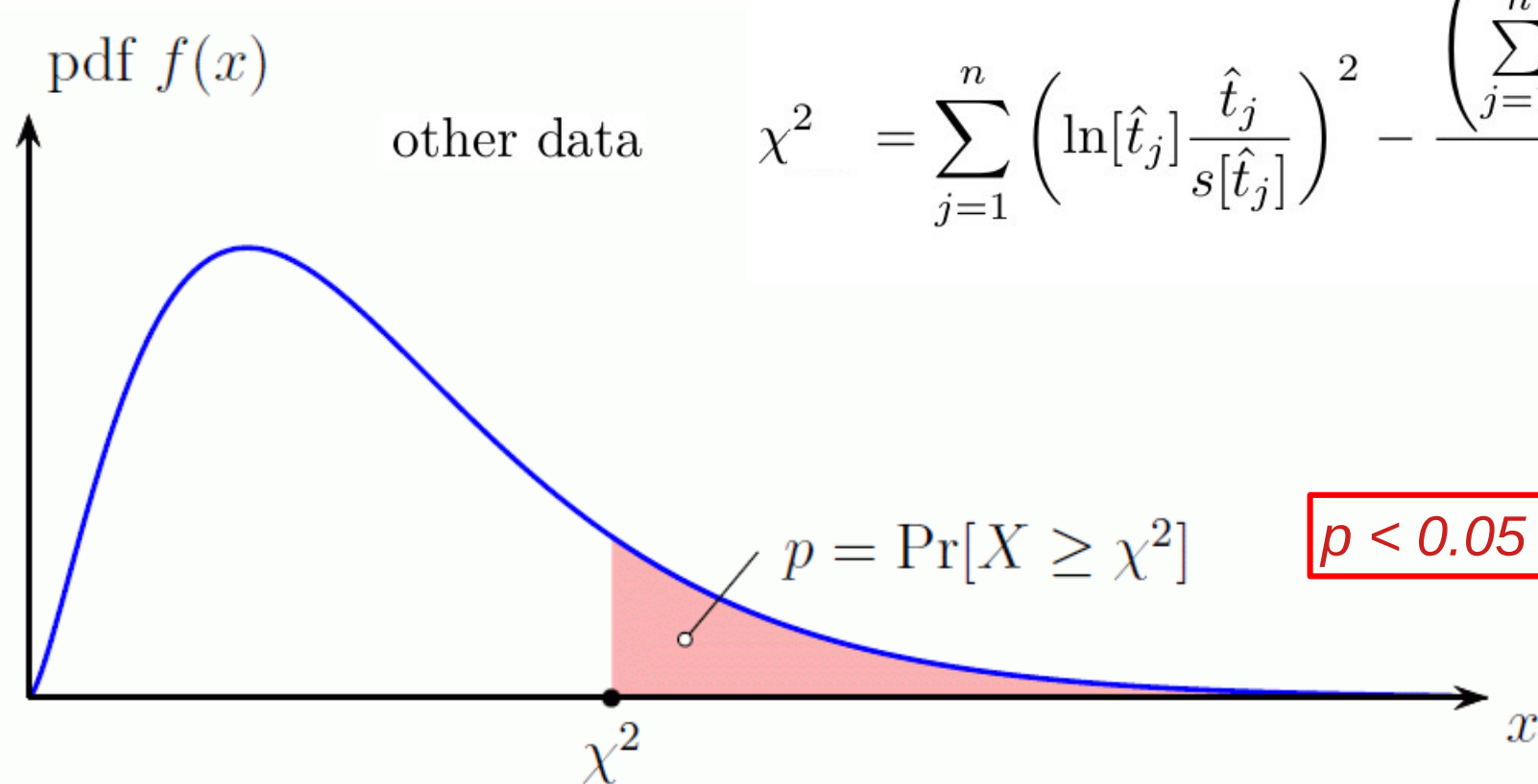
(MSWD > 1.6)

EDM

$$\chi^2 = \frac{1}{N_s N_i} \sum_{j=1}^n \frac{(N_{sj} N_i - N_{ij} N_s)^2}{N_{sj} + N_{ij}}$$

other data

$$\chi^2 = \sum_{j=1}^n \left(\ln[\hat{t}_j] \frac{\hat{t}_j}{s[\hat{t}_j]} \right)^2 - \frac{\left(\sum_{j=1}^n \ln[\hat{t}_j] \left[\frac{s[\hat{t}_j]}{\hat{t}_j} \right]^2 \right)^2}{\sum_{j=1}^n \left(\frac{\hat{t}_j}{s[\hat{t}_j]} \right)^2}$$



$p < 0.05 \Rightarrow \text{overdispersed}$

Research paper

Geoscience Frontiers 7 (2016) 581–589

Strategies towards statistically robust interpretations of *in situ* U–Pb zircon geochronology

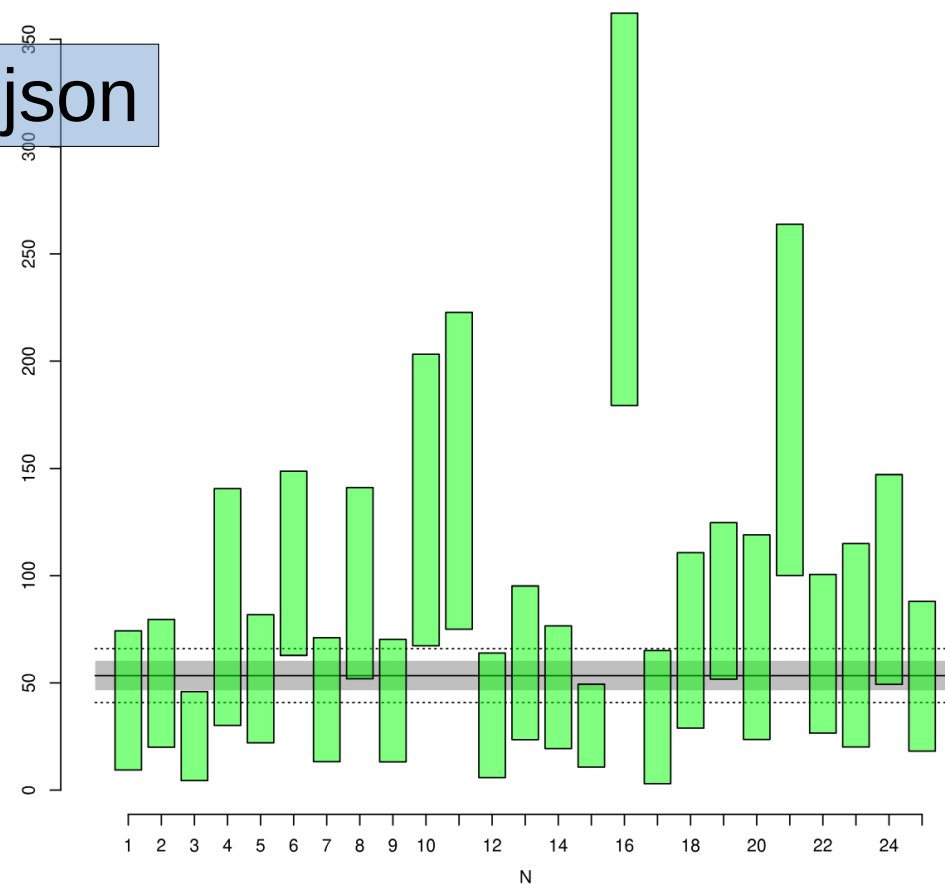
Christopher J. Spencer^{a,*}, Christopher L. Kirkland^{b,c}, Richard J.M. Taylor^a

*“... with an n of 20 the acceptable [MSWD] is 0.35-1.65 [...] Weighted averages with [MSWD] values that fall outside of this range **should not be used**”*

ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²mean = 53.4 ± 6.9 | 13.2 Ma (25/25)
MSWD = 3.3, $p(\chi^2) = 5.9e-08$

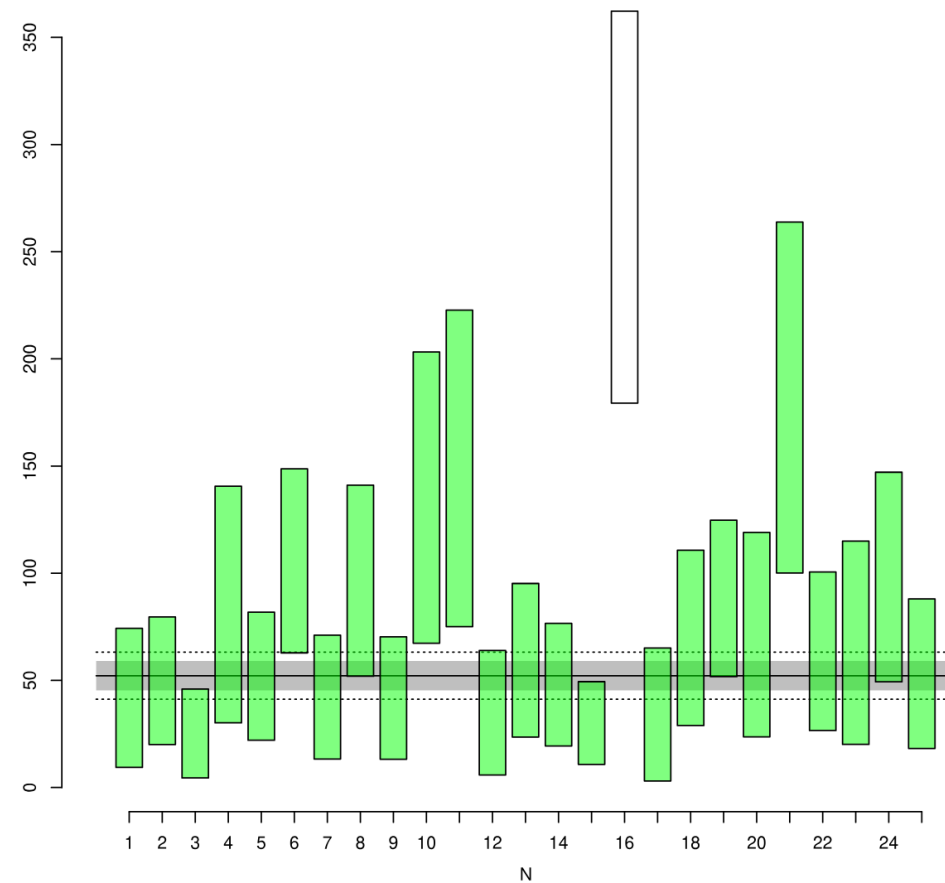
	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	7	73										
2	12	105										
3	6	104										
4	11	56										
5	13	109										
6	29	119										
7	9	93										
8	22	99										
9	9	94										
10	20	64										
11	21	61										
12	6	75										
13	12	88										
14	12	109										
15	10	145										
16	55	87										
17	5	64										
18	13	81										
19	27	133										
20	10	61										
21	27	64										
22	13	89										
23	9	58										
24	19	84										
25	10	82										
26												
27												
28												

FT3.json



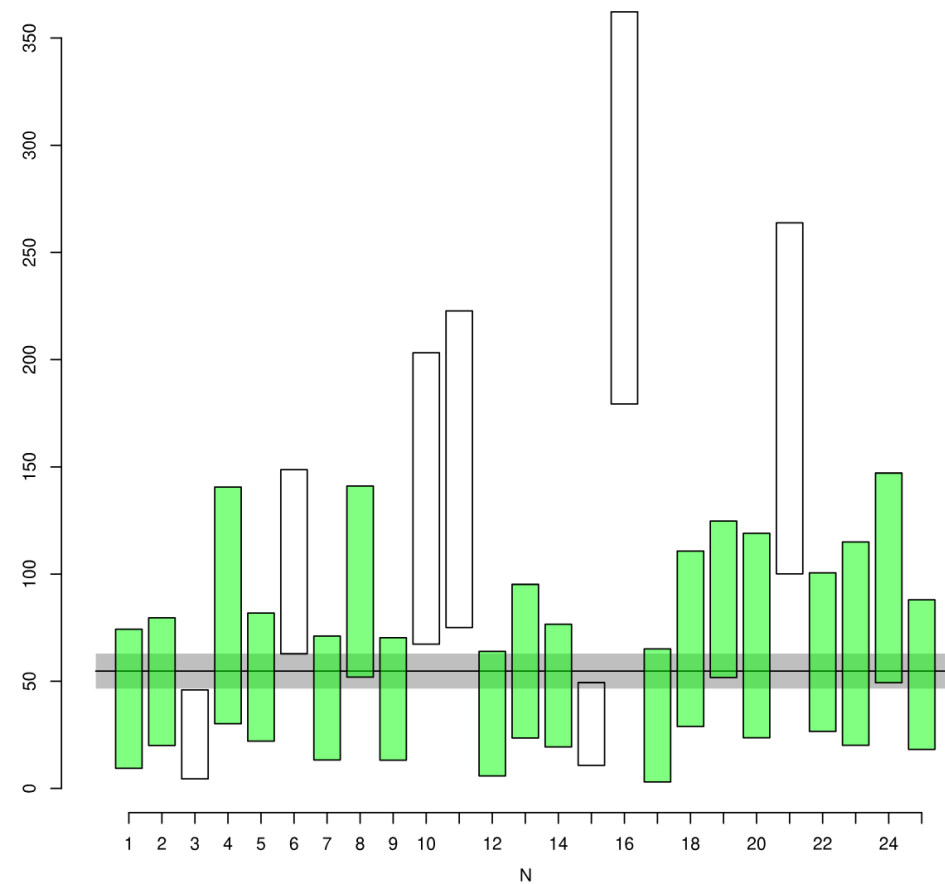
ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²mean = 52.2 ± 6.9 | 11.6 Ma (24/25)
MSWD = 2.5, $p(\chi^2) = 6.8\text{e-}05$

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	7	73										
2	12	105										
3	6	104										
4	11	56										
5	13	109										
6	29	119										
7	9	93										
8	22	99										
9	9	94										
10	20	64										
11	21	61										
12	6	75										
13	12	88										
14	12	109										
15	10	145										
16	55	87	x									
17	5	64										
18	13	81										
19	27	133										
20	10	61										
21	27	64										
22	13	89										
23	9	58										
24	19	84										
25	10	82										
26												
27												
28												



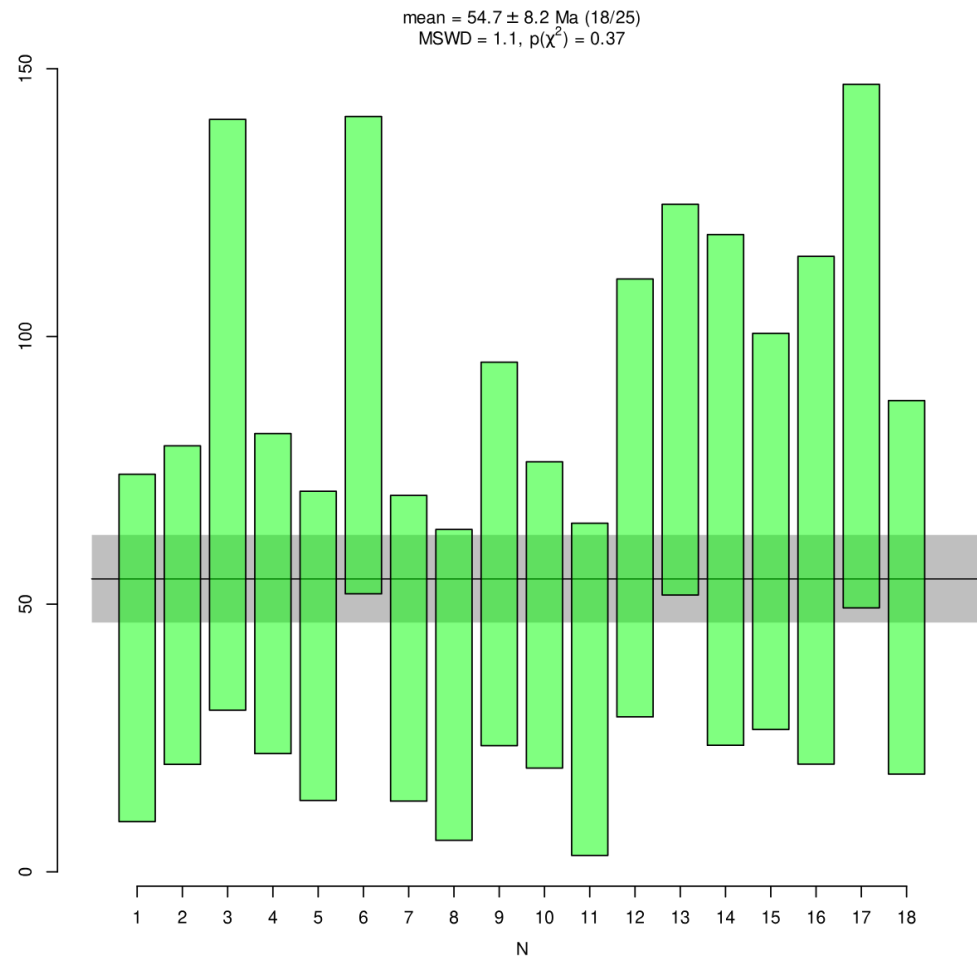
ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²mean = 54.7 ± 8.2 Ma (18/25)
MSWD = 1.1, $p(\chi^2) = 0.37$

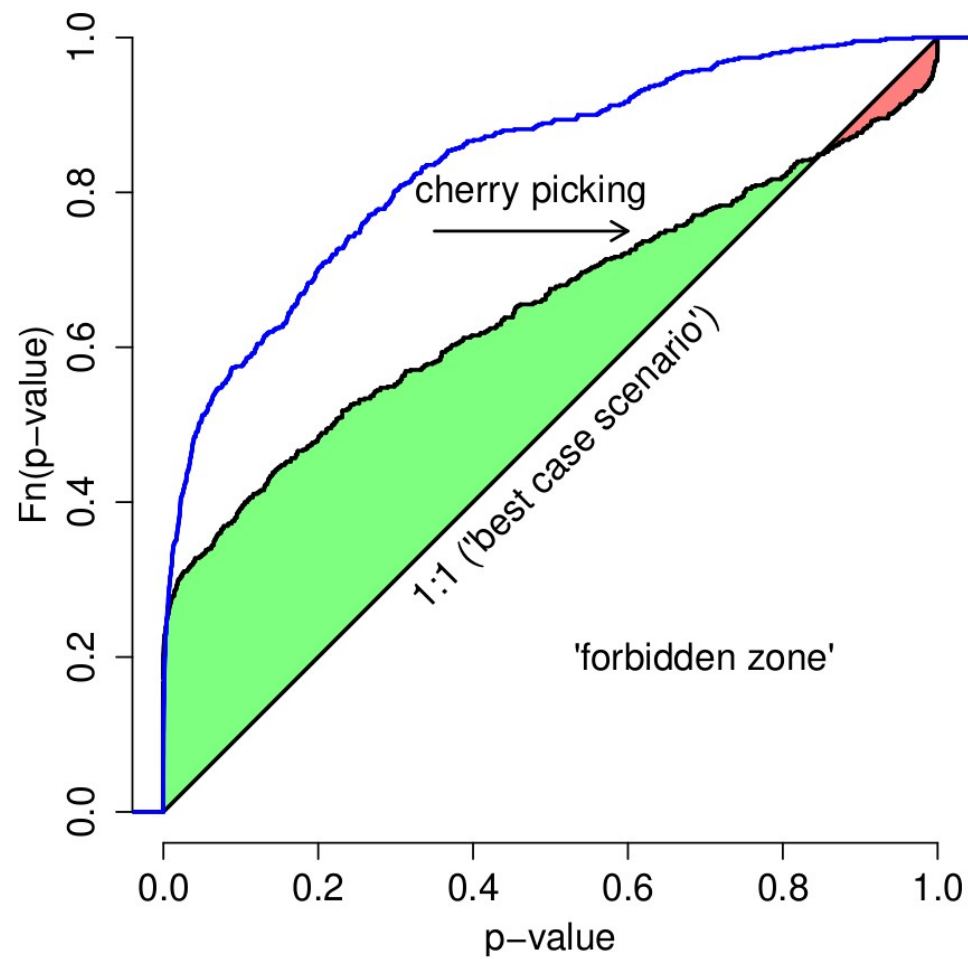
	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	7	73										
2	12	105										
3	6	104		x								
4	11	56										
5	13	109										
6	29	119		x								
7	9	93										
8	22	99										
9	9	94										
10	20	64		x								
11	21	61		x								
12	6	75										
13	12	88										
14	12	109										
15	10	145		x								
16	55	87		x								
17	5	64										
18	13	81										
19	27	133										
20	10	61										
21	27	64		x								
22	13	89										
23	9	58										
24	19	84										
25	10	82										
26												
27												
28												



ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	7	73										
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3	6	104		X								
4	11	56										
5	13	109										
6	29	119		X								
7	9	93										
8	22	99										
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10	20	64		X								
11	21	61		X								
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15	10	145		X								
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Research paper

Geoscience Frontiers 7 (2016) 581–589

Strategies towards statistically robust interpretations of *in situ* U–Pb zircon geochronology

Christopher J. Spencer^{a,*}, Christopher L. Kirkland^{b,c}, Richard J.M. Taylor^a

~~“... with an n of 20 the acceptable [MSWD] is 0.35-1.65 [...] Weighted averages with [MSWD] values that fall outside of this range **should not be used**”~~

PERSPECTIVE

National Science Review

12: nwaf036, 2025

<https://doi.org/10.1093/nsr/nwaf036>

Advance access publication 7 February 2025

EARTH SCIENCES

Special Topic: Geochronology

High MSWDs are not the problem, low ones are

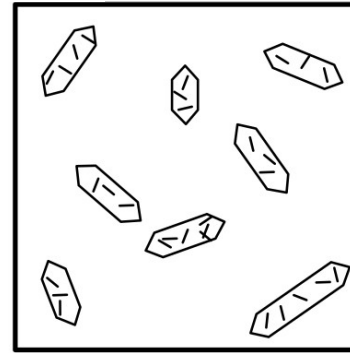
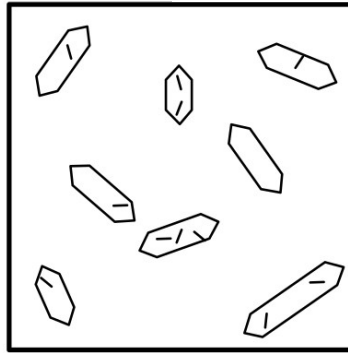
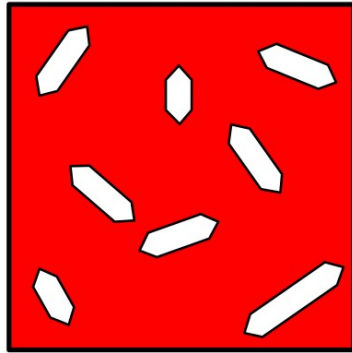
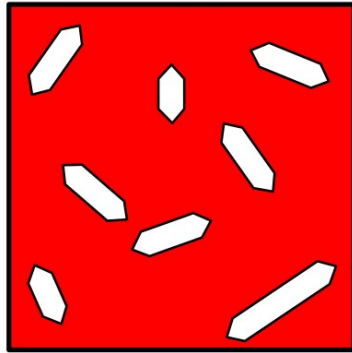
Pieter Vermeesch ¹

Use this paper to defend yourself against misinformed reviewers

What it means to NOT be overdispersed:

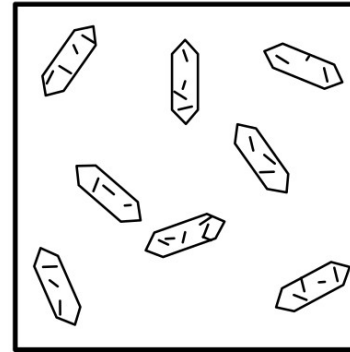
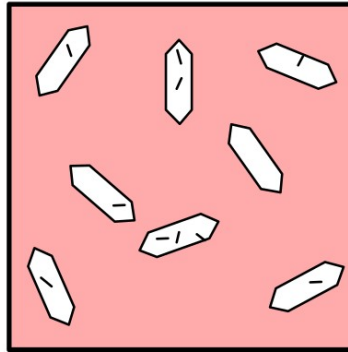
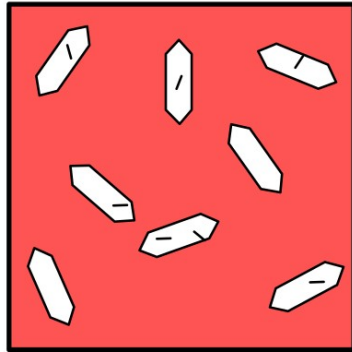
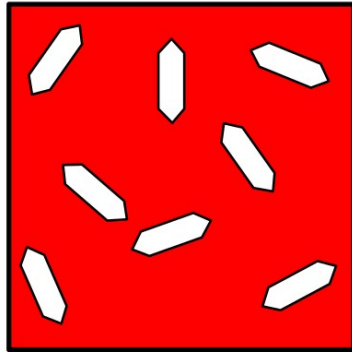


Either:



* Cooled in 1 second

Or:



All grains have the same:

- * chemical composition
- * surface to volume ratio
- * dislocation density
- * fluid inclusion density
- * ...

300 Ma

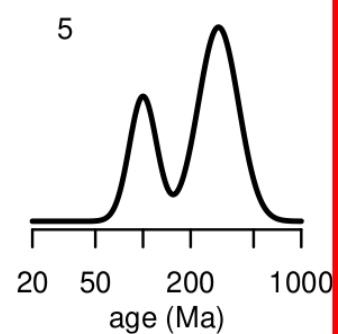
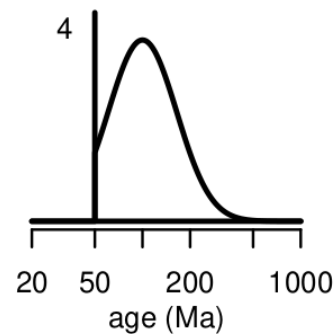
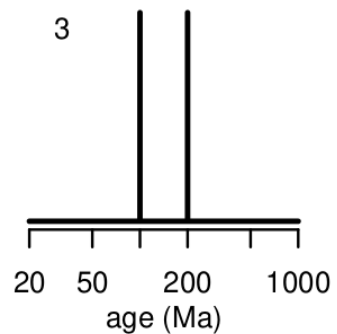
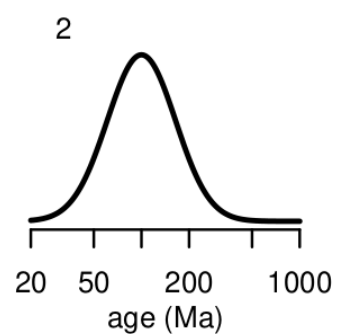
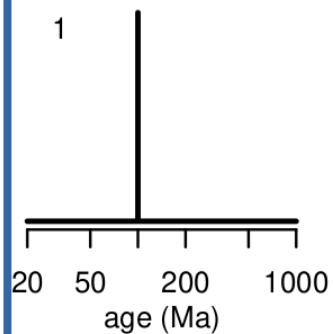
200 Ma

100 Ma

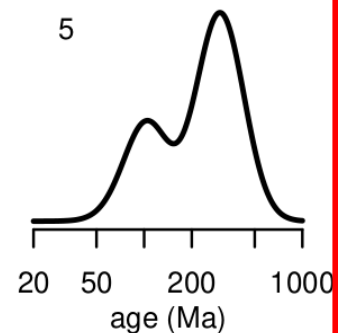
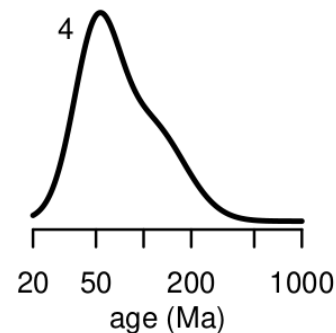
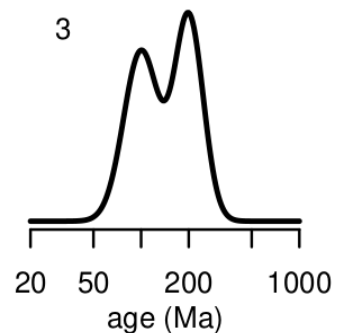
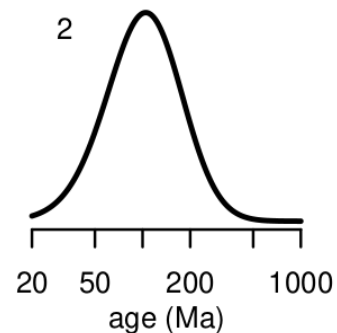
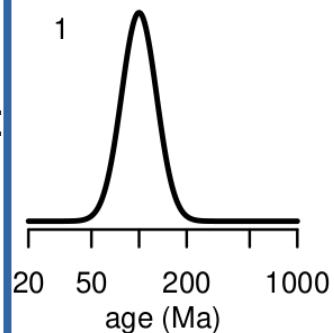
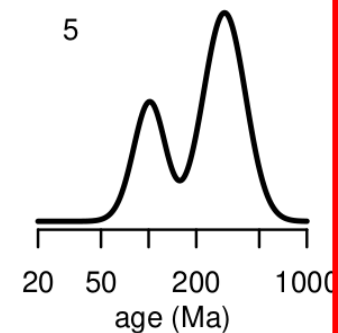
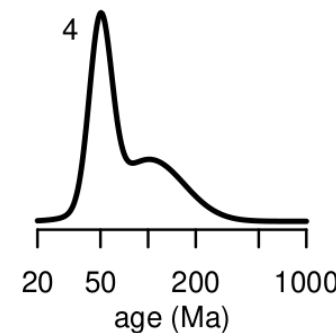
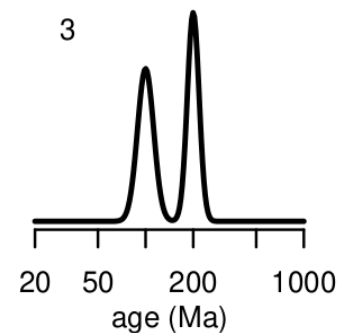
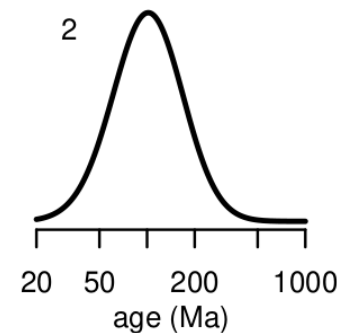
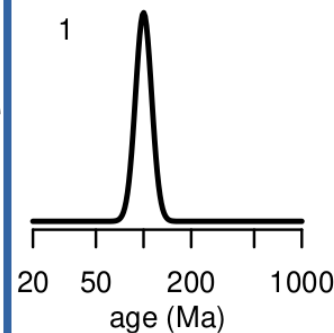
today

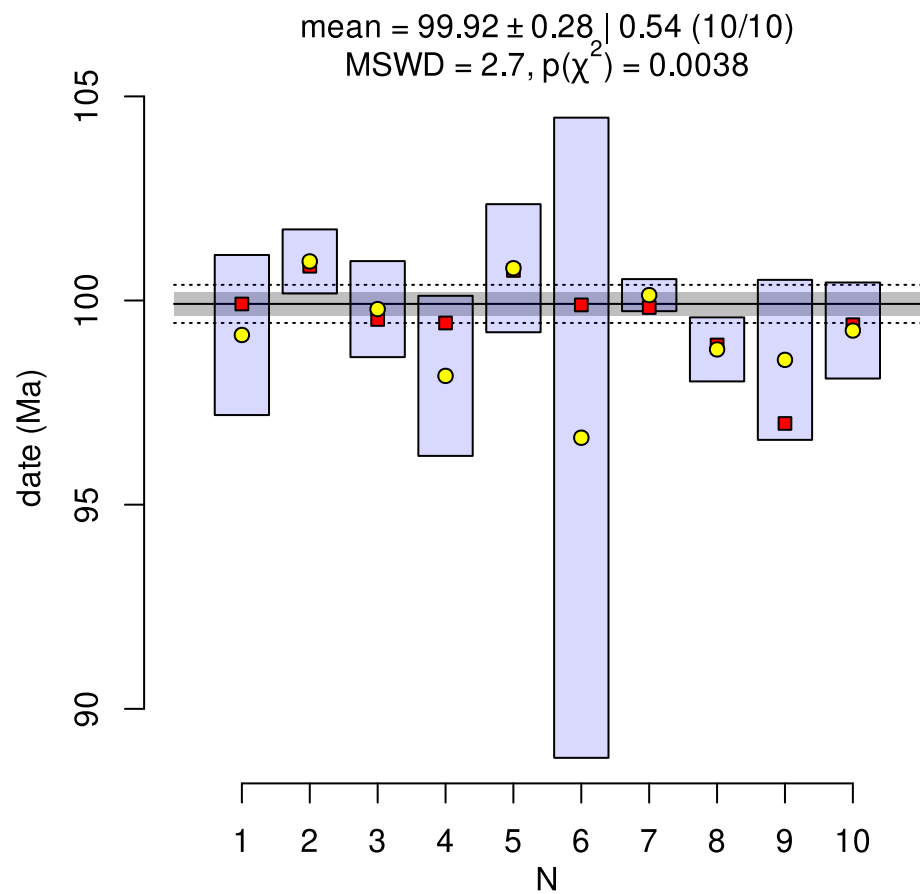
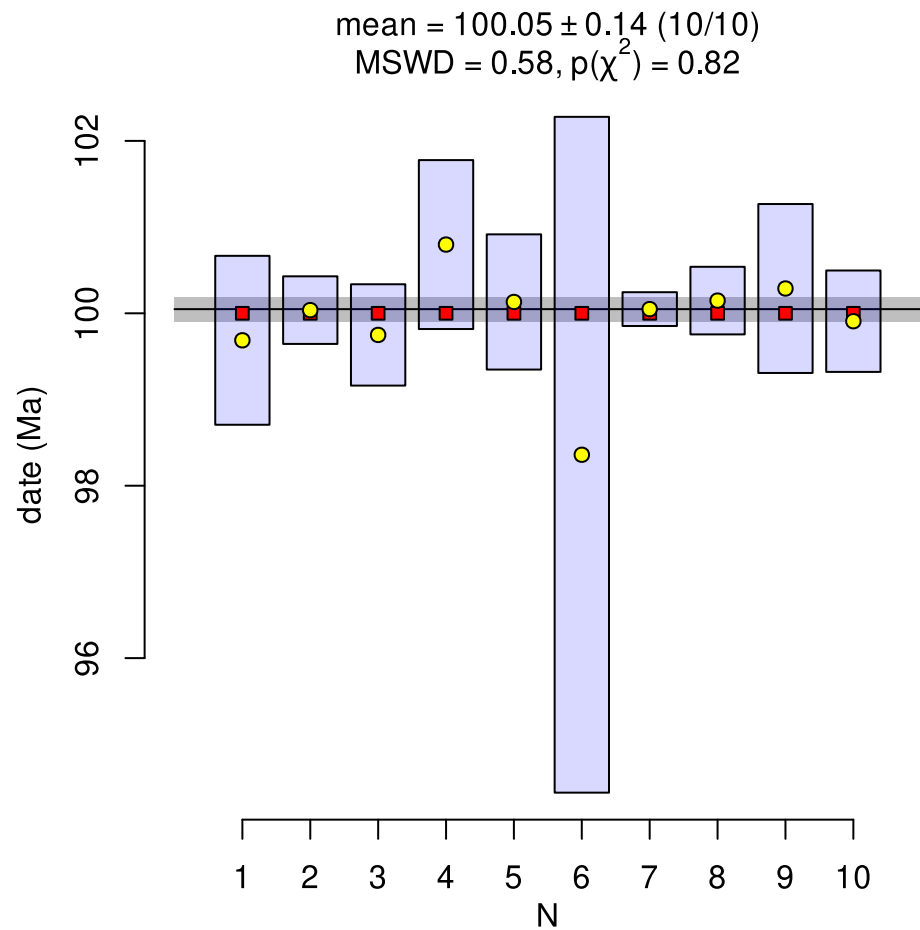
MSWD $\approx$ 1MSWD $\gg$ 1

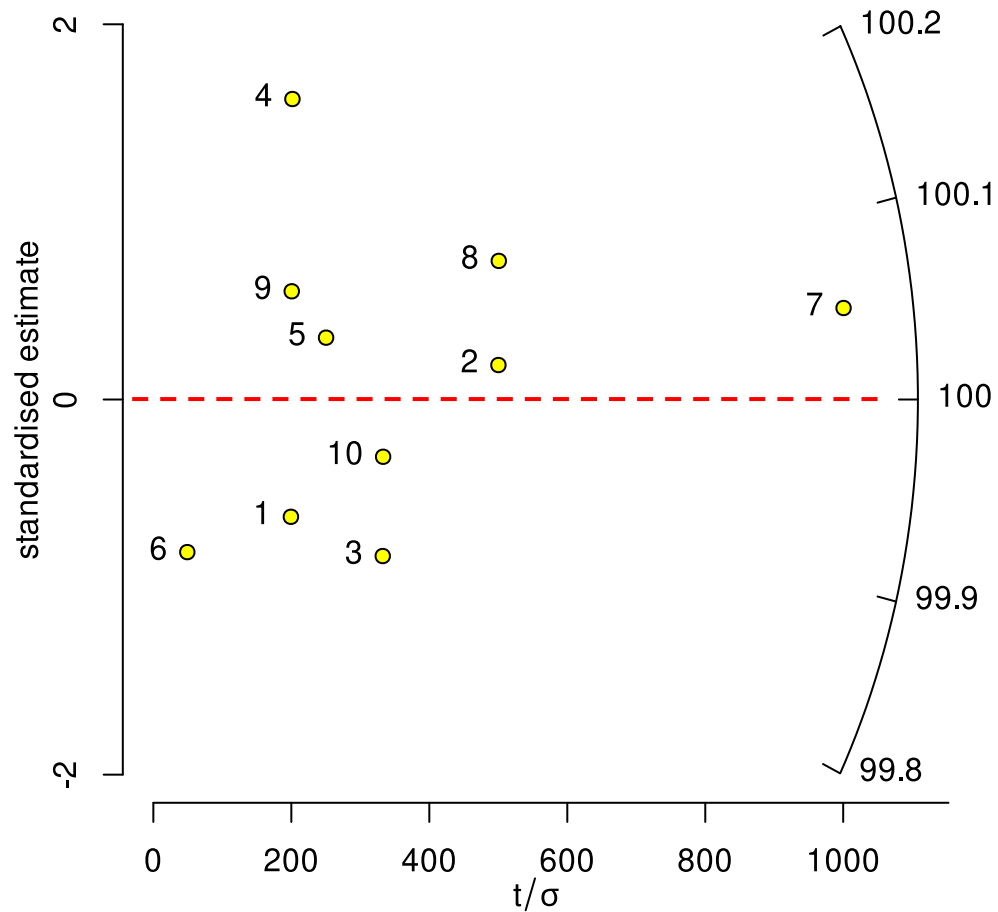
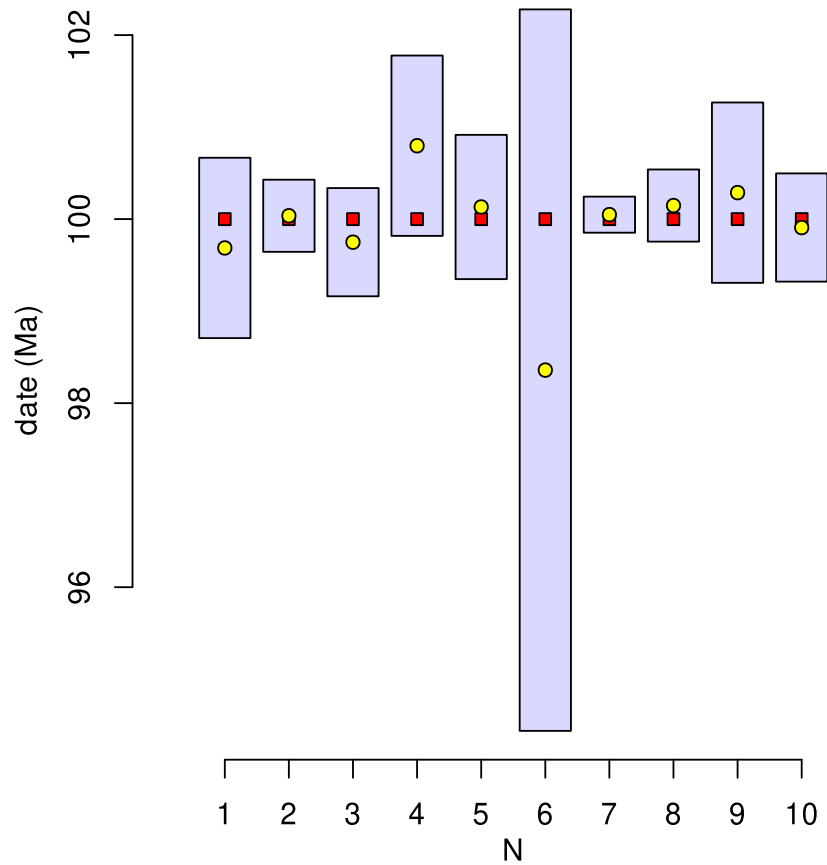
Ages:



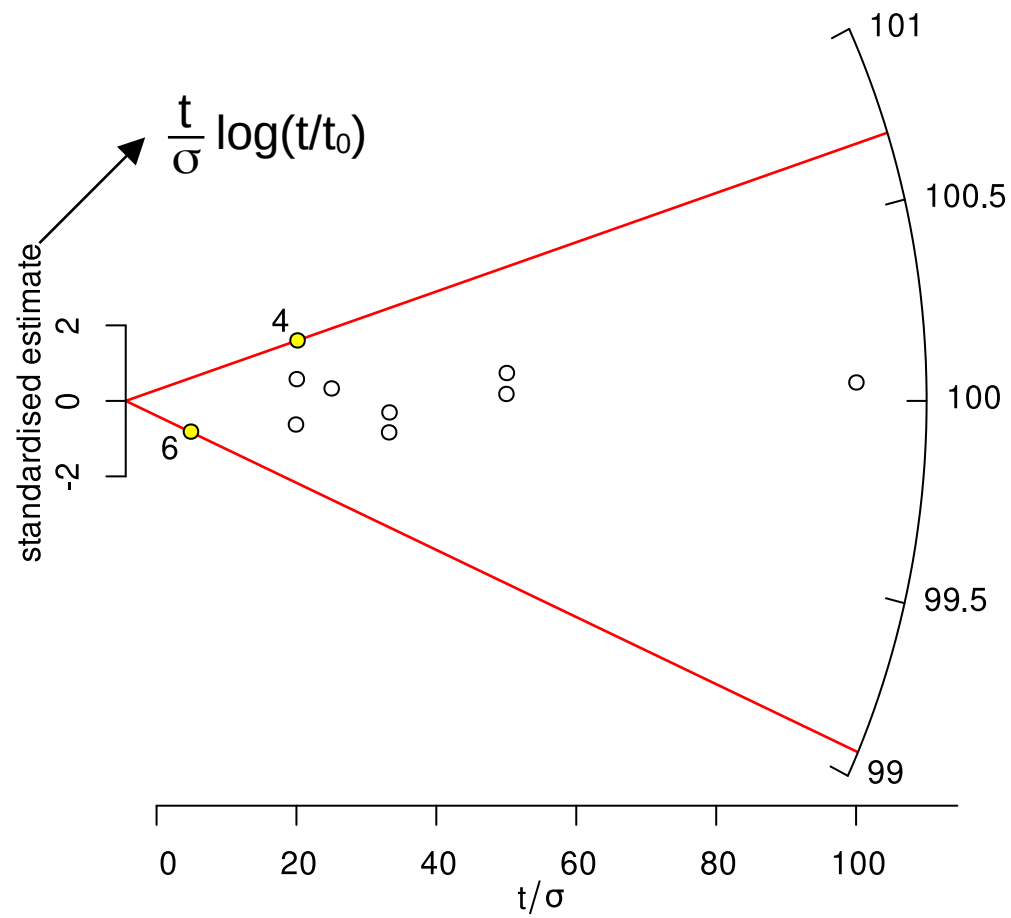
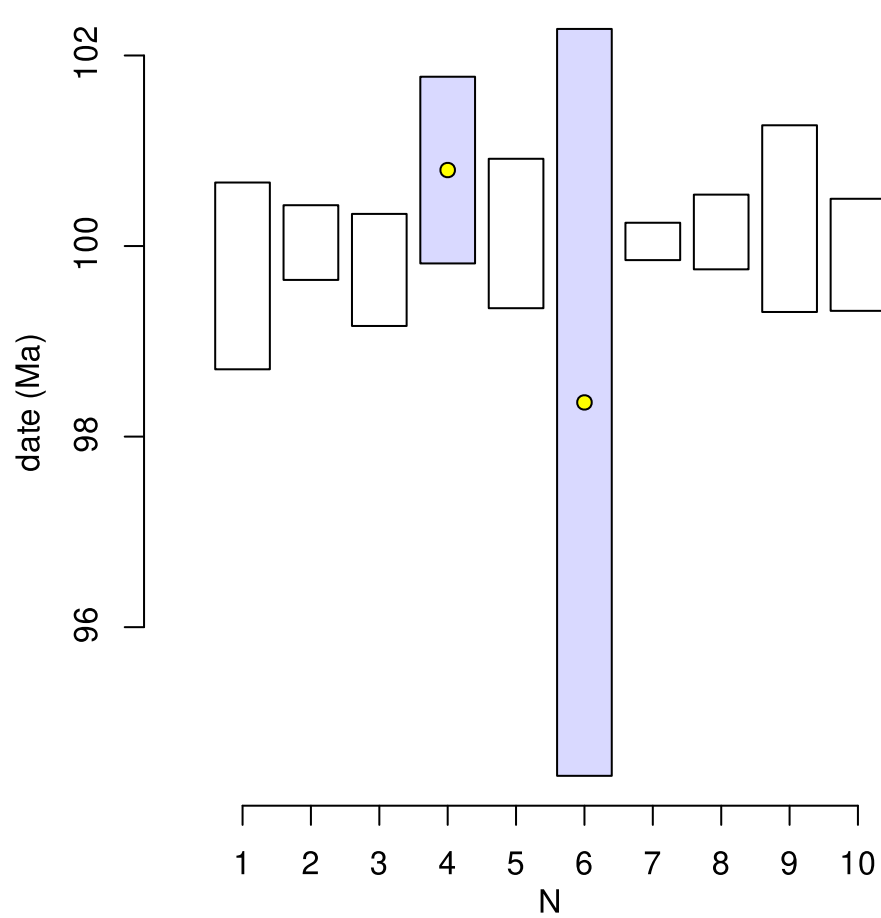
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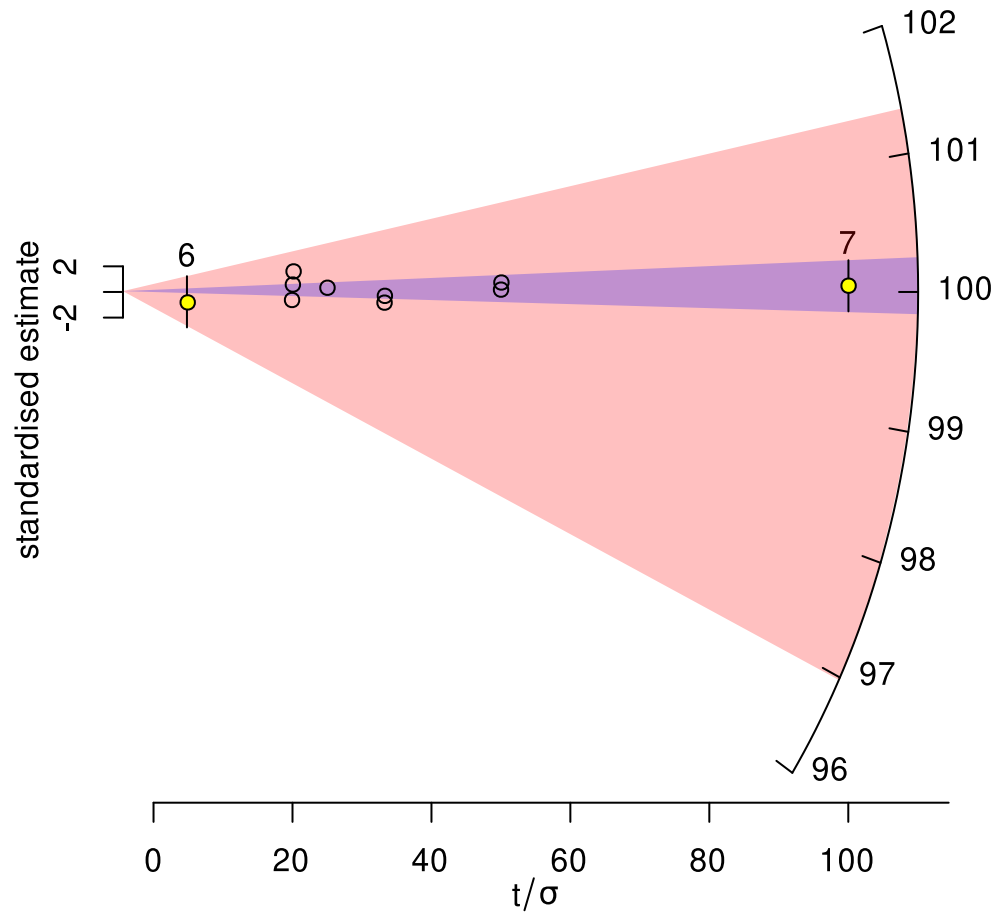
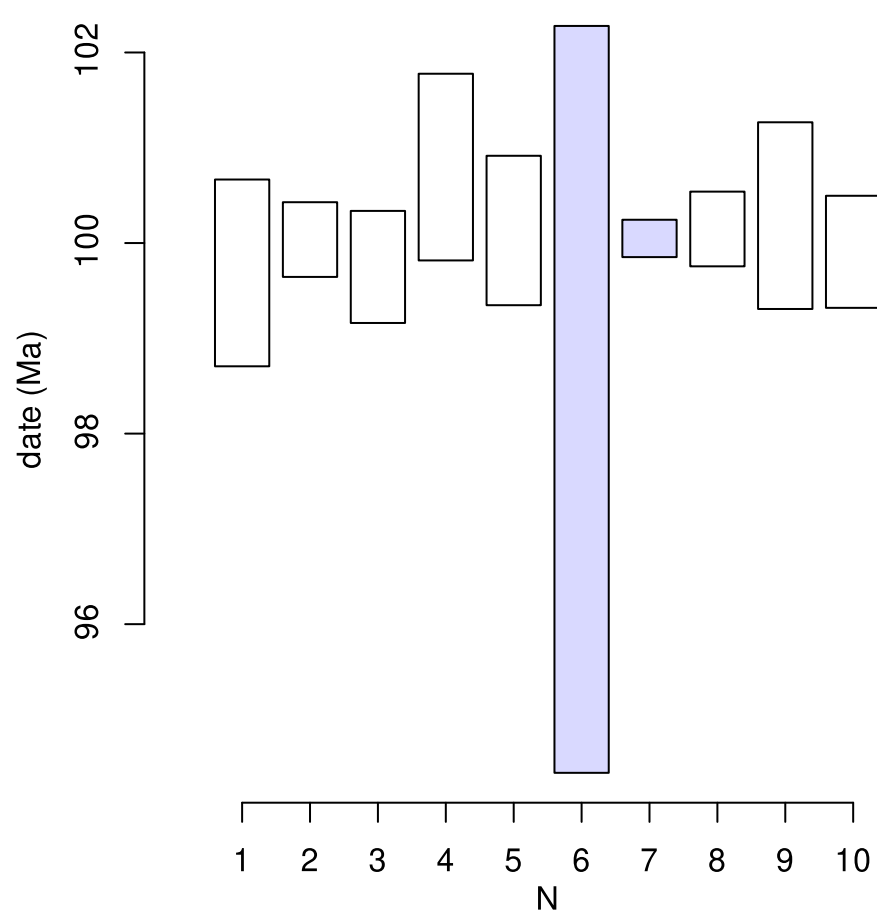
U-Th-He
dates:

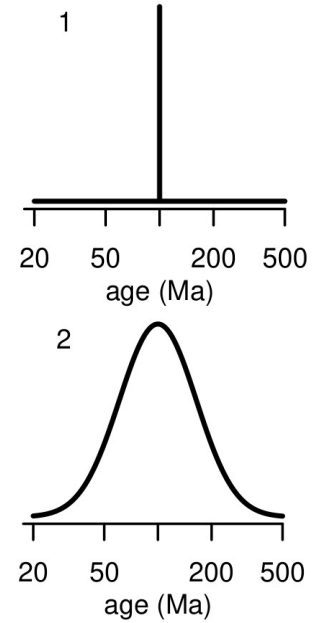
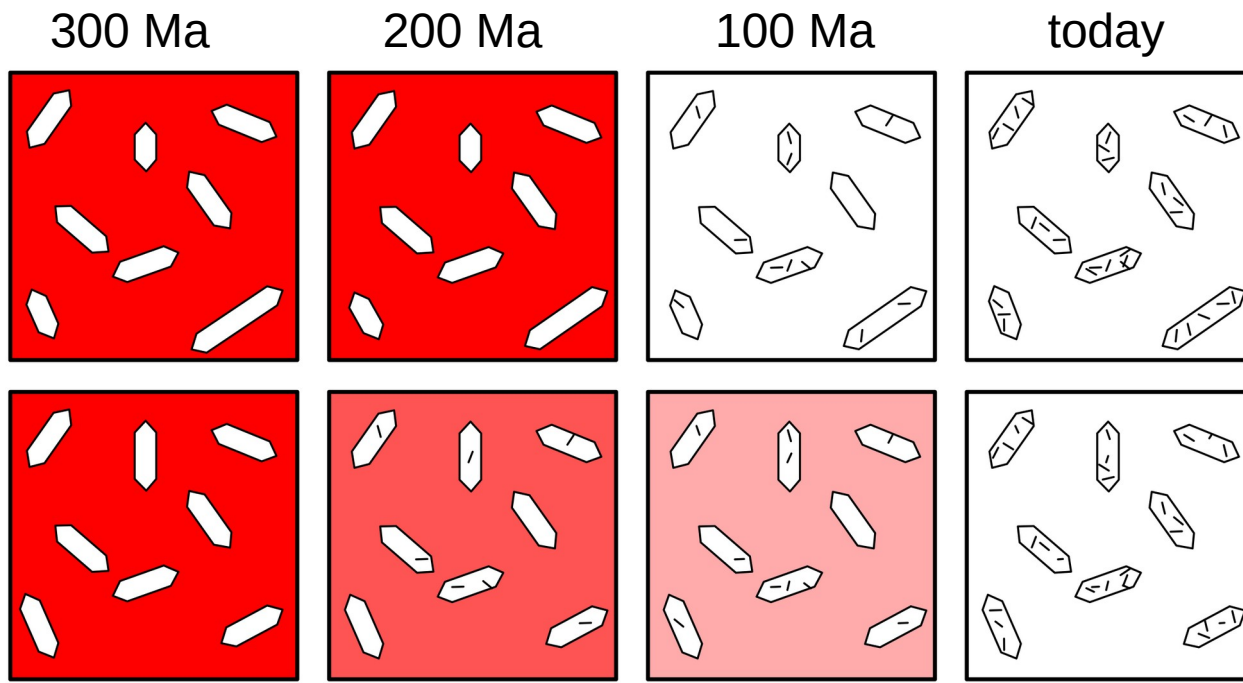




THE RADIAL PLOT: GRAPHICAL ASSESSMENT OF SPREAD IN AGES







EDM

$$\ln[\rho_s/\rho_i] \sim \mathcal{N}(\mu, \sigma^2)$$

other data

$$\ln[t] \sim \mathcal{N}(\mu, \sigma^2)$$

$\hat{\mu}$ and $\hat{\sigma}$

$\exp[\hat{\mu}]$

Central age

dispersion

STATISTICAL MODELS FOR MIXED FISSION TRACK AGES

ζ : 350 ± 10 yr cm² ρ_D : 2500000 ± 10000 1/cm²default
dataset

pooled age = 103.5 ± 7.4 Ma (n=25)
central age = 103.5 ± 9.4 Ma
MSWD = 0.72, $p(\chi^2) = 0.84$

pooled = central

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J
1	38	132								
2	48	187								
3	114	565								
4	13	51								
5	25	122								
6	49	249								
7	39	159								
8	18	50								
9	26	80								
10	24	79								
11	55	217								
12	38	152								
13	61	231								
14	82	347								
15	16	75								
16	38	135								
17	42	186								
18	31	123								
19	15	61								
20	27	107								
21	34	135								
22	6	43								
23	48	215								
24	16	52								
25	31	165								
26										
27										
28										

standardised estimate

2
0
-20 2 4 6 8 10
 t/σ

140

120

100

80

61

ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²

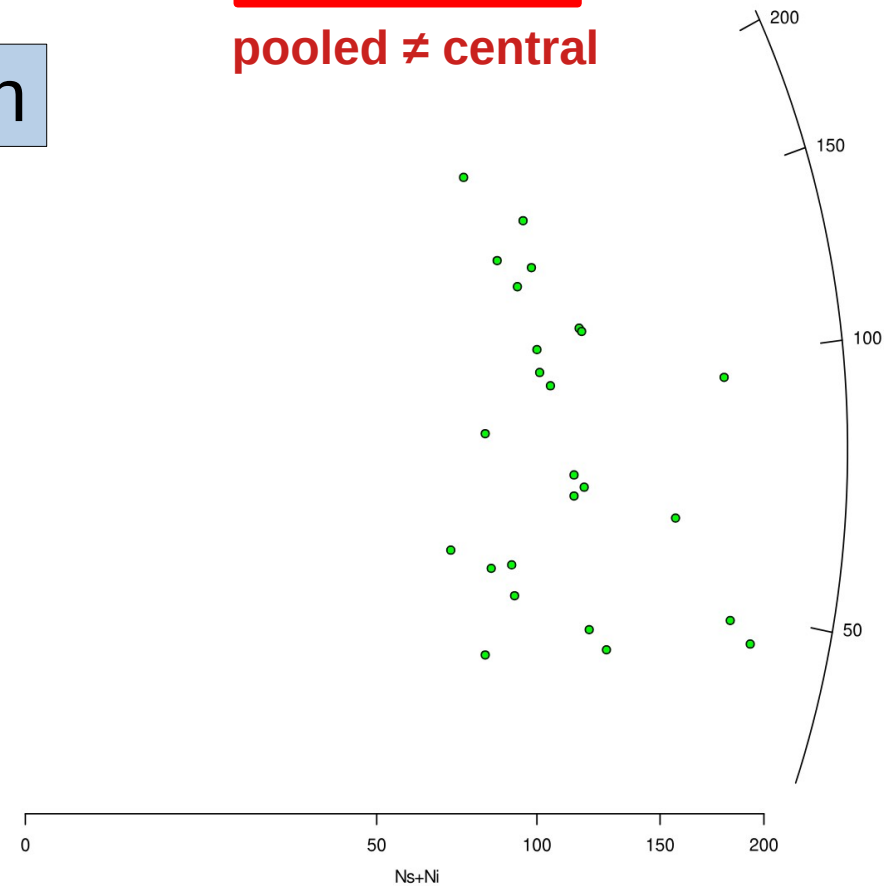
pooled age = 78.9 ± 8.1 Ma (n=25)
central age = 82 ± 16 Ma
MSWD = 3.7, $p(\chi^2) = 1.7\text{e-}09$
dispersion = $43 \pm 17\%$

pooled ≠ central

FT2.json

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	22	64										
2	8	84										
3	16	98										
4	22	71										
5	10	110										
6	13	69										
7	5	77										
8	20	80										
9	24	92										
10	15	99										
11	20	137										
12	19	86										
13	24	74										
14	24	51										
15	19	82										
16	8	76										
17	32	148										
18	10	117										
19	7	64										
20	26	69										
21	18	165										
22	9	82										
23	18	175										
24	16	102										
25	24	93										
26												
27												
28												

standardised estimate



ζ : 9700 ± 50 Myr μm^2 spot size: 35 μm Method: ICP (ζ) ▾

Input errors: 1se (abs) ▾

Output errors: ci (abs) ▾

Probability cutoff: 0.05

☐ Show sample numbers?☐ Propagate external uncertainties?

Transformation: logarithmic ▾

Finite mixtures: 0 (none) ▾

Minimum value: auto

Central value: auto

Maximum value: auto

Maximum precision: auto

Plot character: 21

Significant digits: 2

Fill colour: Custom colour ramp ▾

from:  to: 

Plot symbol magnification: 1

Font magnification: 1

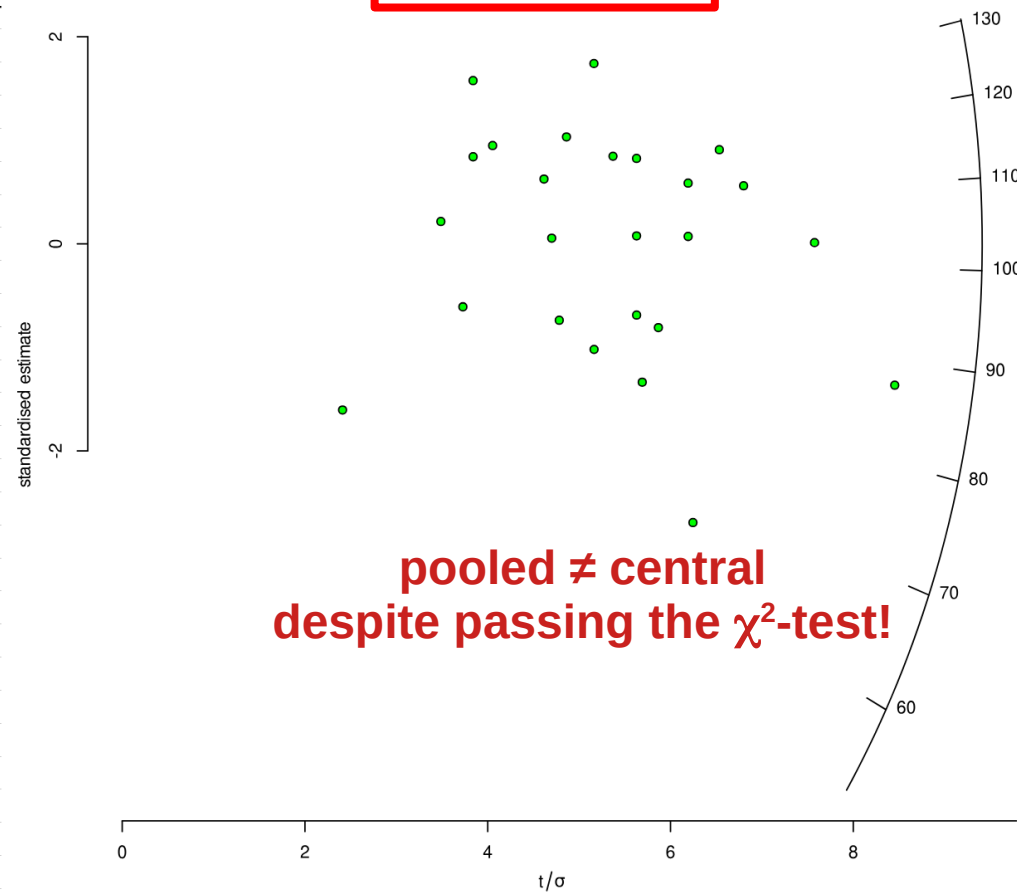
Colour label:

	Ns	A	U1	err[U1]	(U2)	(err[U2])	(U3)	(err[U3])	(U4)	(err[U4])
1	38	4410	0.2848	0.0053	0.4017	0.0063	0.368	0.0061		
2	48	4410	0.4907	0.007	0.4491	0.0067	0.4506	0.0067		
3	114	4410	1.1073	0.0105	1.7847	0.0134	1.4605	0.0121		
4	13	4410	0.1246	0.0035	0.1575	0.004	0.111	0.0033		
5	25	4410	0.3064	0.0055	0.2439	0.0049	0.2414	0.0049		
6	49	4410	0.7553	0.0087	0.8813	0.0094	0.7768	0.0088		
7	39	4410	0.6477	0.008	0.3851	0.0062	0.5782	0.0076		
8	18	1470	0.4525	0.0067						
9	26	2940	0.6265	0.0079	0.3733	0.0061				
10	24	1470	0.6665	0.0082						
11	55	4410	0.565	0.0075	0.3496	0.0059	0.661	0.0081		
12	38	2940	0.5624	0.0075	0.6332	0.008				
13	61	4410	0.5469	0.0074	0.562	0.0075	0.6882	0.0083		
14	82	4410	1.1055	0.0105	0.6209	0.0079	0.9559	0.0098		
15	16	1470	0.3365	0.0058						
16	38	2940	0.826	0.0091	0.5669	0.0075				
17	42	2940	0.749	0.0087	0.7883	0.0089				
18	31	4410	0.2188	0.0047	0.2904	0.0054	0.2018	0.0045		
19	15	1470	0.563	0.0075						
20	27	1470	0.6939	0.0083						
21	34	2940	0.5107	0.0071	0.4176	0.0065				
22	6	1470	0.3732	0.0061						
23	48	1470	1.5108	0.0123						
24	16	1470	0.4085	0.0064						
25	31	4410	0.4958	0.007	0.3928	0.0063	0.3323	0.0058		
26										
27										
28										

$\bar{Z}$: $\pm$ Myr μm^2 spot size: μm

	Ns	A	U1	err[U1]	(U2)	(err[U2])	(U3)	(err[U3])	(U4)	(err)
1	38	4410	0.2848	0.0053	0.4017	0.0063	0.368	0.0061		
2	48	4410	0.4907	0.007	0.4491	0.0067	0.4506	0.0067		
3	114	4410	1.1073	0.0105	1.7847	0.0134	1.4605	0.0121		
4	13	4410	0.1246	0.0035	0.1575	0.004	0.111	0.0033		
5	25	4410	0.3064	0.0055	0.2439	0.0049	0.2414	0.0049		
6	49	4410	0.7553	0.0087	0.8813	0.0094	0.7768	0.0088		
7	39	4410	0.6477	0.008	0.3851	0.0062	0.5782	0.0076		
8	18	1470	0.4525	0.0067						
9	26	2940	0.6265	0.0079	0.3733	0.0061				
10	24	1470	0.6665	0.0082						
11	55	4410	0.565	0.0075	0.3496	0.0059	0.661	0.0081		
12	38	2940	0.5624	0.0075	0.6332	0.008				
13	61	4410	0.5469	0.0074	0.562	0.0075	0.6882	0.0083		
14	82	4410	1.1055	0.0105	0.6209	0.0079	0.9559	0.0098		
15	16	1470	0.3365	0.0058						
16	38	2940	0.826	0.0091	0.5669	0.0075				
17	42	2940	0.749	0.0087	0.7883	0.0089				
18	31	4410	0.2188	0.0047	0.2904	0.0054	0.2018	0.0045		
19	15	1470	0.563	0.0075						
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22	6	1470	0.3732	0.0061						
23	48	1470	1.5108	0.0123						
24	16	1470	0.4085	0.0064						
25	31	4410	0.4958	0.007	0.3928	0.0063	0.3323	0.0058		
26										
27										
28										

pooled age = 99.7 ± 6.4 Ma (n=25)
central age = 102.4 ± 7.8 Ma
MSWD = 1.2, $p(\chi^2) = 0.26$



Contents

1 Introduction to R

- 1.1 RStudio
- 1.2 IsoplotR
- 1.3 Basic R

2 Ages vs. dates

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- 2.2 Populations

3 Plotting data

- 3.1 Kernel Density Estimates
- 3.2 Cumulative Age Distributions
- 3.3 Radial plots
- 3.4 Helioplots

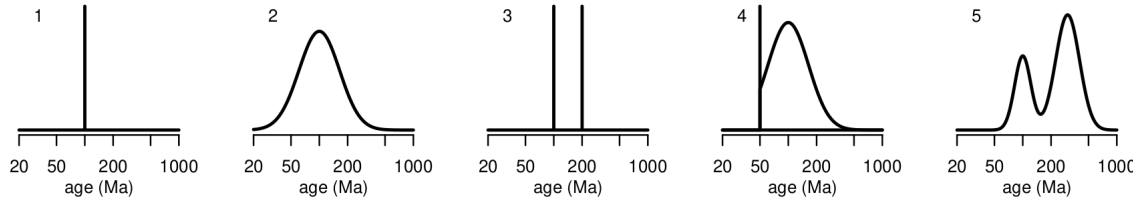
4 Mixture models

- 4.1 The χ^2 test for homogeneity
- 4.2 The random effects model
- 4.3 Finite mixtures
- 4.4 How many components to fit?
- 4.5 The minimum age model
- 4.6 Logratios

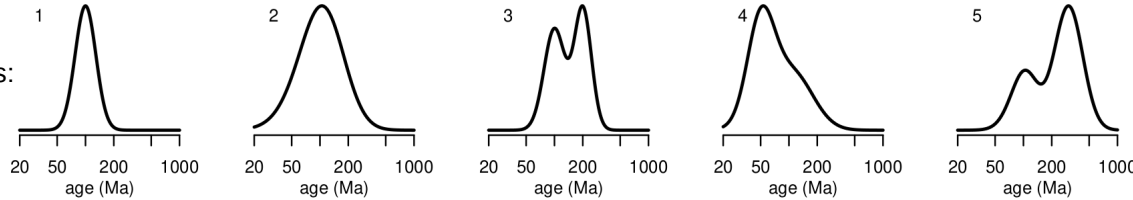
5 Mineral fertility estimation with provenance

- 5.1 Grain size estimation
- 5.2 Source Rock Density (SRD) correction

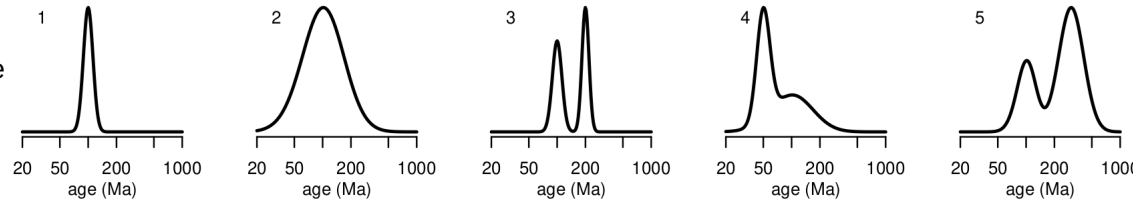
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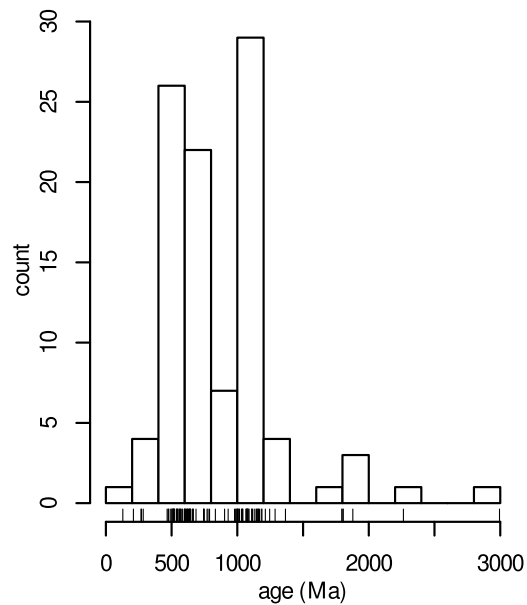
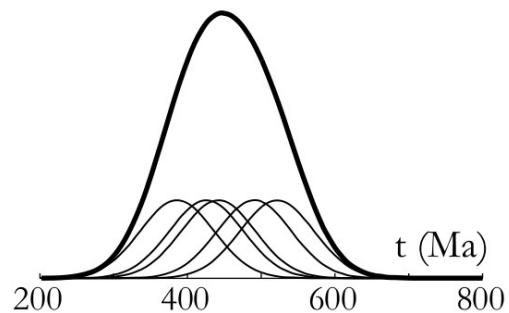


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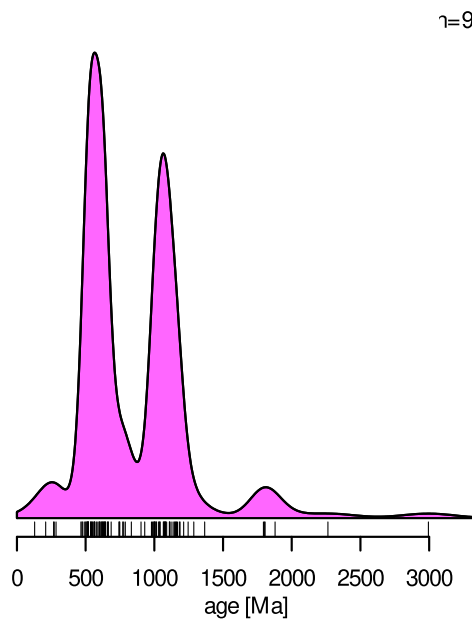


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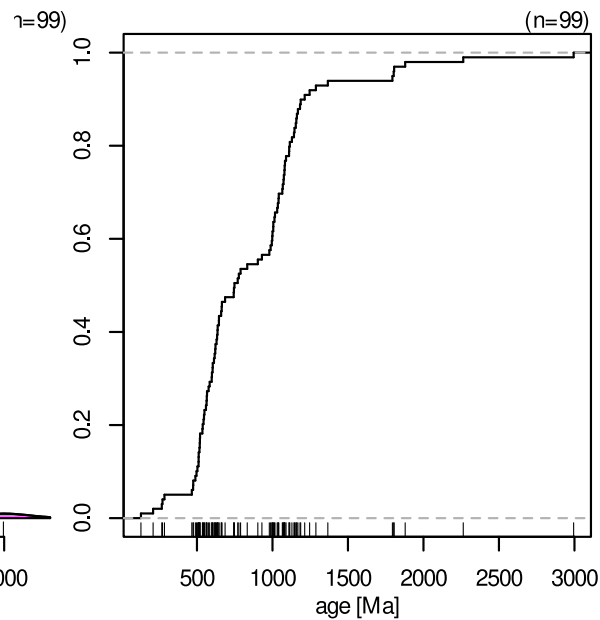




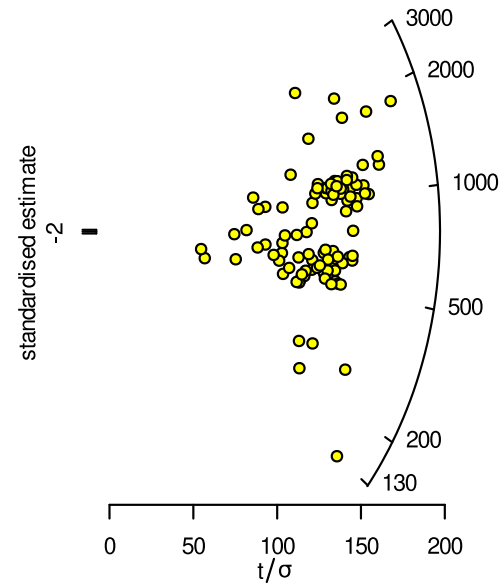
histogram



KDE

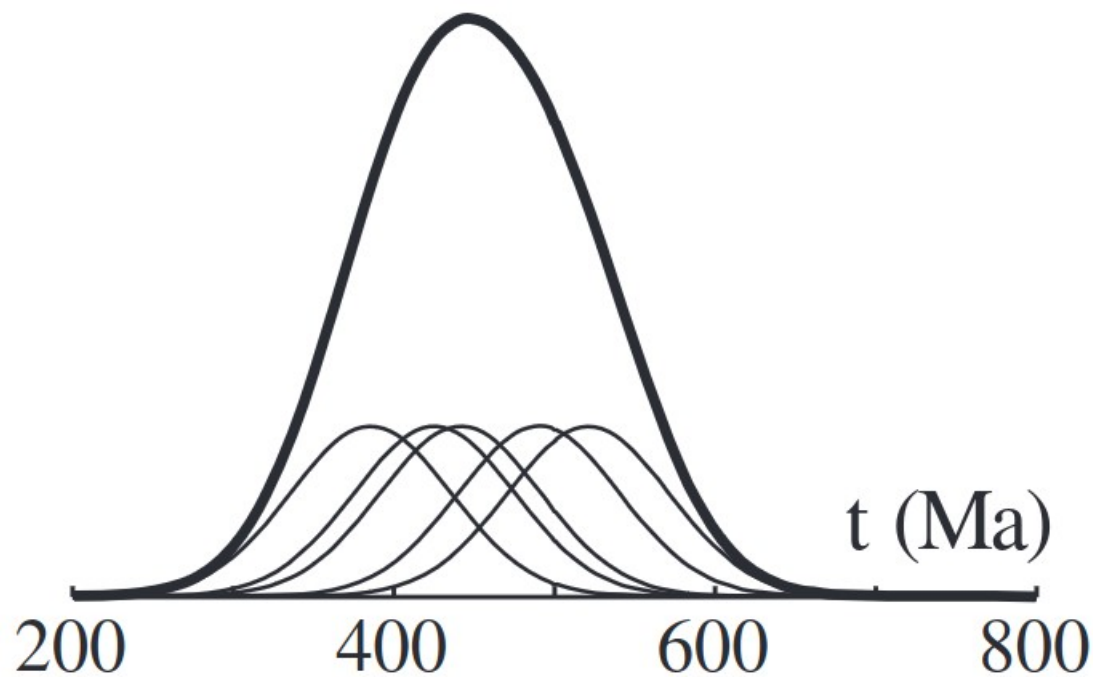


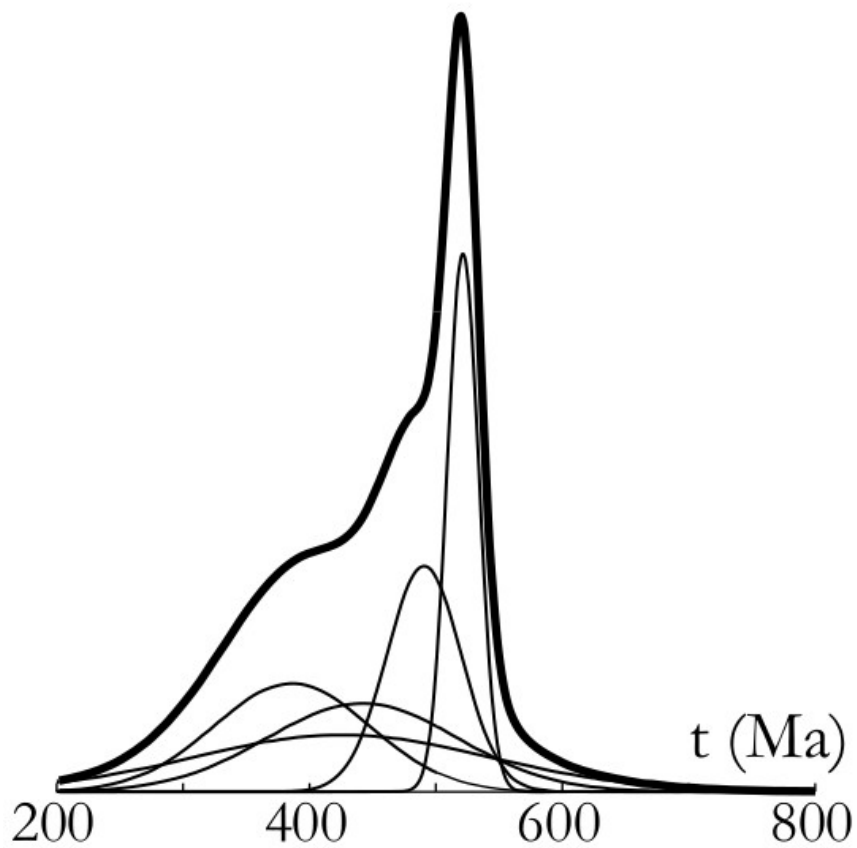
ecdf/CAD



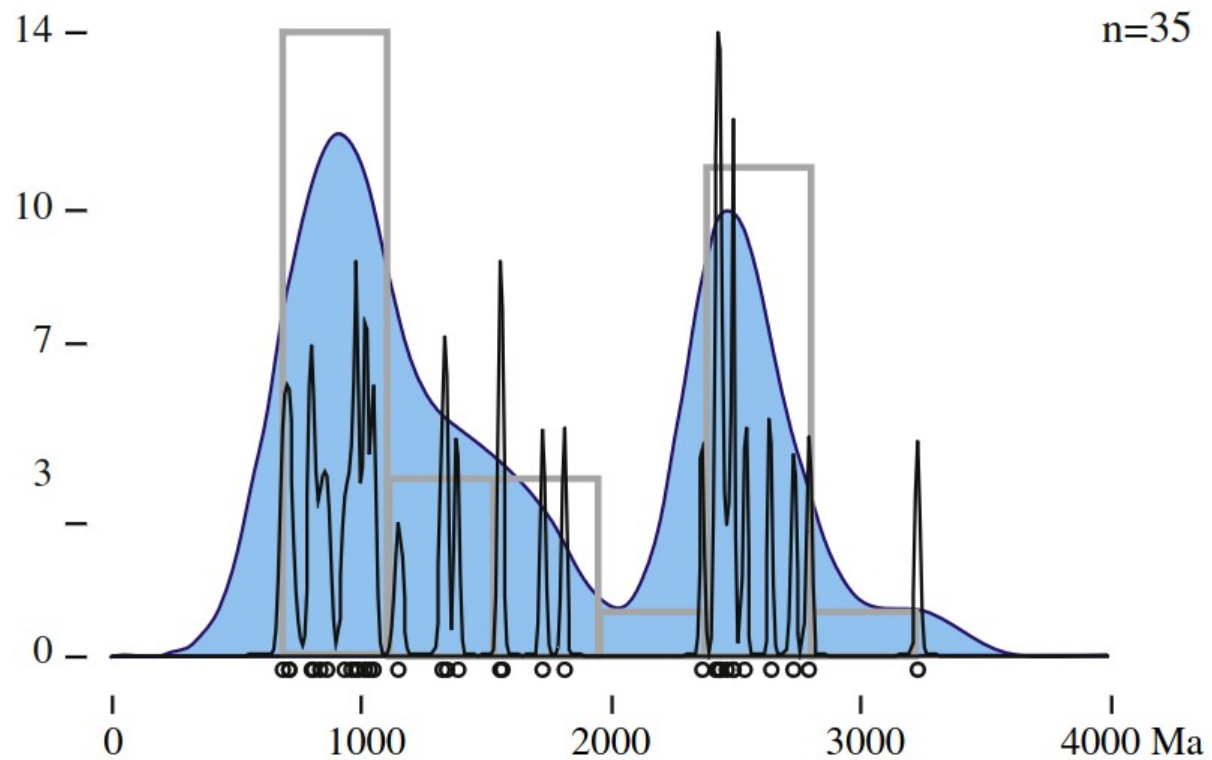
radial plot

$$\text{KDE}(x) = \frac{1}{nh\sqrt{2\pi}} \sum_{j=1}^n \exp \left[-\frac{1}{2} \left(\frac{x - \hat{t}_j}{h} \right)^2 \right]$$

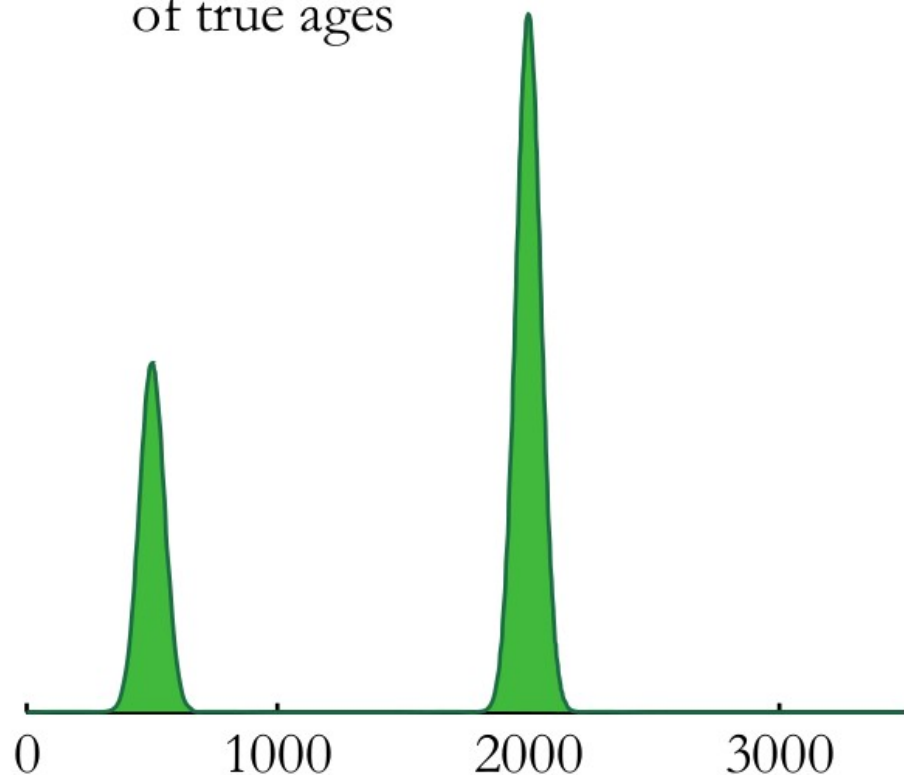




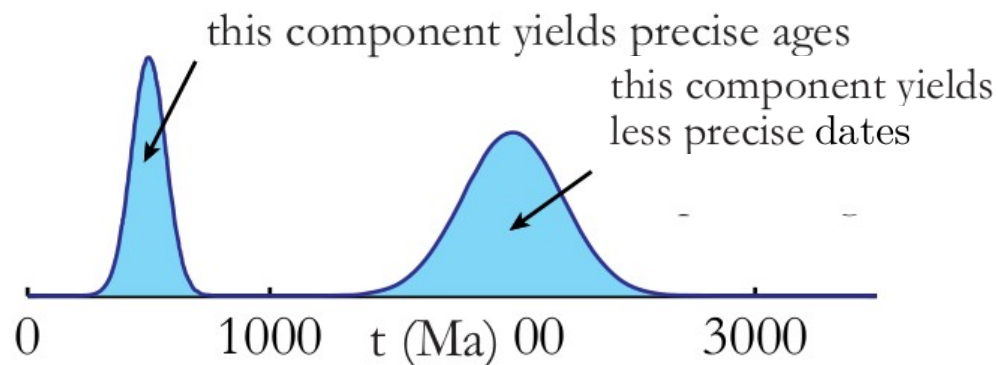
$$PDP(x) = \sum_{i=1}^N \frac{1}{e_i \sqrt{2\pi}} \exp^{-\frac{(t-x_i)^2}{2e_i^2}}$$



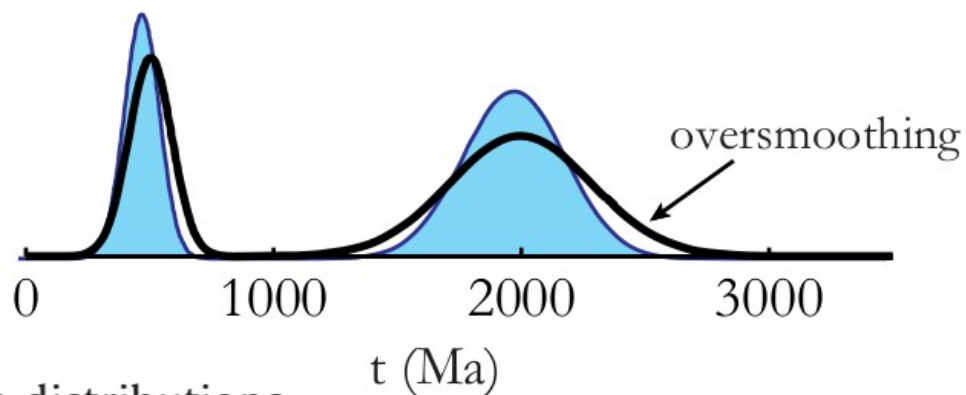
distribution
of true ages



distribution of the dates

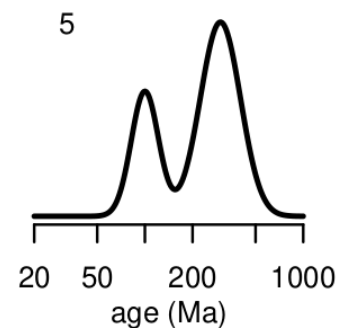
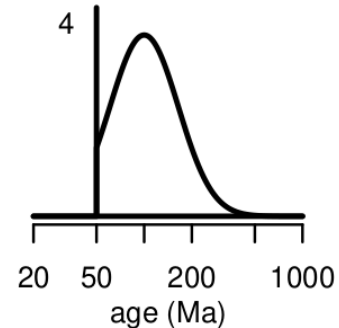
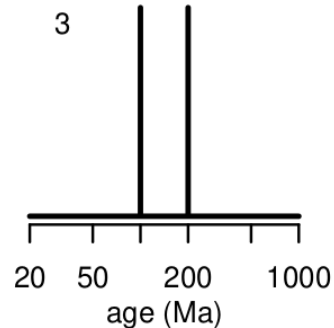
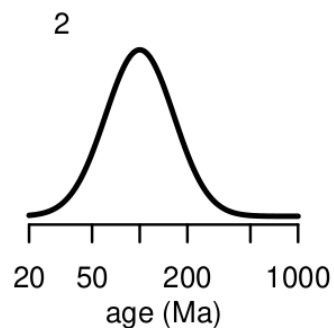
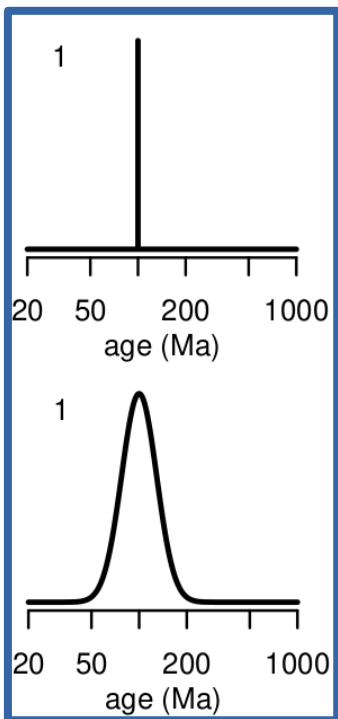


probability density plot

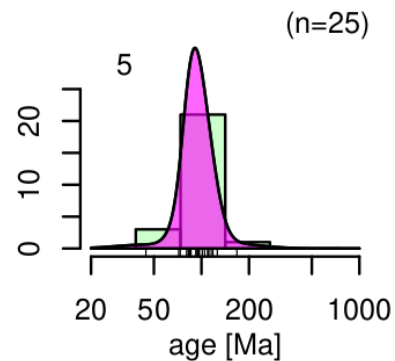
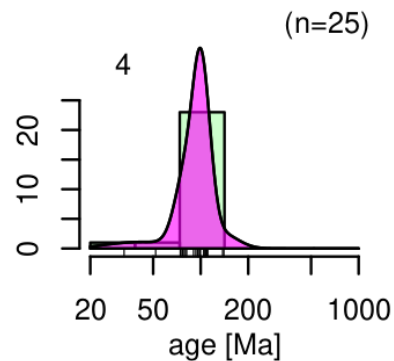
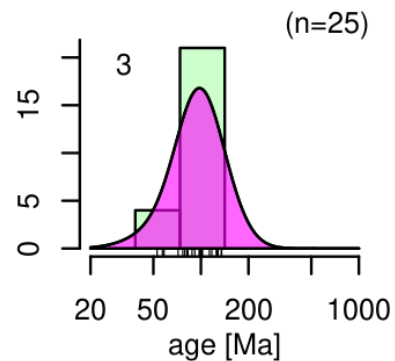
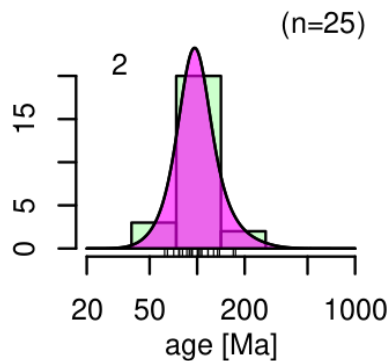
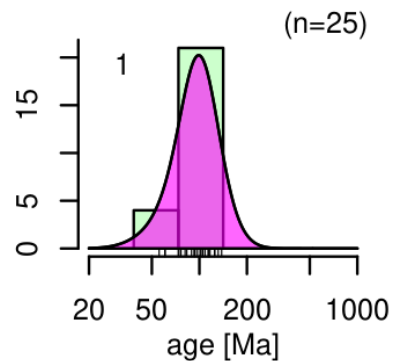
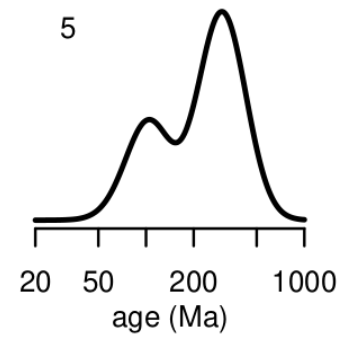
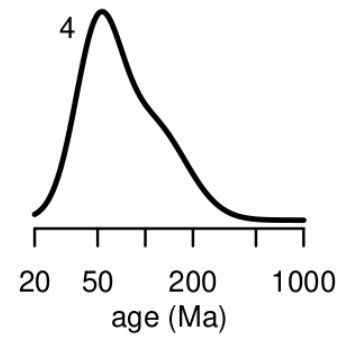
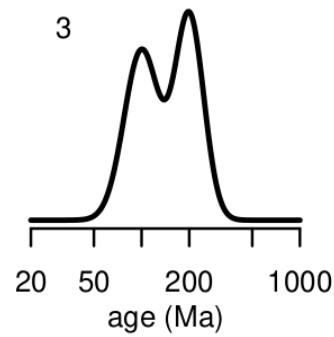
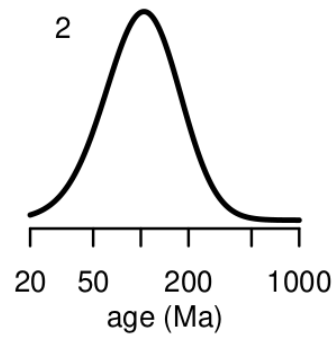
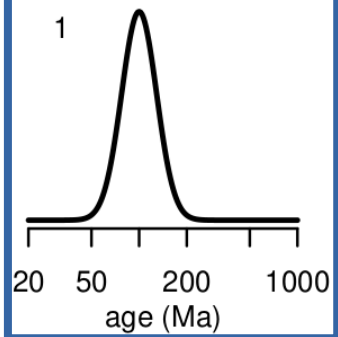


On the visualisation of detrital age distributions

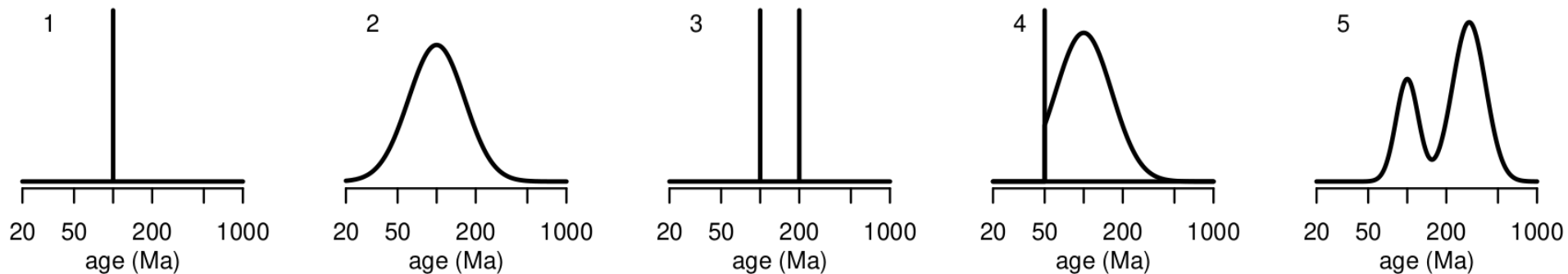
Ages:



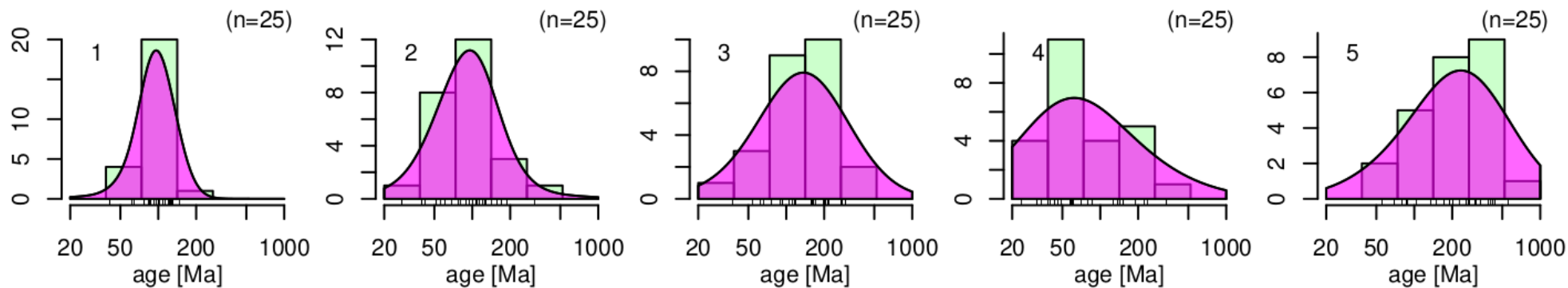
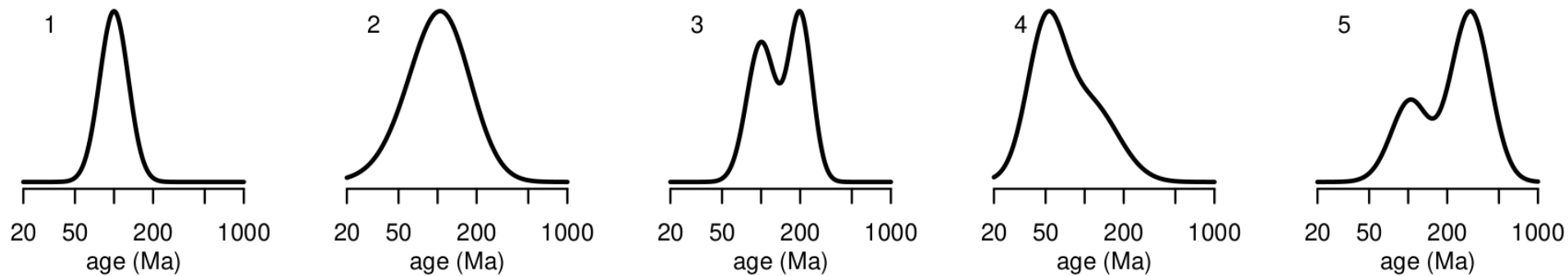
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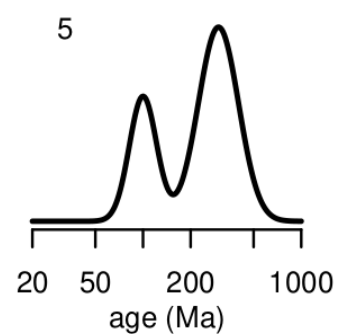
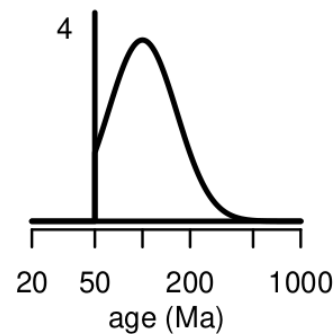
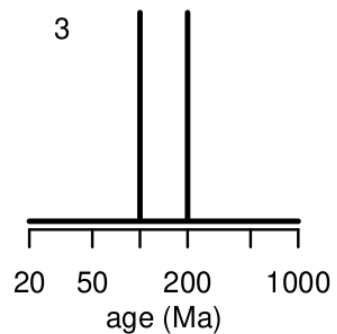
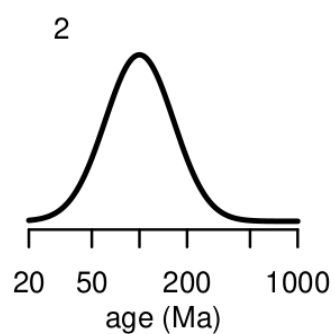
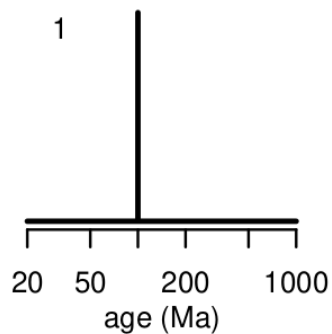
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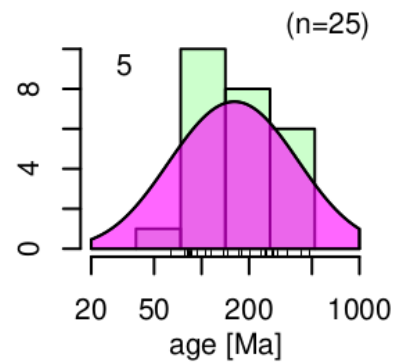
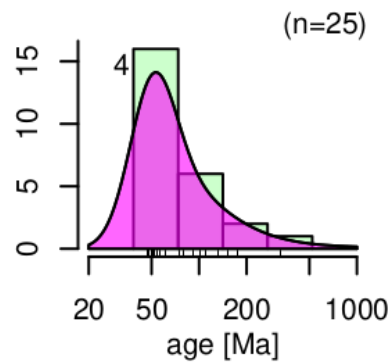
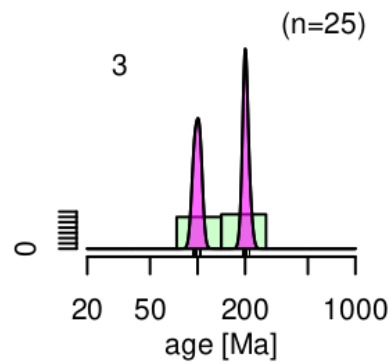
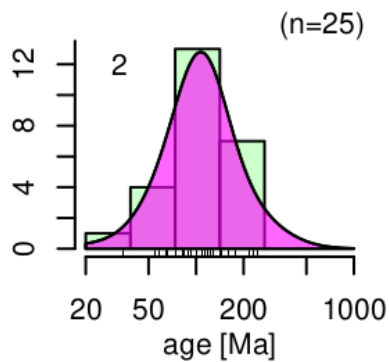
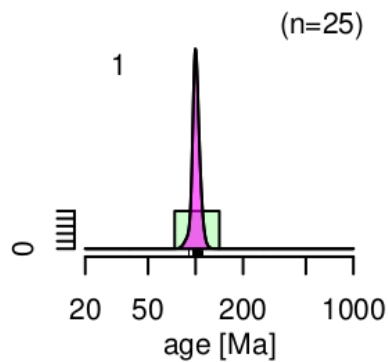
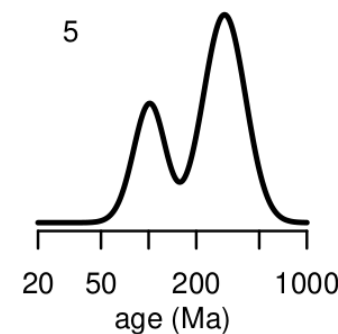
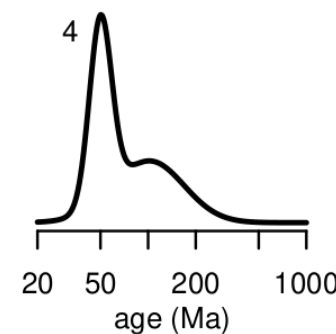
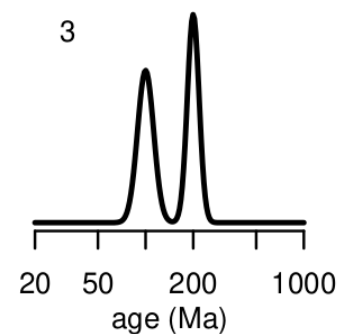
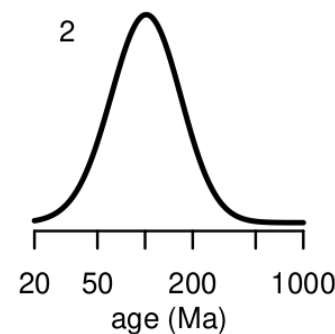
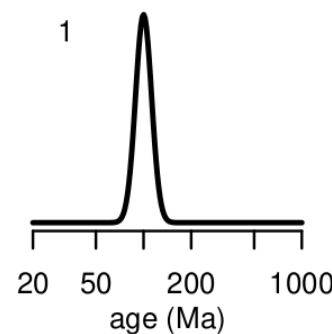
FT dates:



Ages:

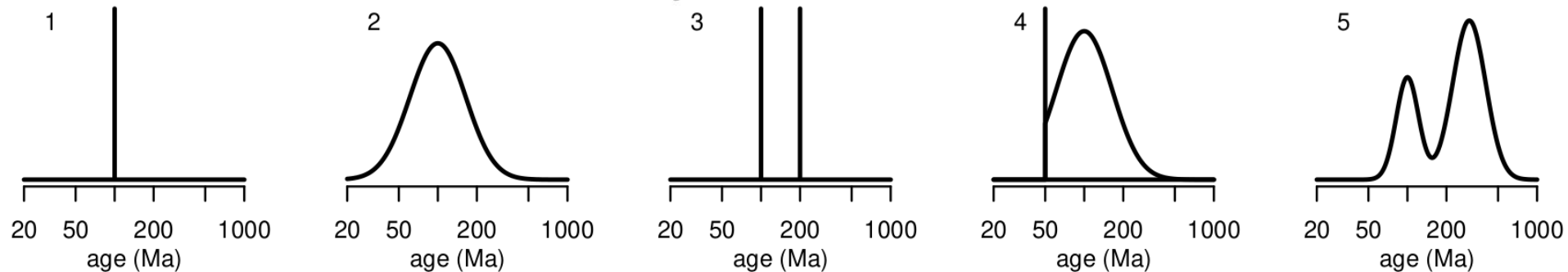


U-Th-He
dates:

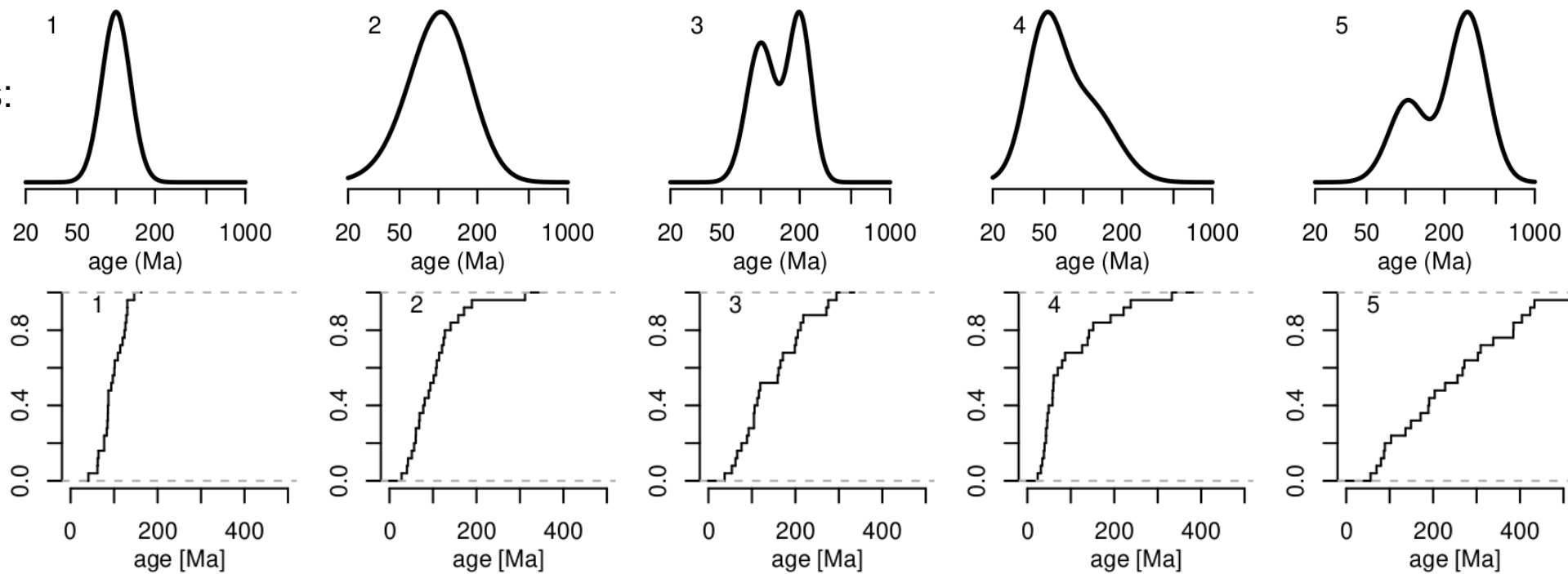


$$\text{CAD}(x) = \frac{1}{n} \sum_{j=1}^n I(\hat{t}_j < x)$$

Ages:

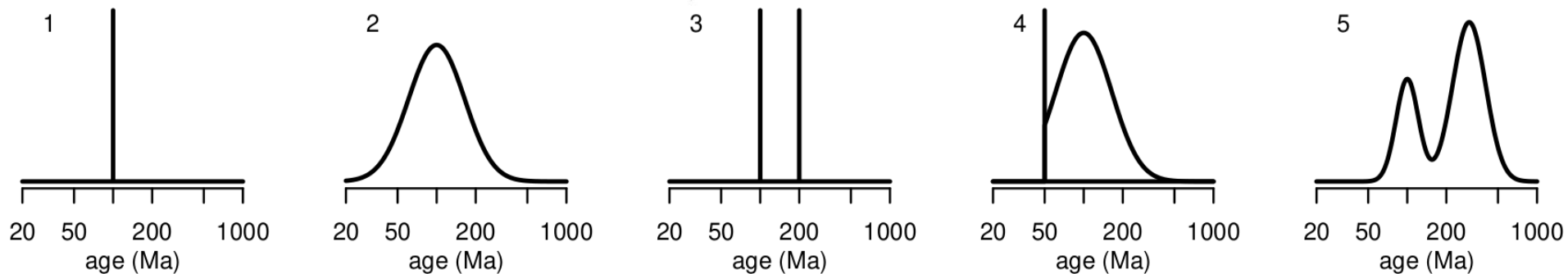


FT dates:

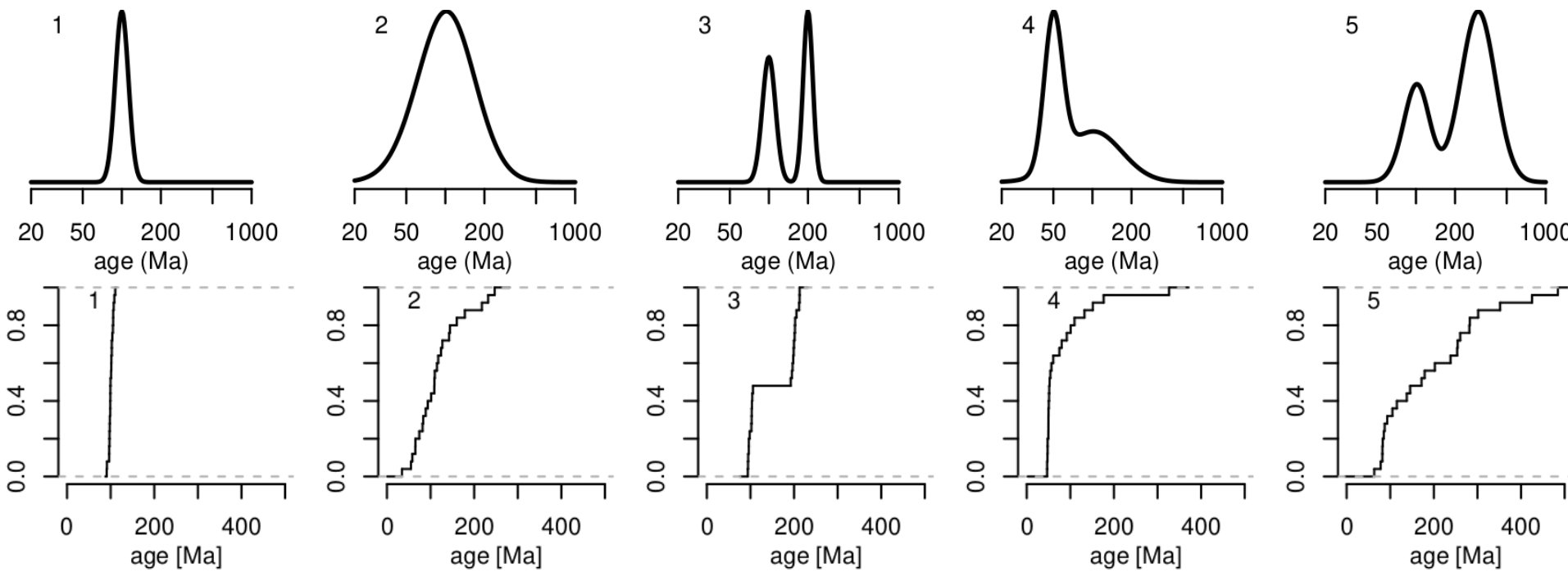


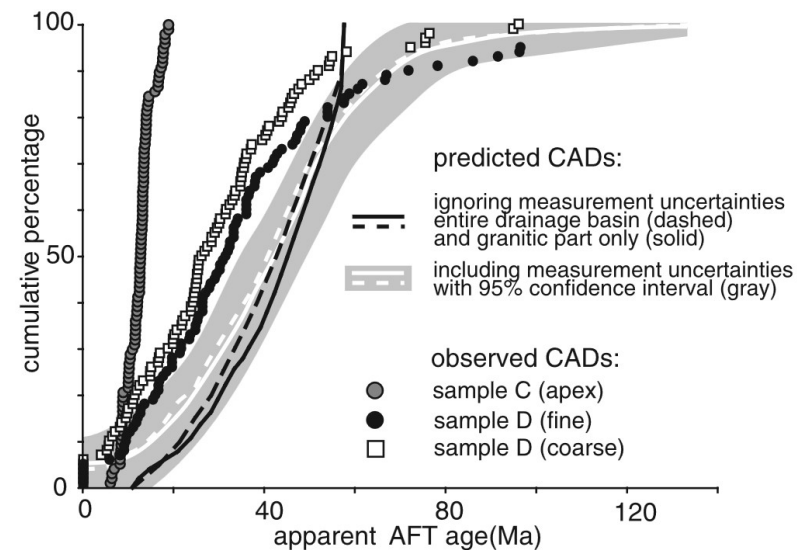
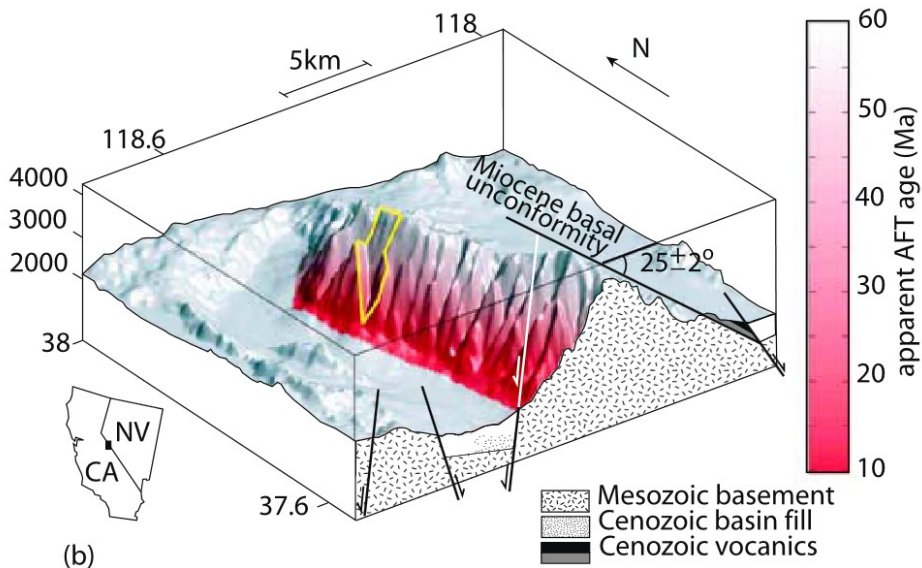
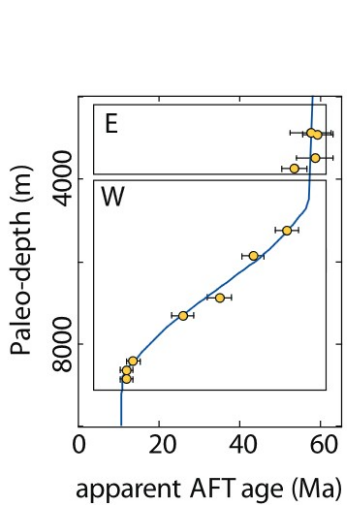
$$\text{CAD}(x) = \frac{1}{n} \sum_{j=1}^n I(\hat{t}_j < x)$$

Ages:



U-Th-He
dates:

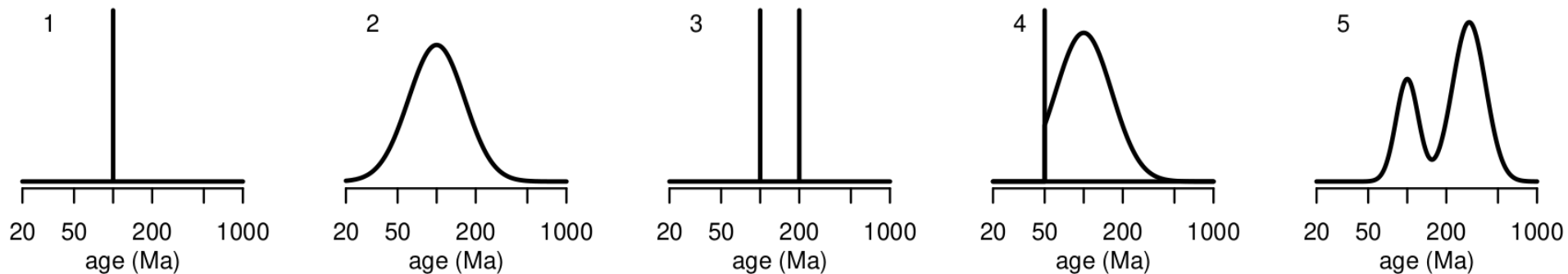




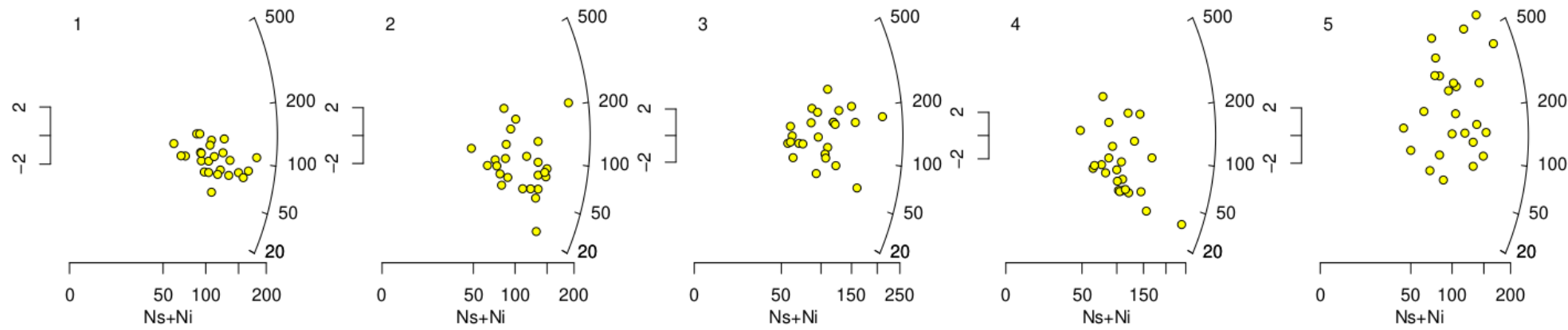
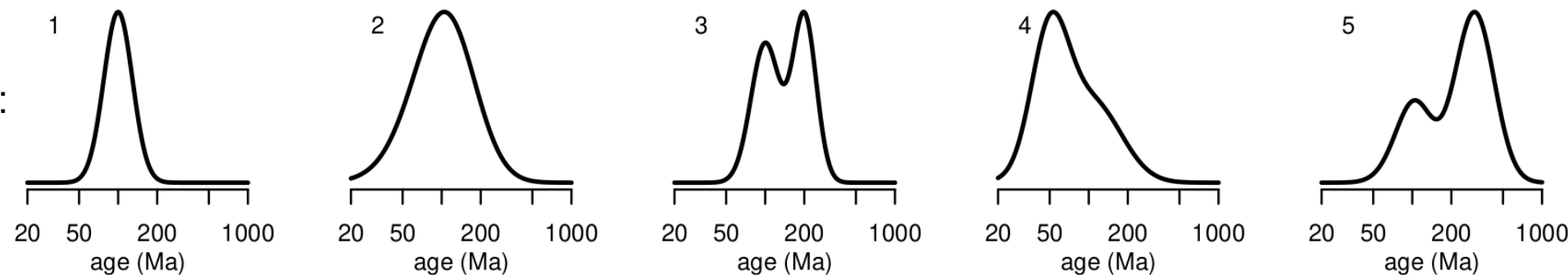
Quantitative geomorphology of the White Mountains (California) using detrital apatite fission track thermochronology

Pieter Vermeesch¹

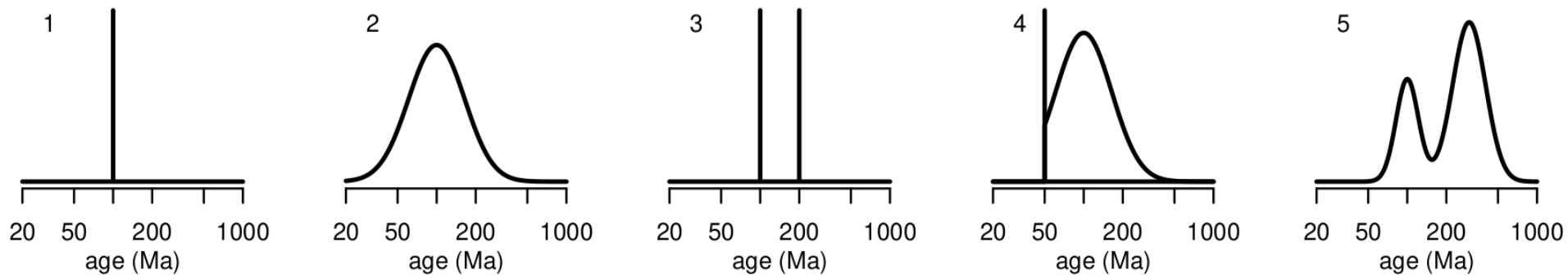
Ages:



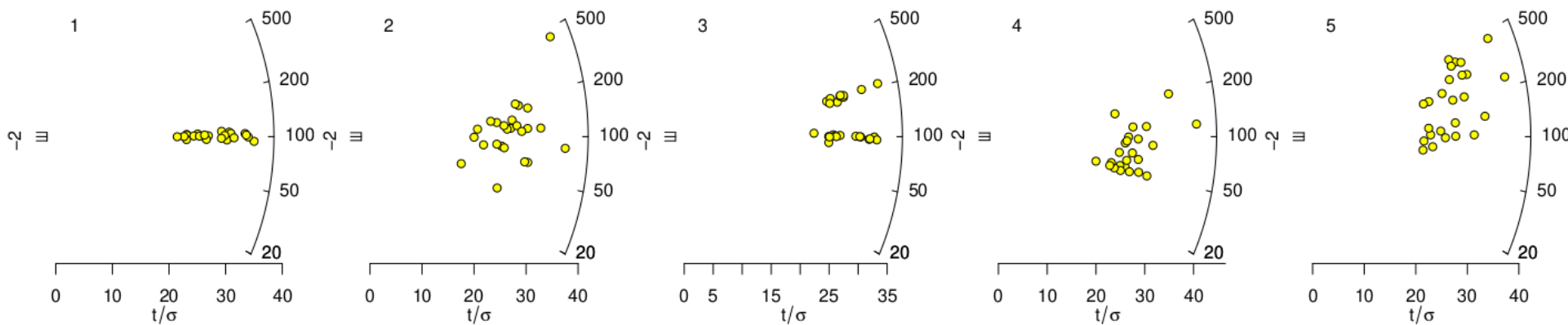
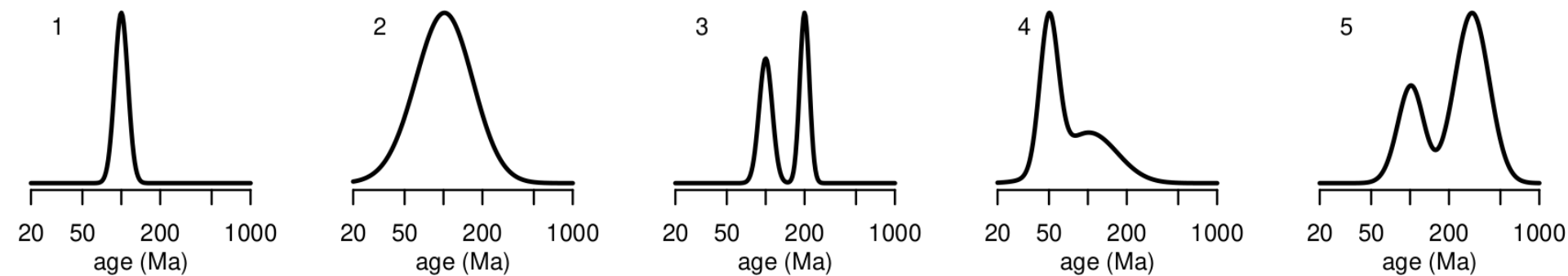
FT dates:



Ages:



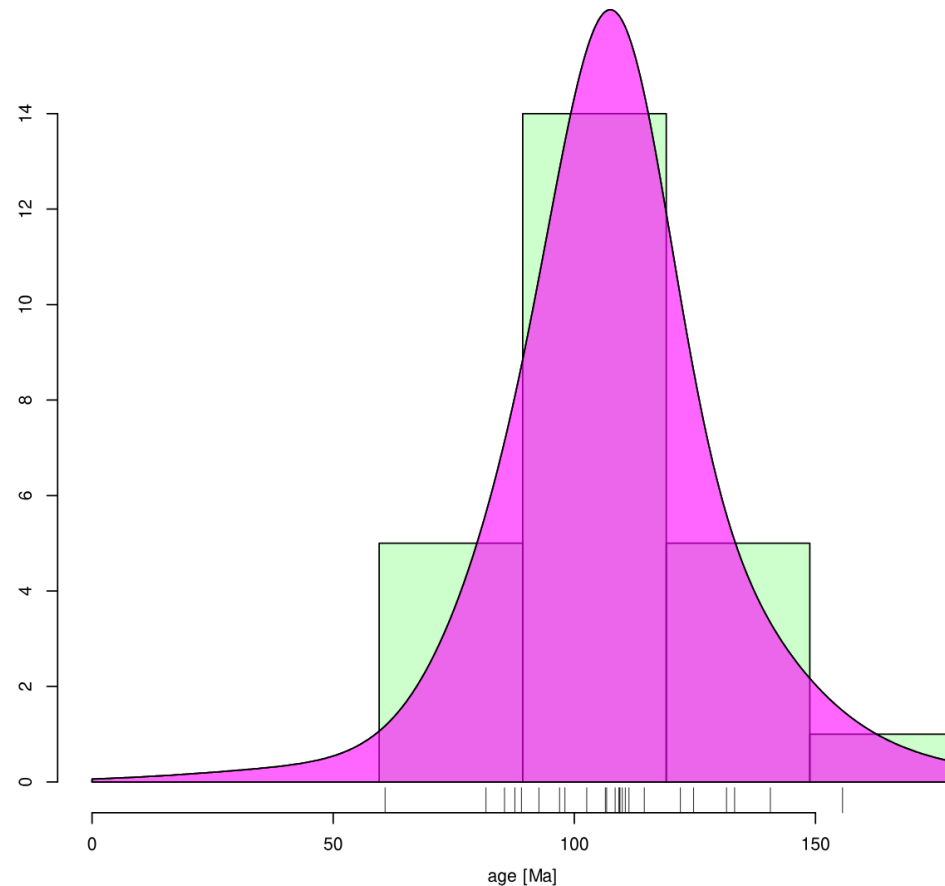
U-Th-He
dates:



ζ : $\pm$ yr cm² ρ_D : $\pm$ 1/cm²

(n=25)

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	38	132										
2	48	187										
3	114	565										
4	13	51										
5	25	122										
6	49	249										
7	39	159										
8	18	50										
9	26	80										
10	24	79										
11	55	217										
12	38	152										
13	61	231										
14	82	347										
15	16	75										
16	38	135										
17	42	186										
18	31	123										
19	15	61										
20	27	107										
21	34	135										
22	6	43										
23	48	215										
24	16	52										
25	31	165										
26												
27												
28												



ζ : 350 ± 10 yr cm² ρ_D : 2500000 ± 10000 1/cm²

radial plot
weighted mean
KDE
CAD
get ζ
ages

method: EDM ▾

put errors: 1se (abs) ▾

minimum value: auto

maximum value: auto

Kernel bandwidth: auto

Histogram binwidth: auto

Font magnification: 1

Significant digits: 2

Label 0 ▾ modes

☒ Take logarithms?☒ Show histogram?☒ Adaptive KDE?☒ Add rug plot?

Try the different
options, view the
contextual help

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I
1	38	132							
2	48	187							
3	114	565							
4	13	51							
5	25	122							
6	49	249							
7	39	159							
8	18	50							
9	26	80							
10	24	79							
11	55	217							
12	38	152							
13	61	231							
14	82	347							
15	16	75							
16	38	135							
17	42	186							
18	31	123							
19	15	61							
20	27	107							
21	34	135							
22	6	43							
23	48	215							
24	16	52							
25	31	165							
26									
27									
28									

```
library(IsoplotR)

source("helper.R")

FT <- FTsamp(pop=1,ng=50)

kde(FT)

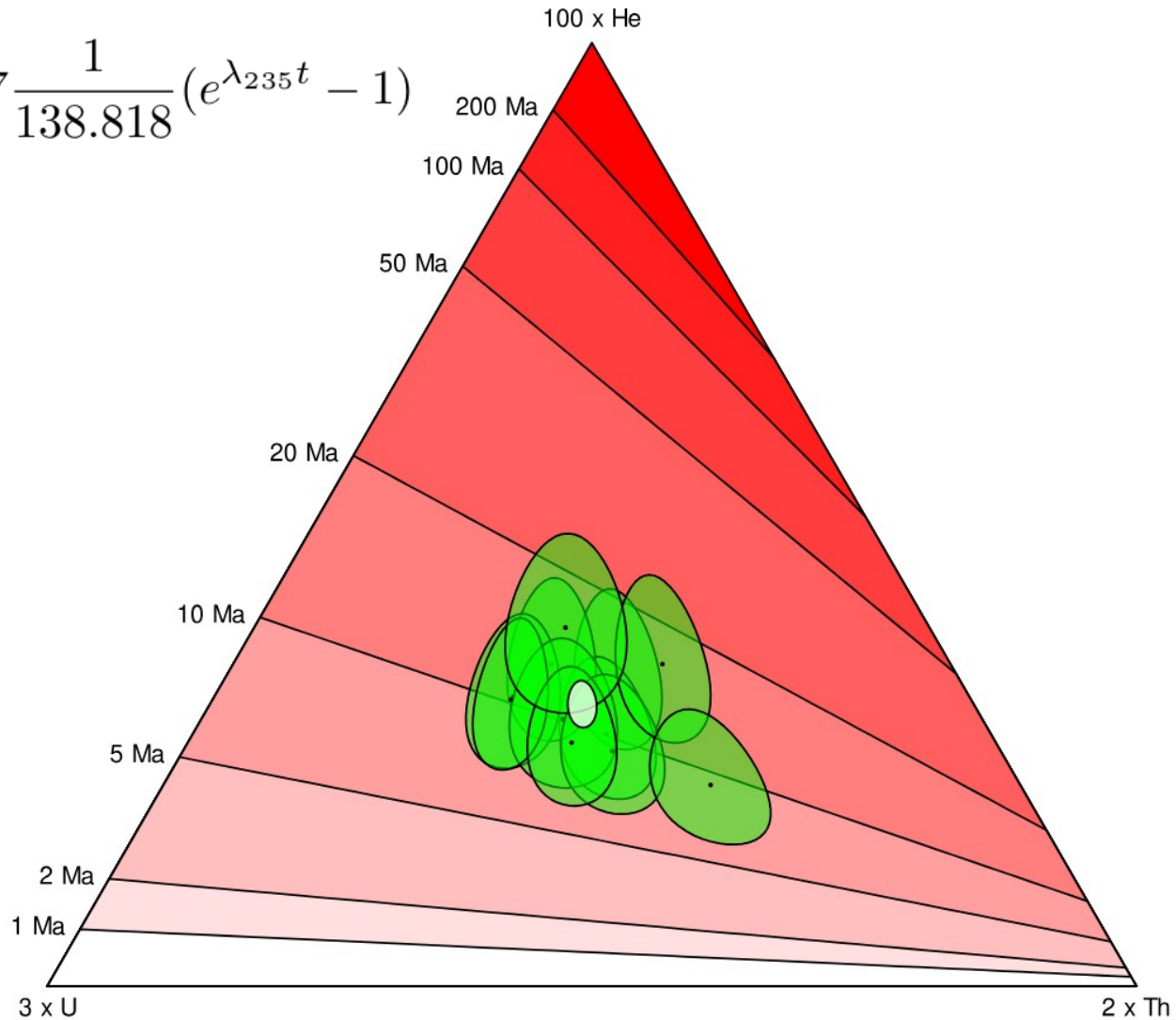
cad(FT)

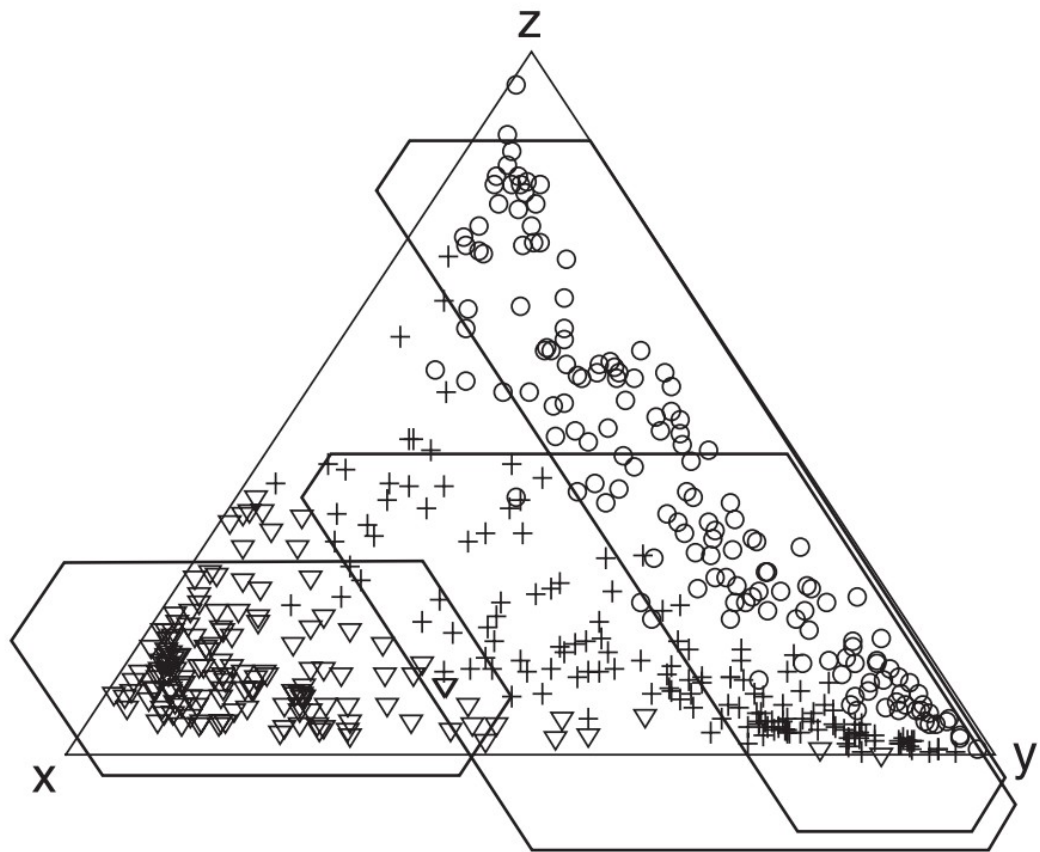
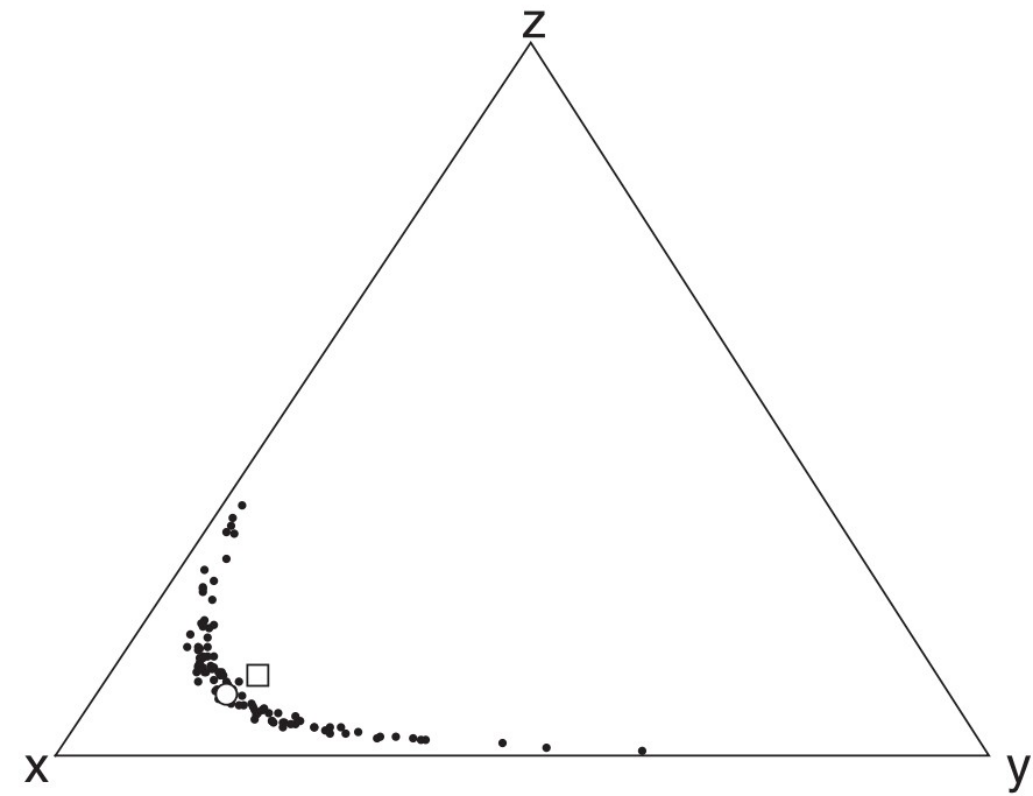
radialplot(FT)
```

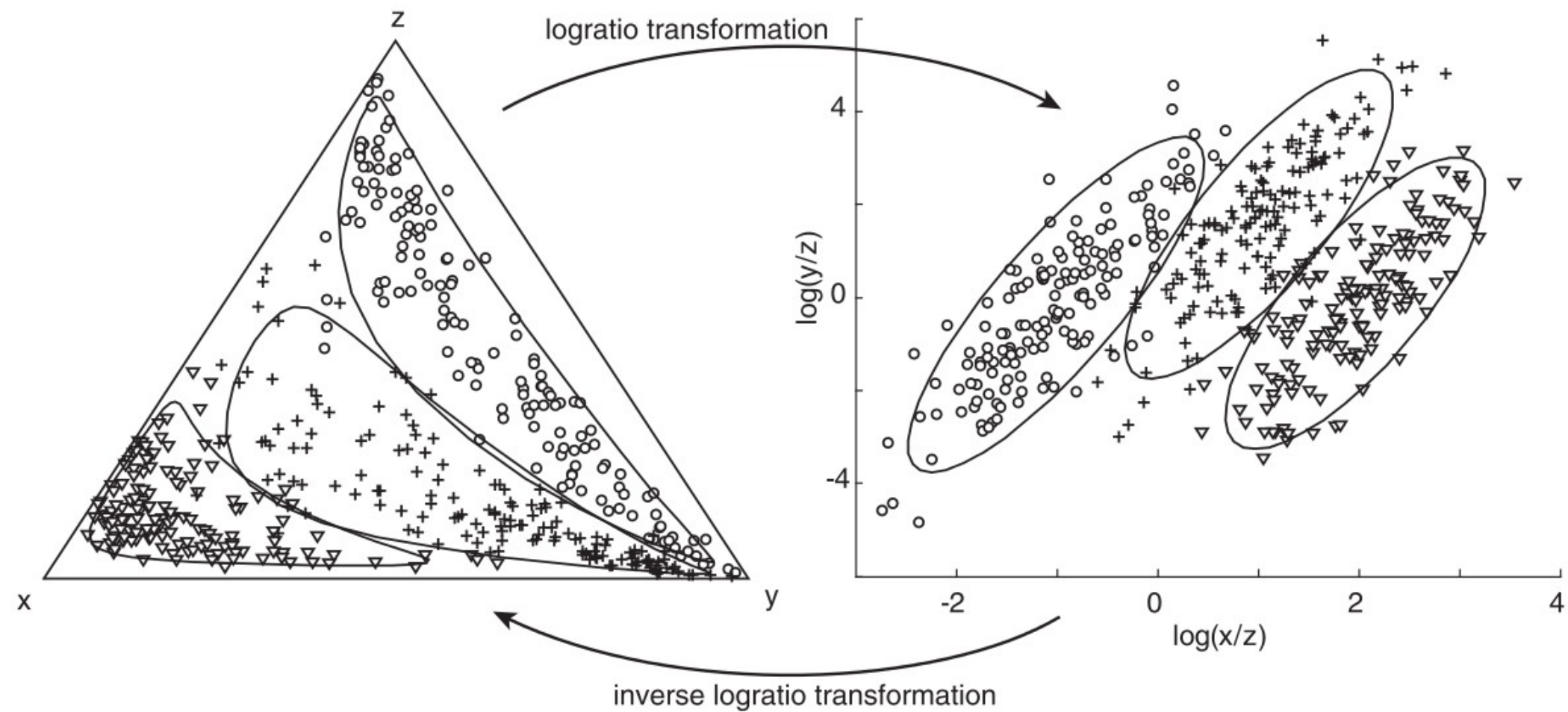
$$[\text{He}] = a(t)[\text{U}] + b(t)[\text{Th}]$$

where $a(t) = 8 \frac{137.818}{138.818} (e^{\lambda_{238} t} - 1) + 7 \frac{1}{138.818} (e^{\lambda_{235} t} - 1)$

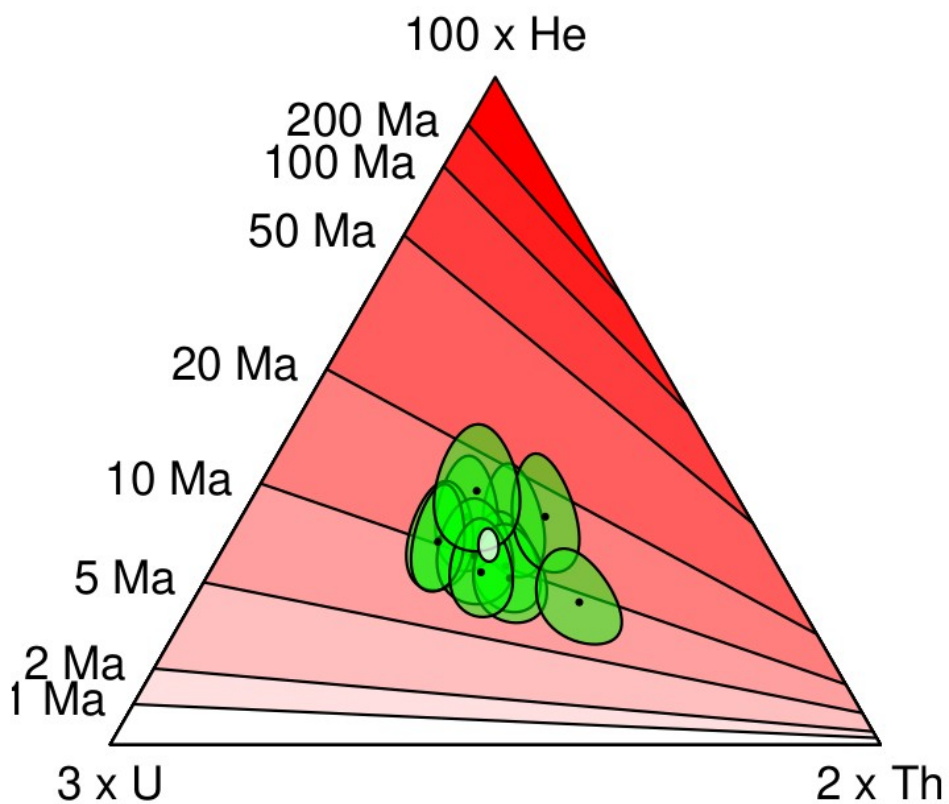
and $b(t) = 6(e^{\lambda_{232} t} - 1)$







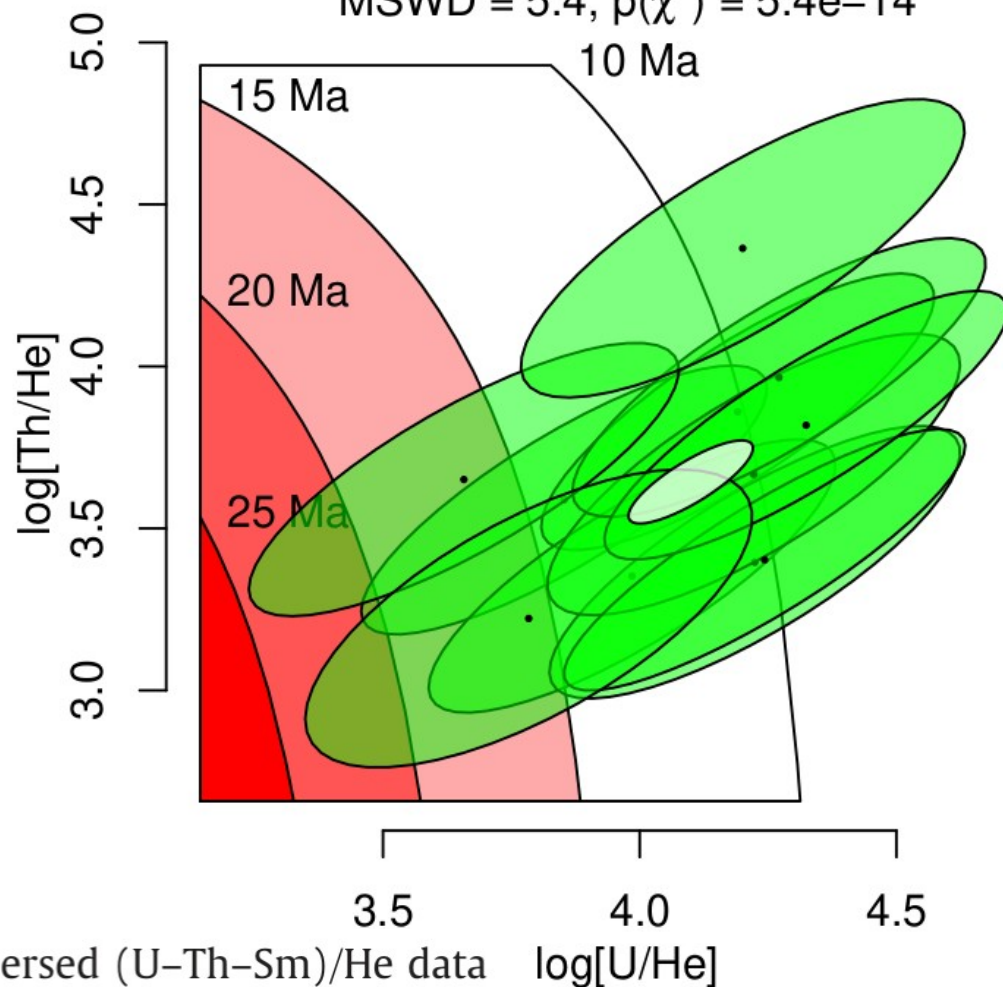
barycentric age = 11.27 ± 1.08 | 2.67 Ma (n=11)
 MSWD = 5.4, $p(\chi^2) = 5.4e-14$



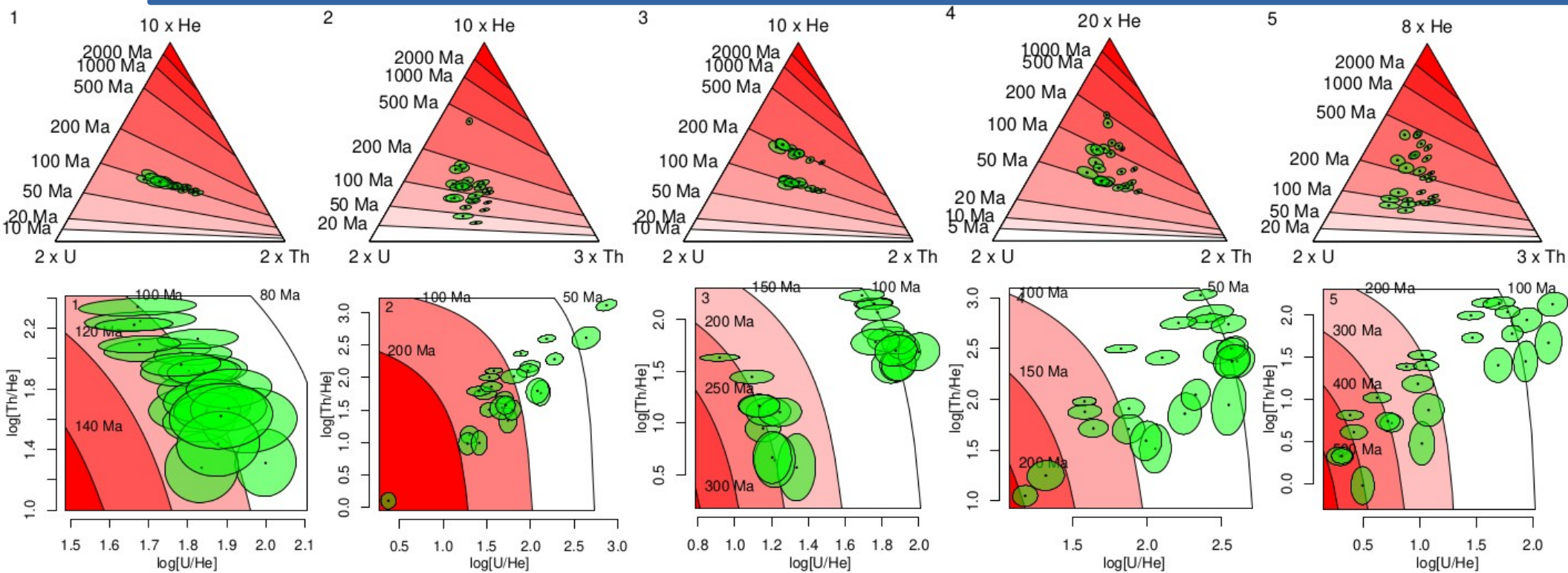
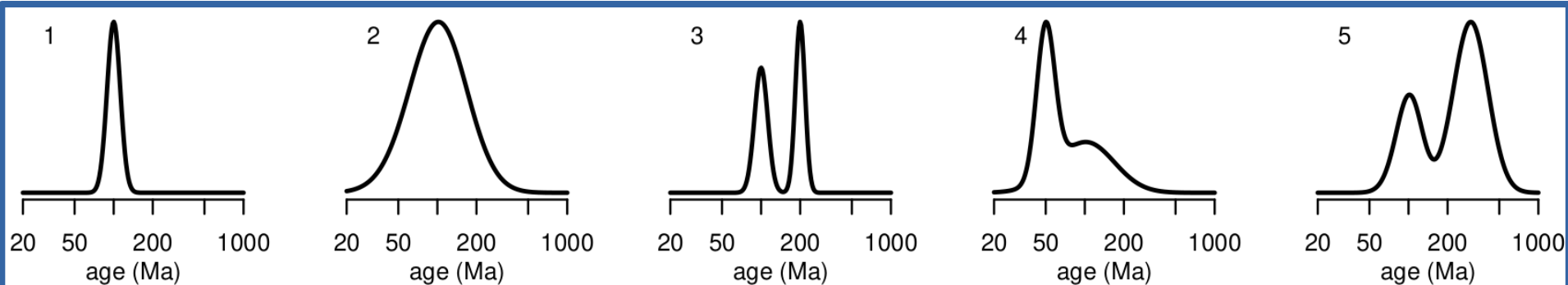
HelioPlot, and the treatment of overdispersed (U–Th–Sm)/He data

Pieter Vermeesch

barycentric age = 11.27 ± 1.08 | 2.67 Ma (n=11)
 MSWD = 5.4, $p(\chi^2) = 5.4e-14$



U-Th-He
dates:



```
He <- UThHesamp(pop=1,ng=20)  
helioplot(He)  
helioplot(He,logratio=FALSE)
```

Contents

1 Introduction to R

- 1.1 RStudio
- 1.2 IsoplotR
- 1.3 Basic R

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- 2.2 Populations

3 Plotting data

- 3.1 Kernel Density Estimates
- 3.2 Cumulative Age Distributions
- 3.3 Radial plots
- 3.4 Helioplots

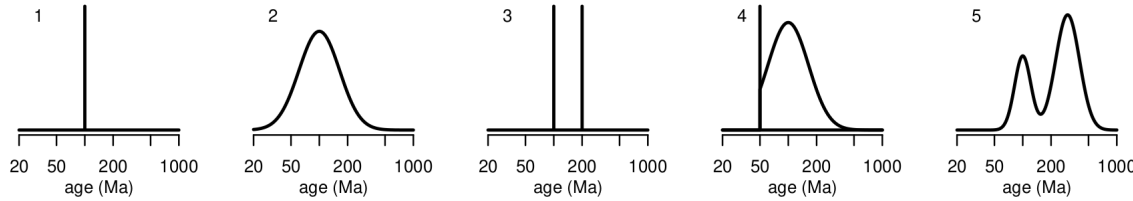
4 Mixture models

- 4.1 The χ^2 test for homogeneity
- 4.2 The random effects model
- 4.3 Finite mixtures
- 4.4 How many components to fit?
- 4.5 The minimum age model
- 4.6 Logratios

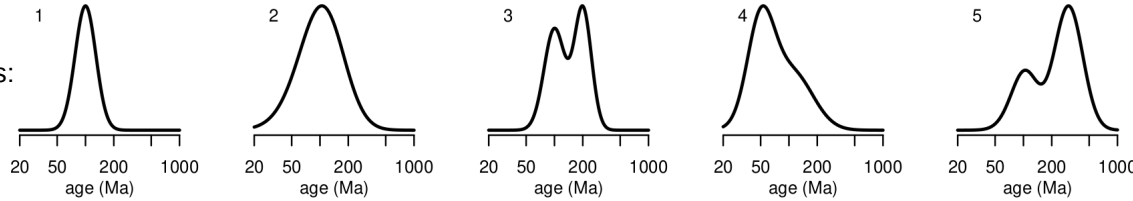
5 Mineral fertility estimation with provenance

- 5.1 Grain size estimation
- 5.2 Source Rock Density (SRD) correction

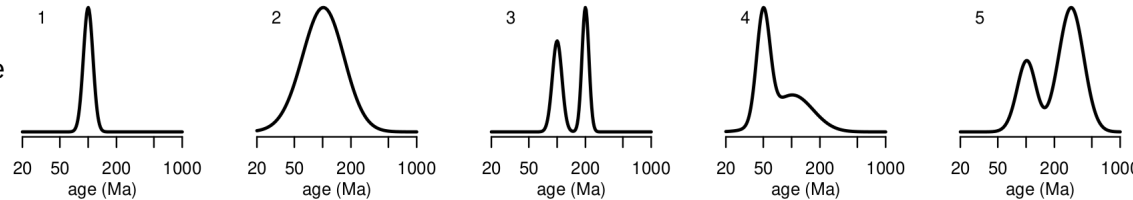
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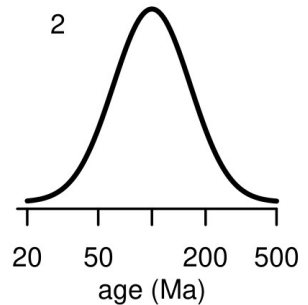
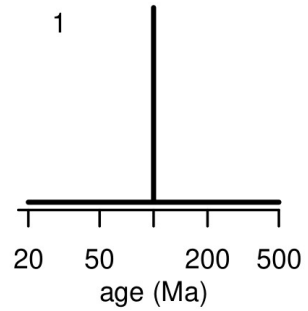
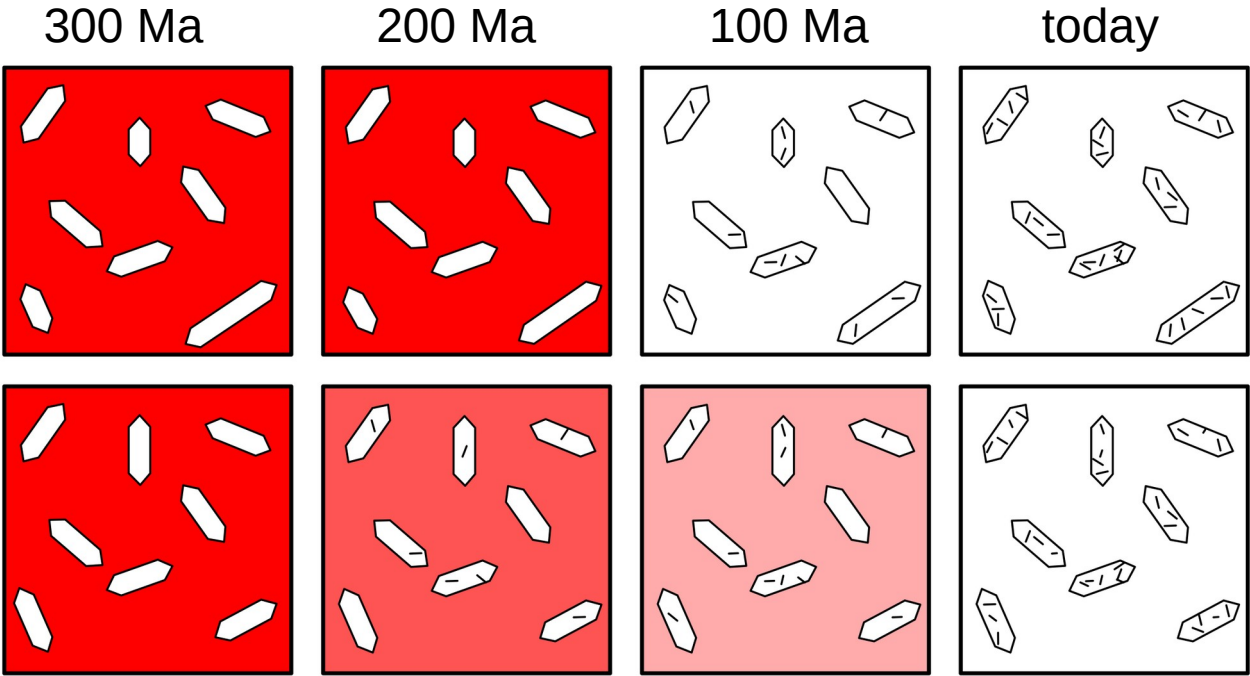
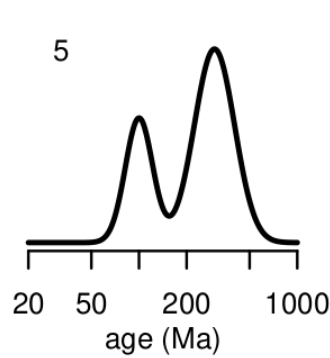
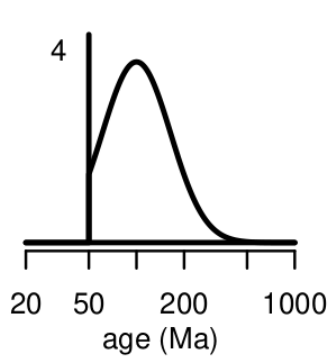
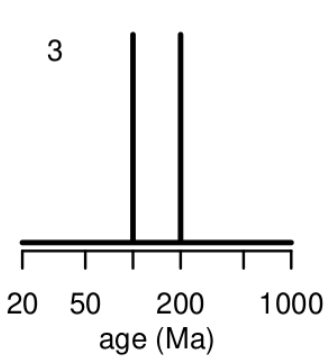
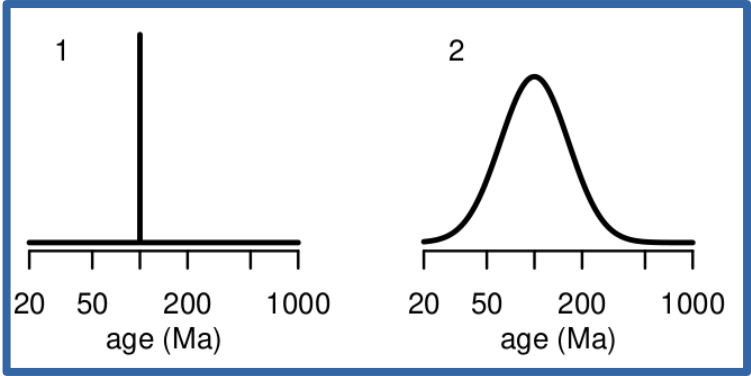
FT dates:



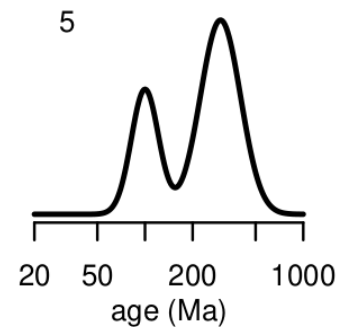
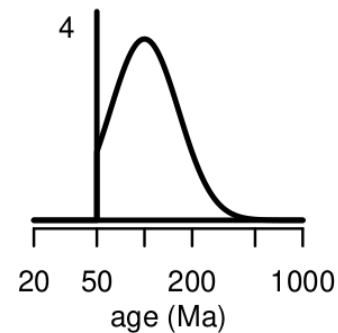
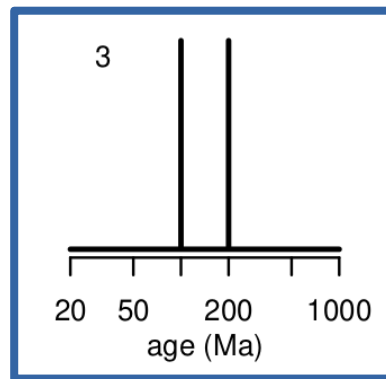
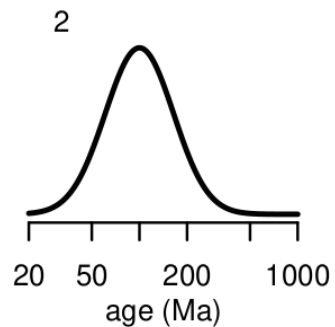
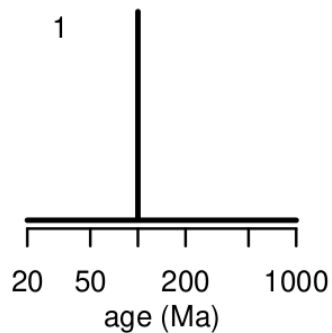
U-Th-He
dates:



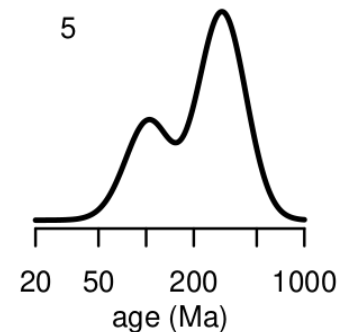
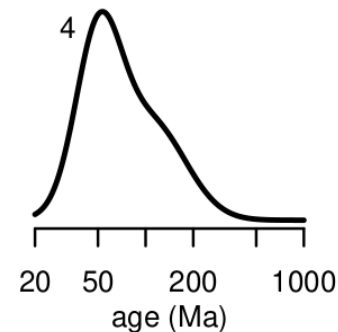
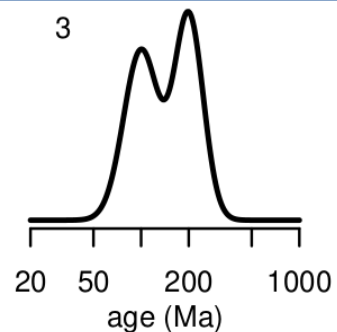
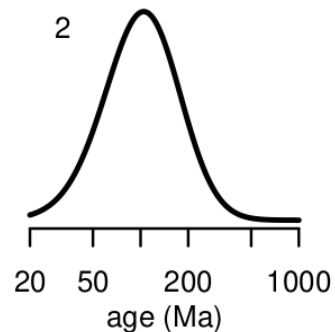
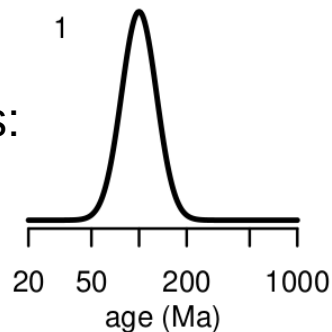
Ages:



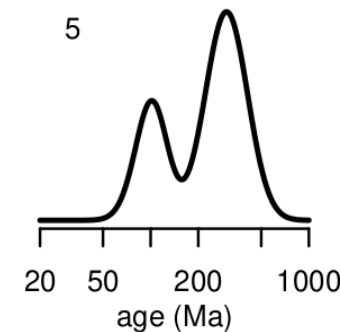
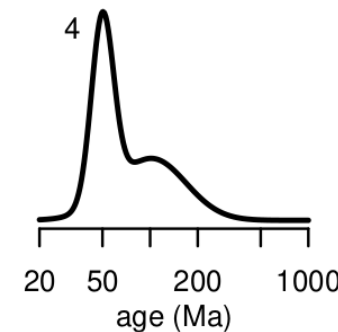
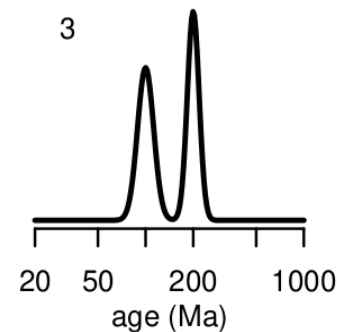
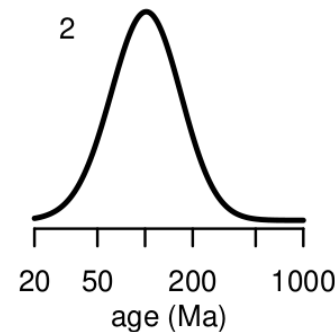
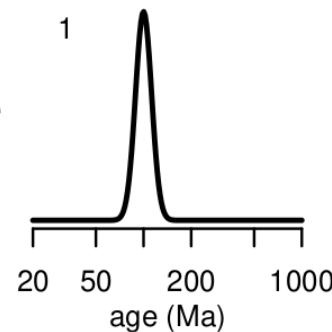
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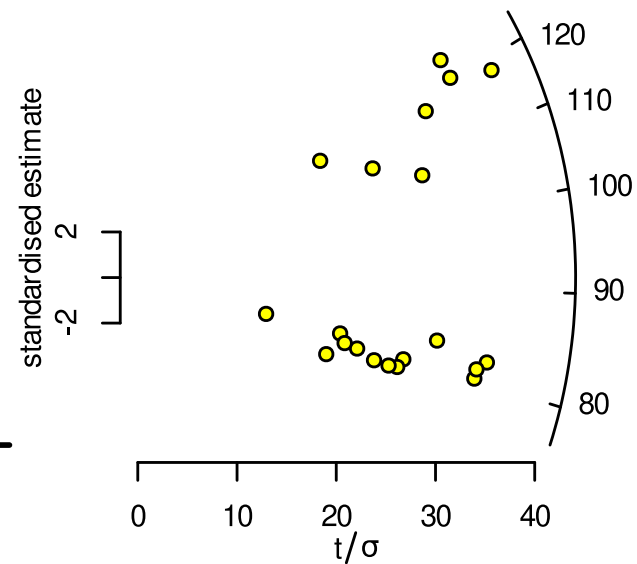
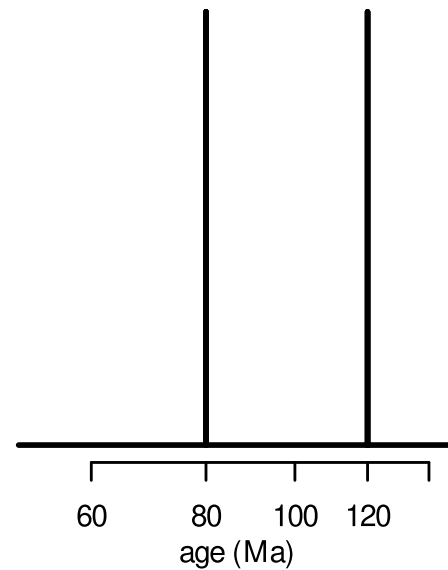
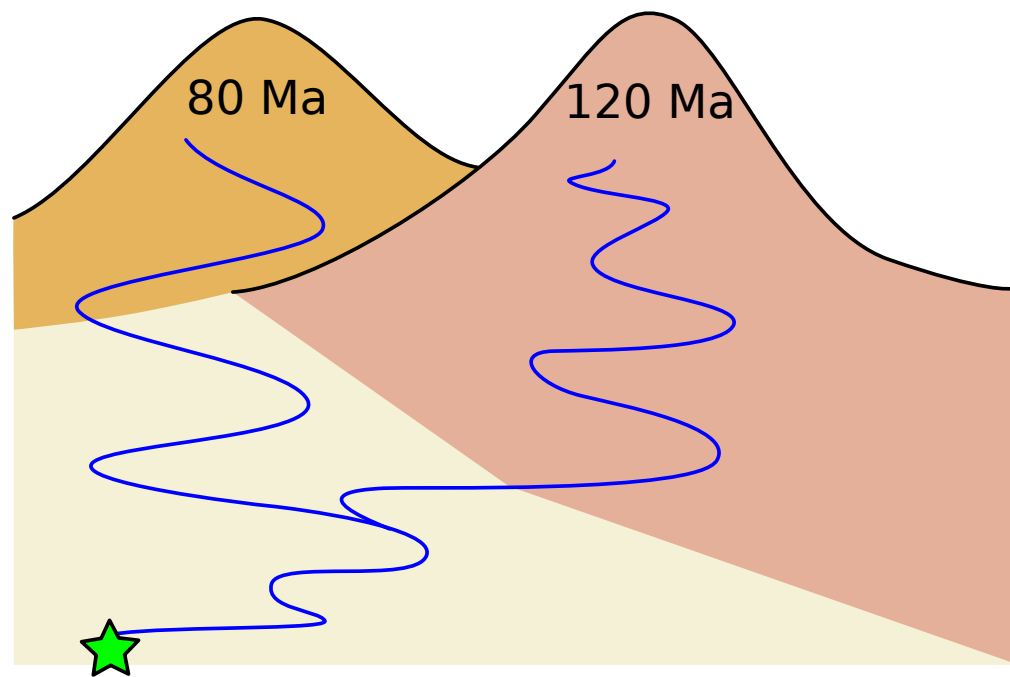


FT dates:



U-Th-He
dates:





Maximise

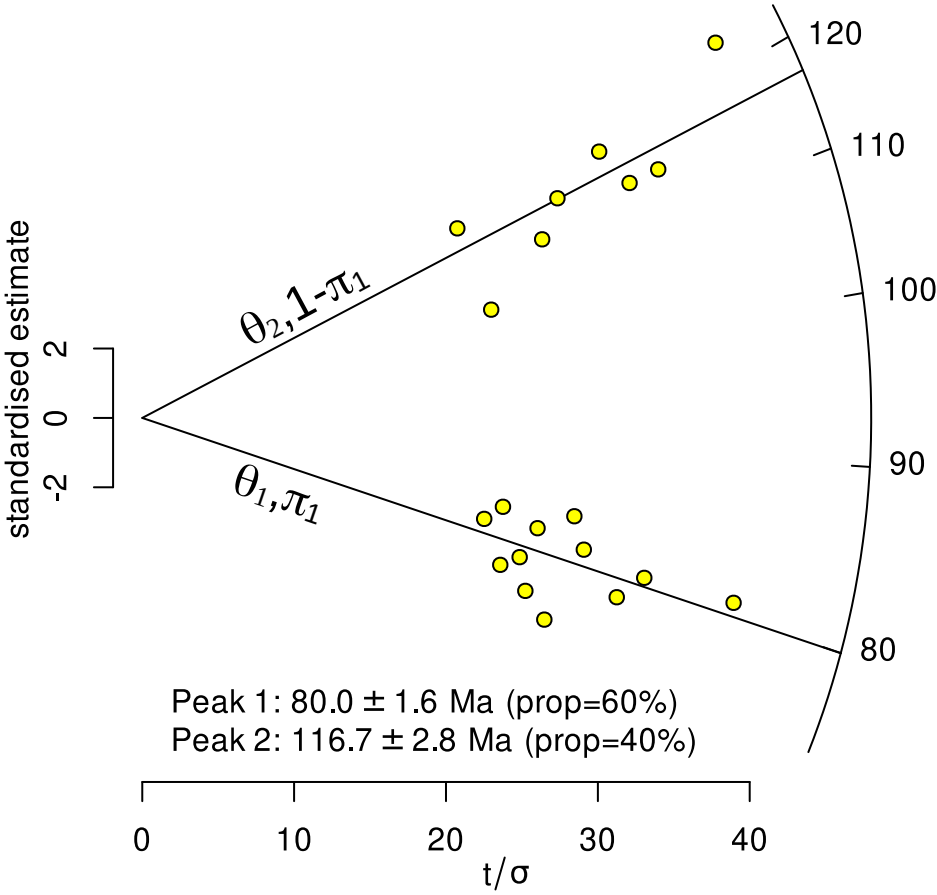
$$\mathcal{L}(\pi_k, \theta_k, k = 1 \dots K) = \sum_{i=1}^n \ln \left[\sum_{k=1}^K \pi_k f(\theta_k | z_i, s[z_i]) \right]$$

where $\pi_K = 1 - \sum_{k=1}^{K-1} \pi_k$

$$z_i = \ln(t_i)$$

$$s[z_i] = \frac{s[t_i]}{t_i}, \text{ and}$$

$$f(\theta_k | z_i, s[z_i]) = \frac{1}{\sqrt{2 \pi} s[z_i]} \exp \left[-\frac{(z_i - \theta_k)^2}{2 s[z_i]^2} \right]$$

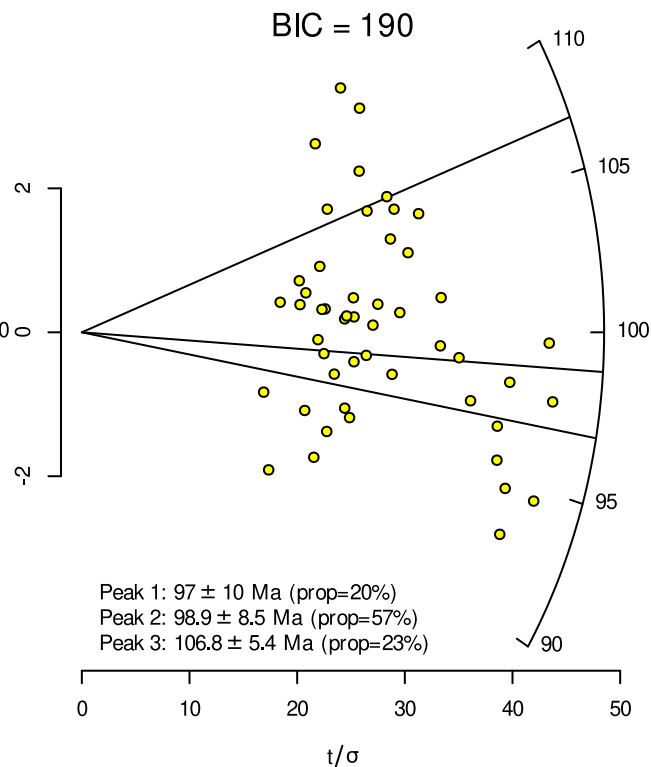
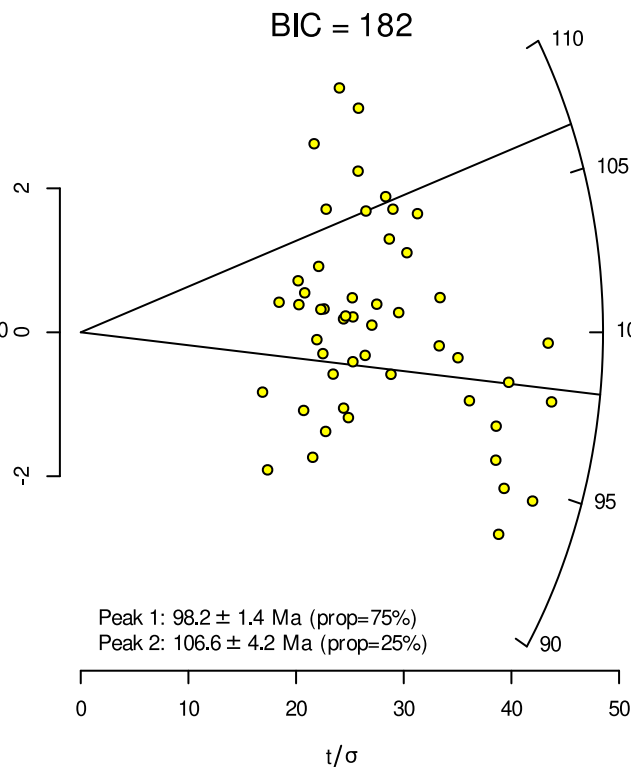
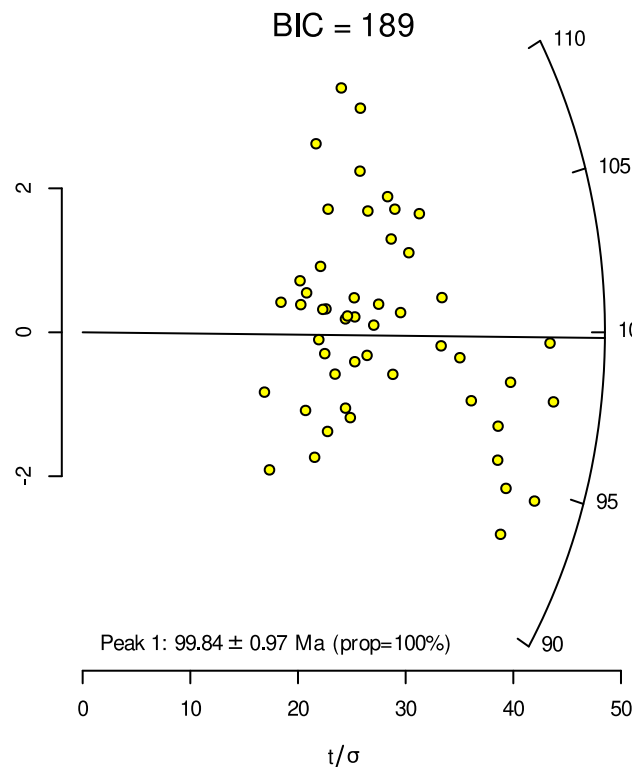


STATISTICAL MODELS FOR MIXED FISSION TRACK AGES

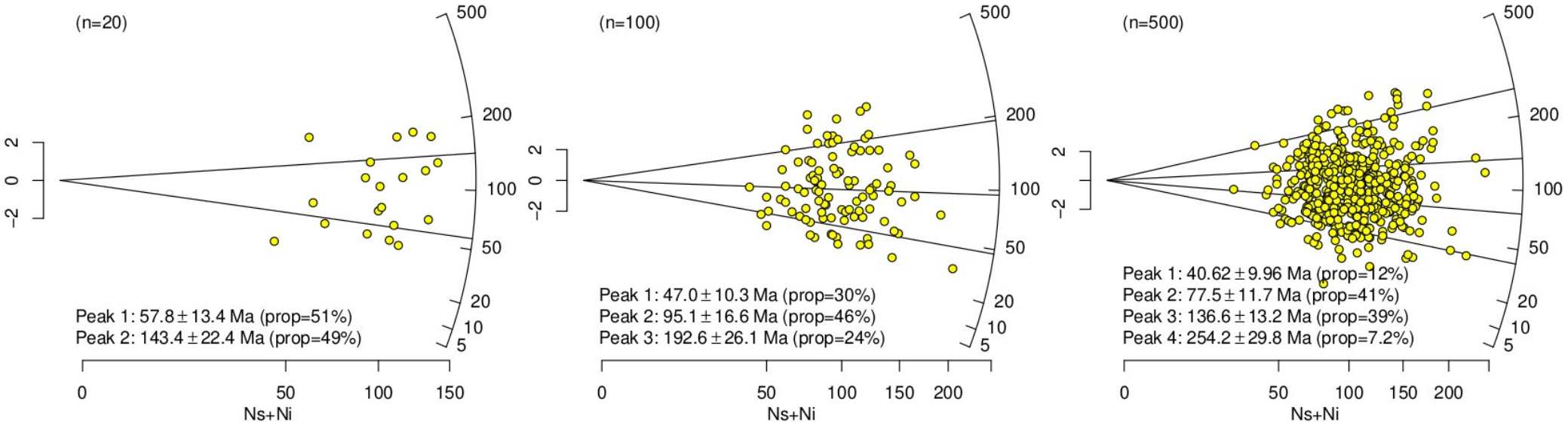
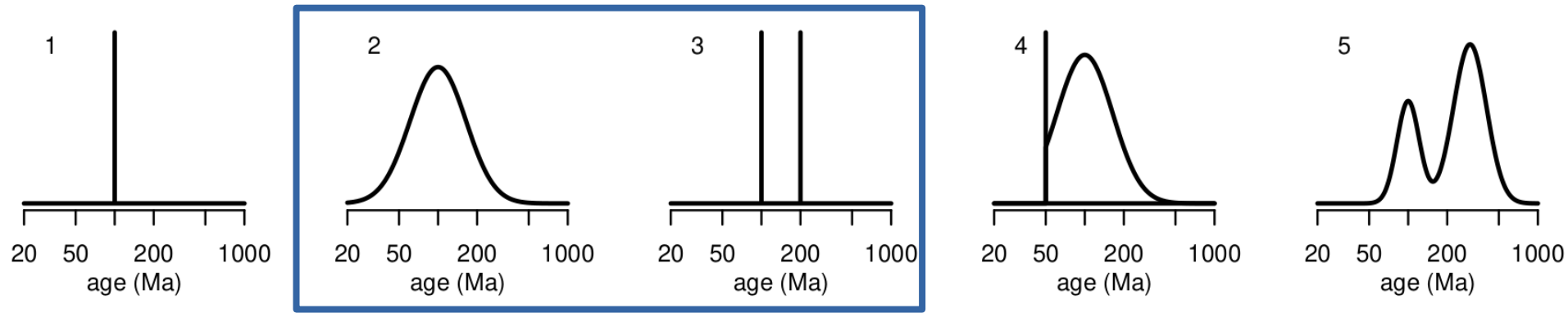
How many peaks to fit?

$$\mathcal{L}(\pi_k, \theta_k, k = 1 \dots K) = \sum_{i=1}^n \ln \left[\sum_{k=1}^K \pi_k f(\theta_k | z_i, s[z_i]) \right]$$

$$\text{BIC} = -2\mathcal{L}_{\max} + p \ln(n)$$

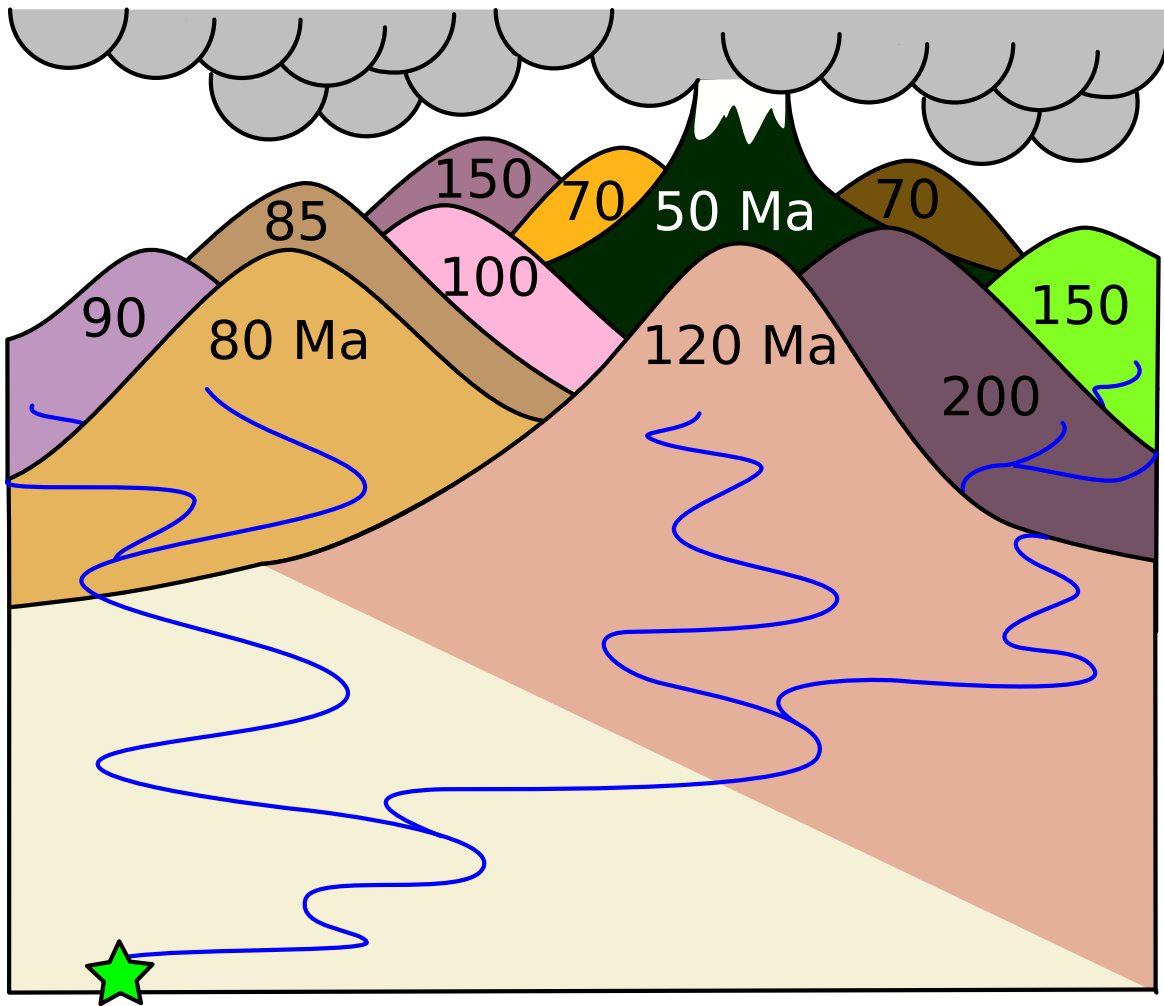


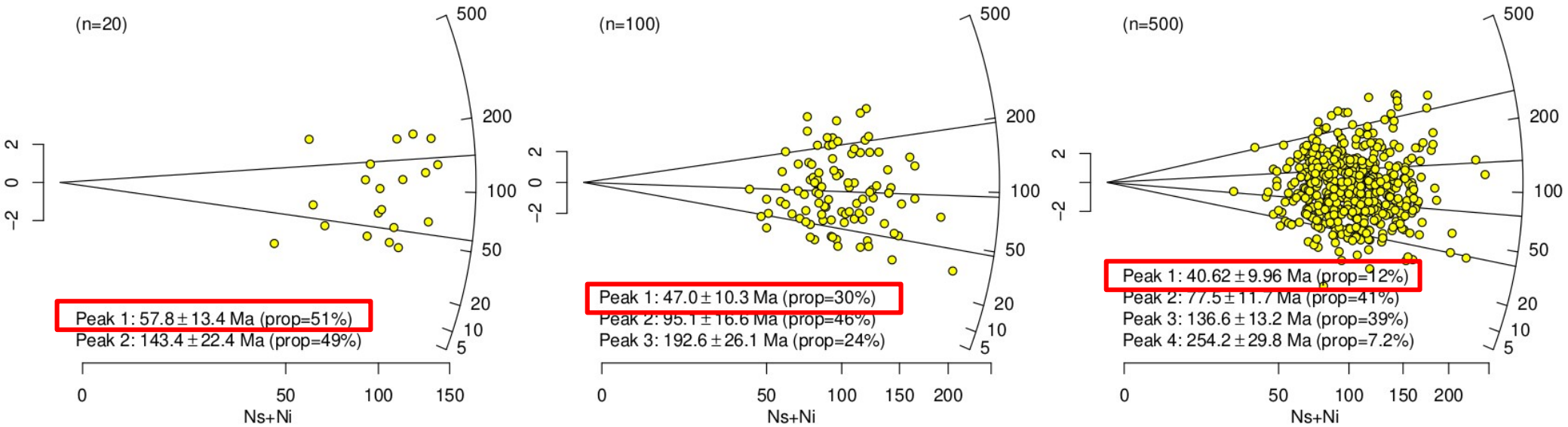
Ages:



Ghost age components in detrital thermochronology

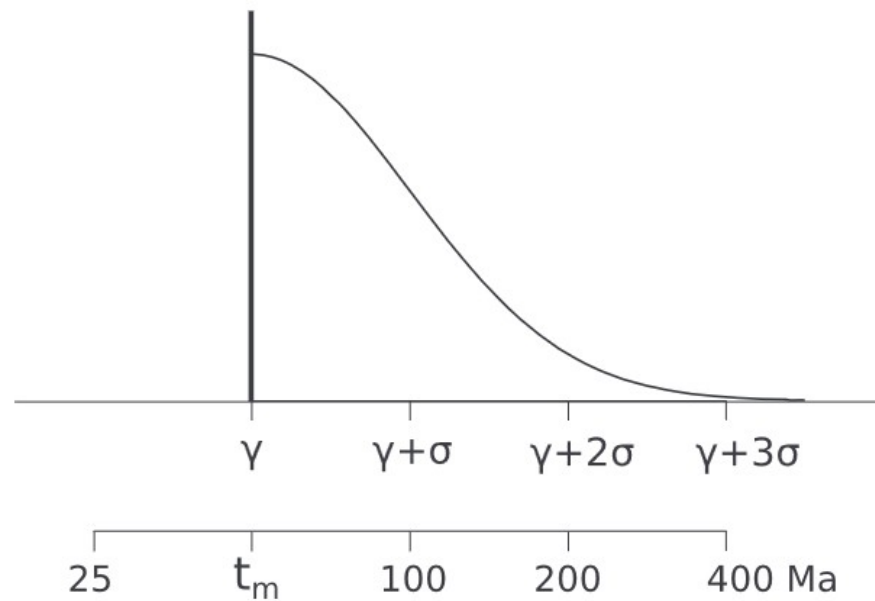
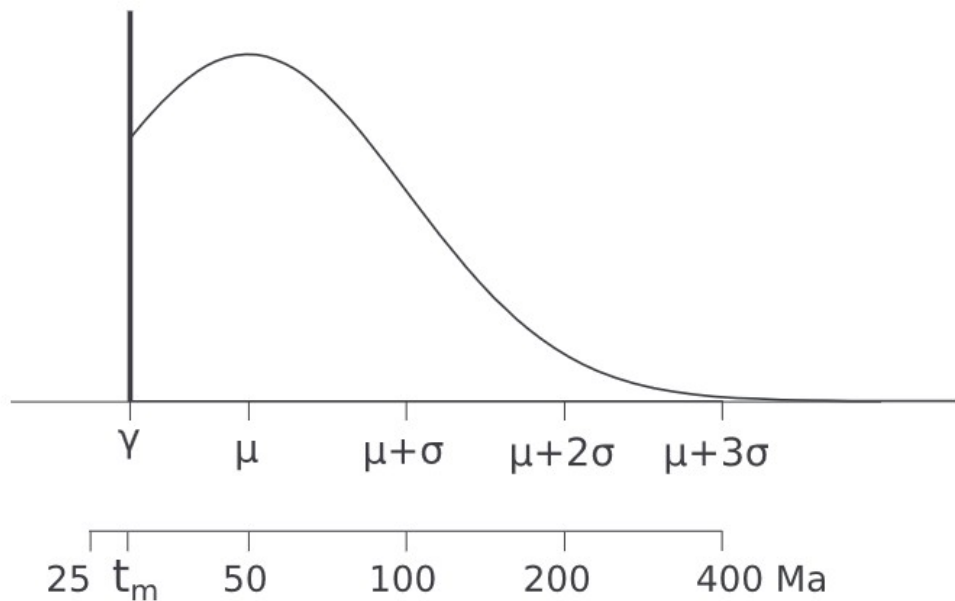
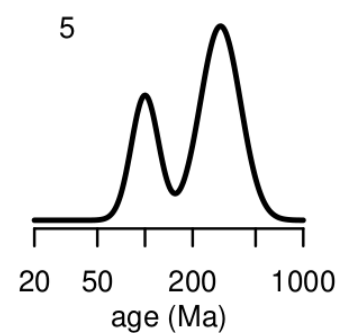
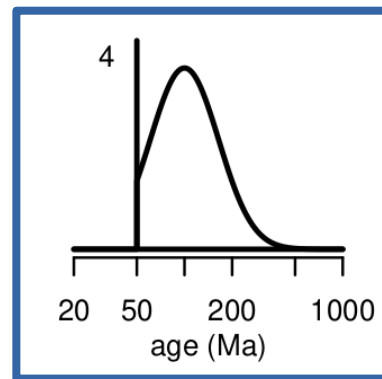
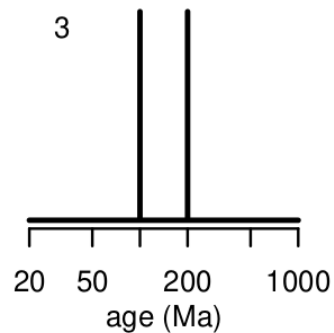
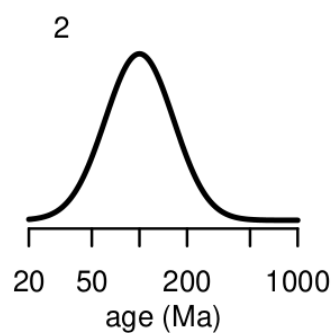
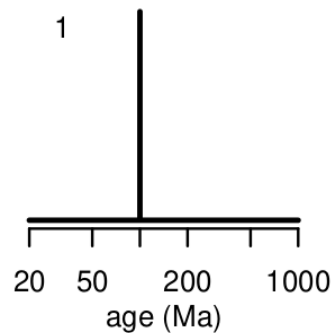
B. Härtel^{a,*}, Pieter Vermeesch^b, Eva Enkelmann^a, Stijn Glorie^c





The minimum age drifts to younger ages with increasing sample size

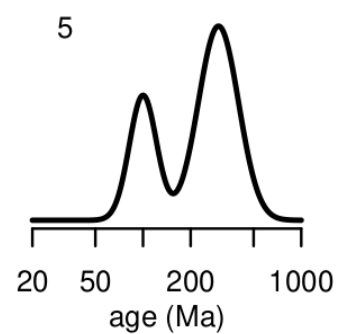
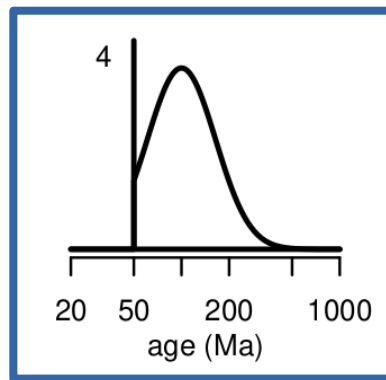
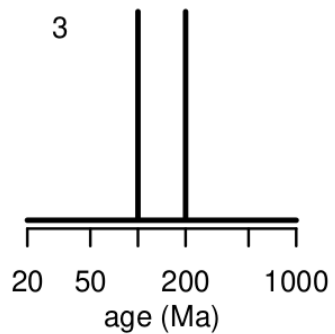
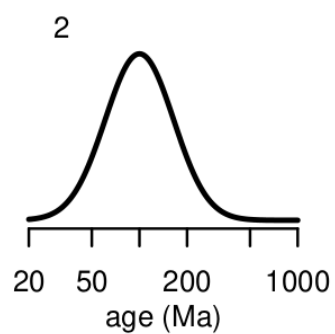
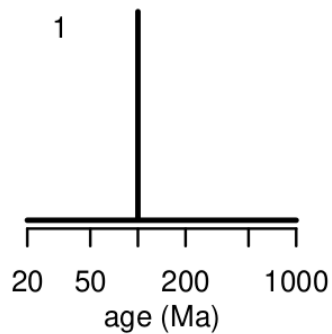
Ages:



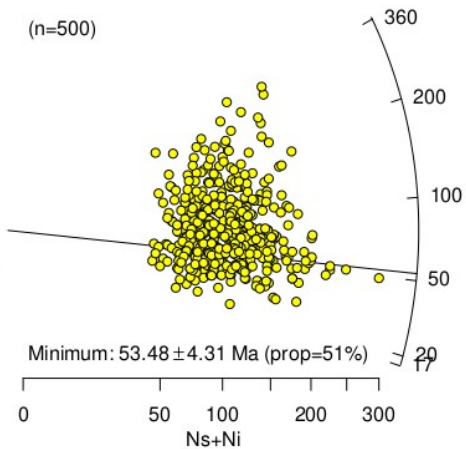
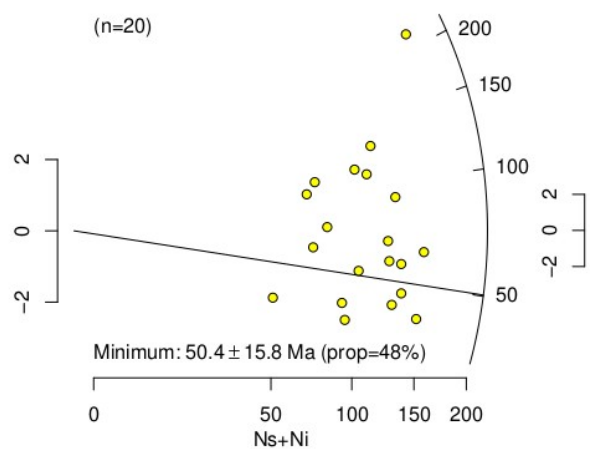
STATISTICAL MODELS FOR MIXED FISSION TRACK AGES

Maximum depositional age estimation revisited

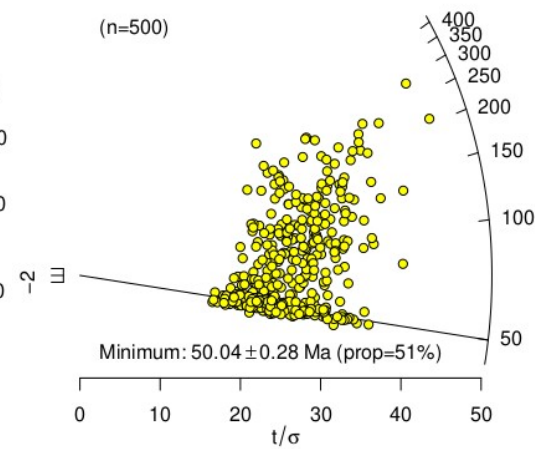
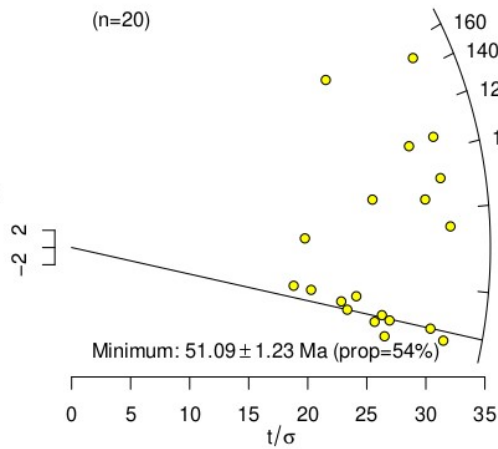
Ages:



Fission tracks



U-Th-He



ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²

Method: EDM ▾

Input errors: 1se (abs) ▾

Output errors: ci (abs) ▾

Probability cutoff: 0.05

☐ Show sample numbers?☐ Propagate external uncertainties?

Transformation: arcsin ▾

Finite mixtures: 2 ▾

Minimum value: 1 auto

Central value: 2 auto

Maximum value: 3 auto

Maximum precision: 4 auto

Plot character: 5 1

Significant digits: auto

Fill colour: minimum Custom colour ramp ▾

from:  to: 

Plot symbol magnification: 1

Font magnification: 1

Colour label:

FT3.json

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	22	64										
2	8	84										
3	16	98										
4	22	71										
5	10	110										
6	13	69										
7	5	77										
8	20	80										
9	24	92										
10	15	99										
11	20	137										
12	19	86										
13	24	74										
14	24	51										
15	19	82										
16	8	76										
17	32	148										
18	10	117										
19	7	64										
20	26	69										
21	18	165										
22	9	82										
23	18	175										
24	16	102										
25	24	93										
26												
27												
28												

Defaults

Clear

Open

Save

PLOT

PDF

ζ : 350 ± 2 yr cm² ρ_D : 2500000 ± 50000 1/cm²

pooled age = 67.5 ± 7.5 Ma (n=25)
central age = 69 ± 16 Ma
MSWD = 4, $p(\chi^2) = 2e-10$
dispersion = 49 ± 20%

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	16	76										
2	12	85										
3	15	73										
4	14	92										
5	15	122										
6	8	67										
7	24	138										
8	7	85										
9	13	49										
10	12	107										
11	7	75										
12	20	91										
13	13	75										
14	27	134										
15	8	58										
16	4	53										
17	13	136										
18	11	81										
19	27	65										
20	6	63										
21	14	150										
22	10	109										
23	13	108										
24	14	147										
25	33	56										
26												
27												
28												

FT4.json

standardised estimate

2
0
-2

Minimum: 54 ± 22 Ma

0

Ns+Ni

50

100

150

200

250

50

100

150

200

250

ζ : 350 ± 2 yr cm²
 ρ_D : 2500000 ± 50000 1/cm²

Method: EDM

Input errors: 1se (abs)

Output errors: ci (abs)

Probability cutoff: 0.05

- ☐ Show sample numbers?
- ☐ Propagate external uncertainties?

Transformation: arcsin

Finite mixtures: minimum ☐ 4 parameters

Minimum value: 1 auto

Central value: 2 auto

Maximum value: 3 auto

Maximum precision: 4 auto

Plot character: 5 1

Significant digits: auto

Fill colour: minimum Custom colour ramp

from:  to: 

Plot symbol magnification: 1

Font magnification: 1

Colour label:

	Ns	Ni	(C)	(omit)	(comment)	F	G	H	I	J	K	L
1	16	76										
2	12	85										
3	15	73										
4	14	92										
5	15	122										
6	8	67										
7	24	138										
8	7	85										
9	13	49										
10	12	107										
11	7	75										
12	20	91										
13	13	75										
14	27	134										
15	8	58										
16	4	53										
17	13	136										
18	11	81										
19	27	65										
20	6	63										
21	14	150										
22	10	109										
23	13	108										
24	14	147										
25	33	56										
26												
27												
28												

```
FT2a <- FTsamp(pop=2,ng=20)
```

```
radialplot(FT2a,k=2)
```

```
radialplot(FT2a,k="auto")
```

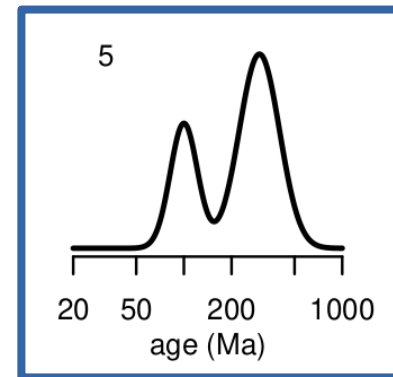
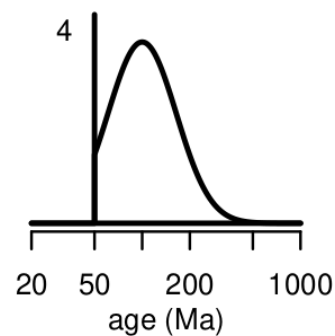
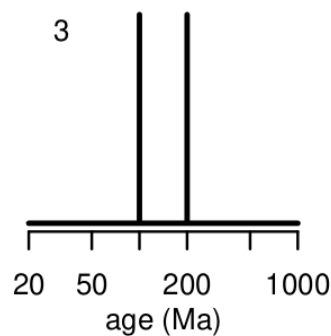
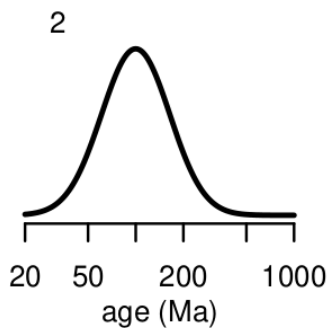
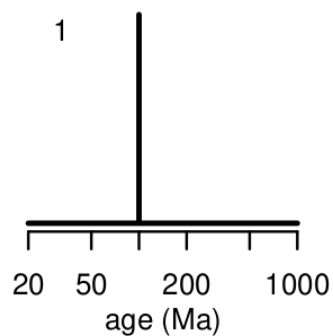
```
FT2b <- FTsamp(pop=2,ng=200)
```

```
radialplot(FT2b,k="auto")
```

```
FT3 <- FTsamp(pop=3,ng=20)
```

```
radialplot(FT3,k="min")
```

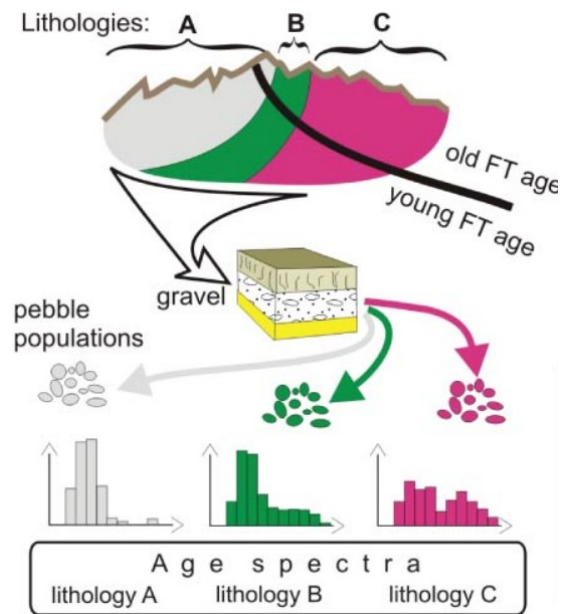
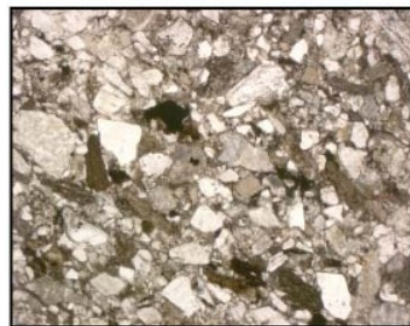
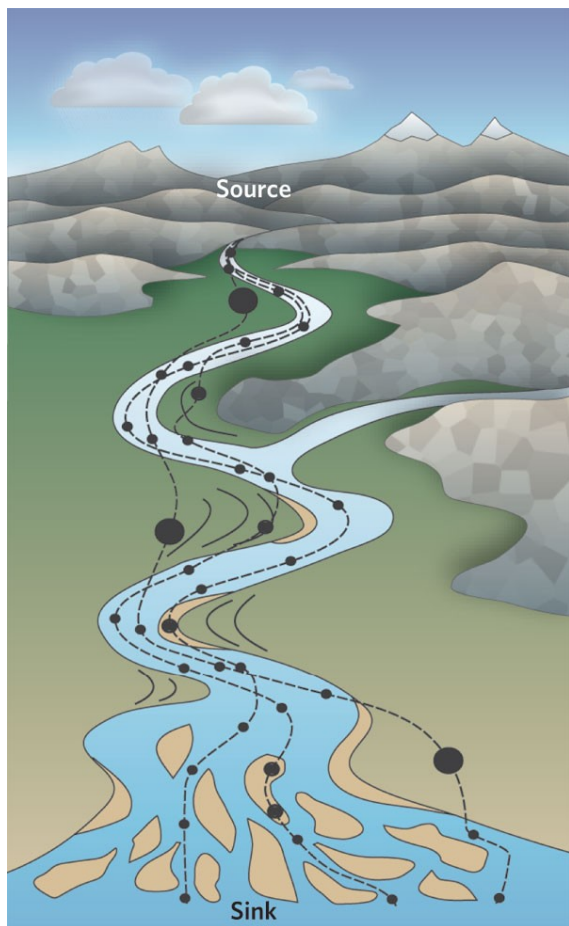
Ages:



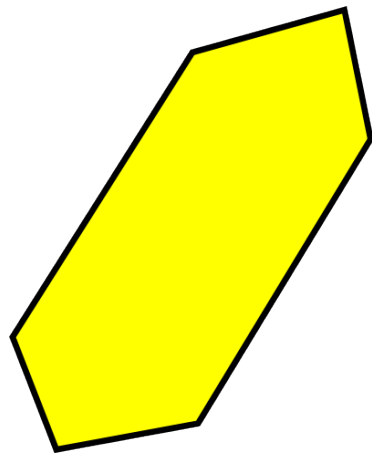
Bayesian Mixture Modelling in Geochronology via Markov Chain Monte Carlo¹

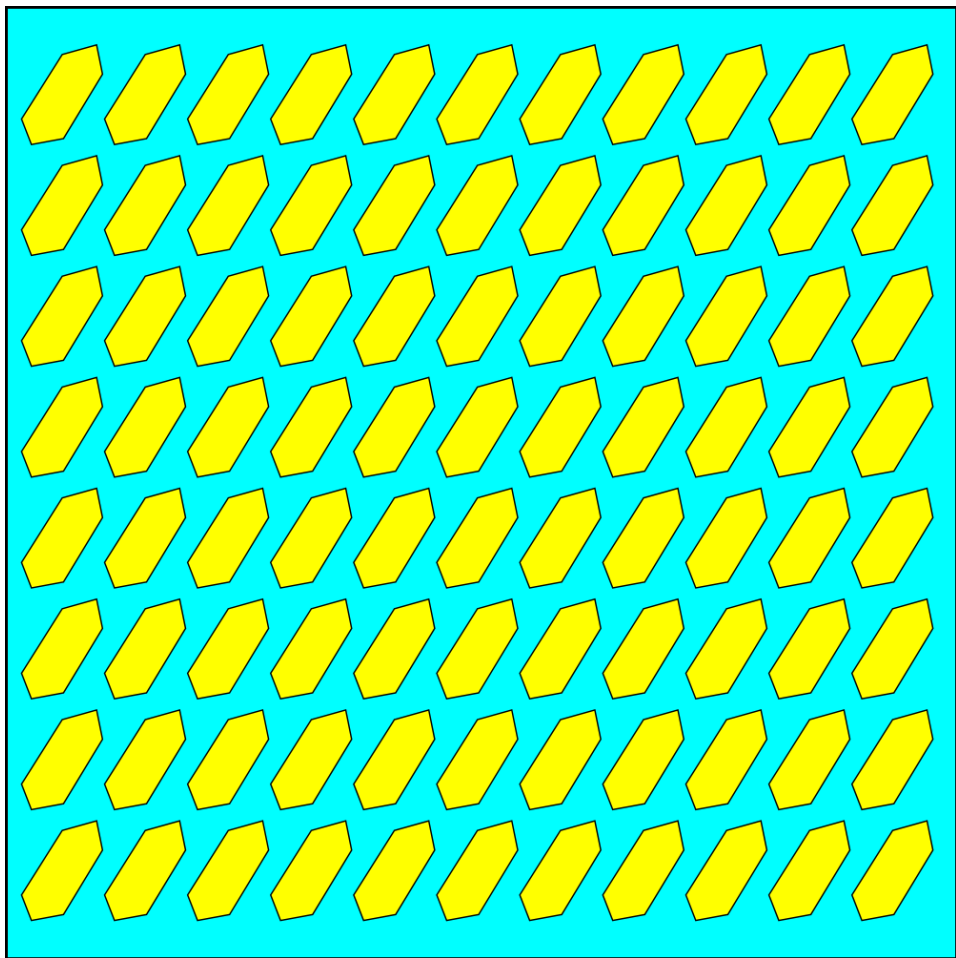
Ajay Jasra,² David A. Stephens,² Kerry Gallagher,³
and Christopher C. Holmes^{4,5}

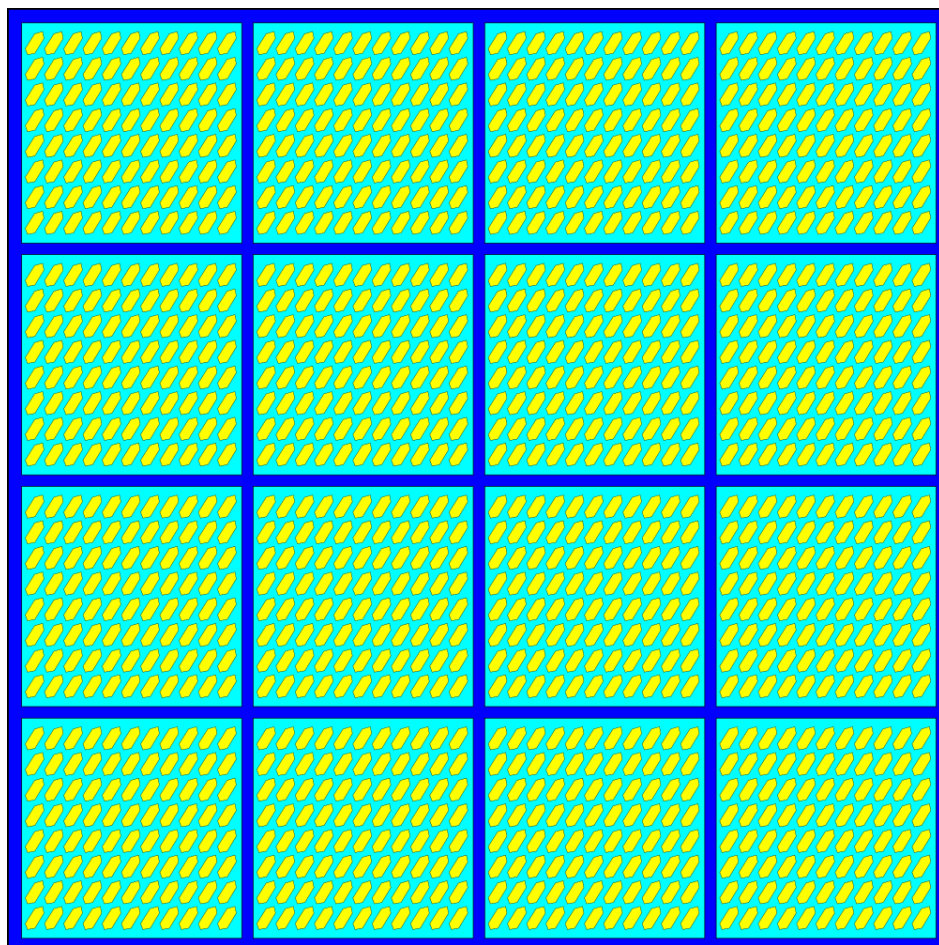
3p-1 parameters, including p σ s $\rightarrow$ requires very large datasets

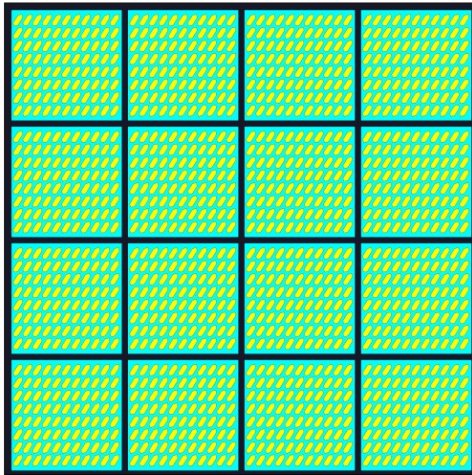
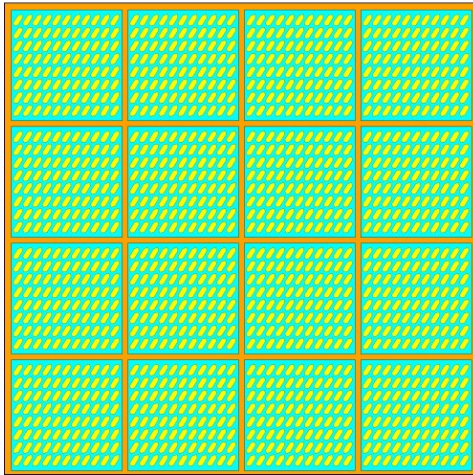
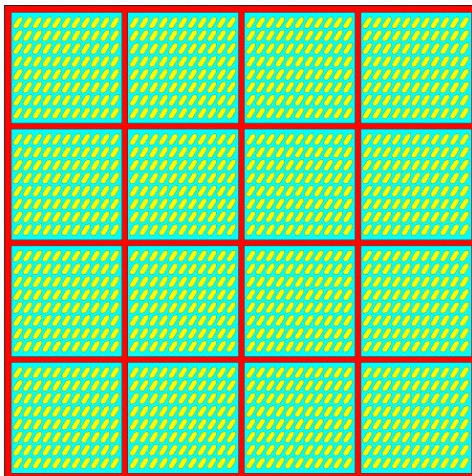
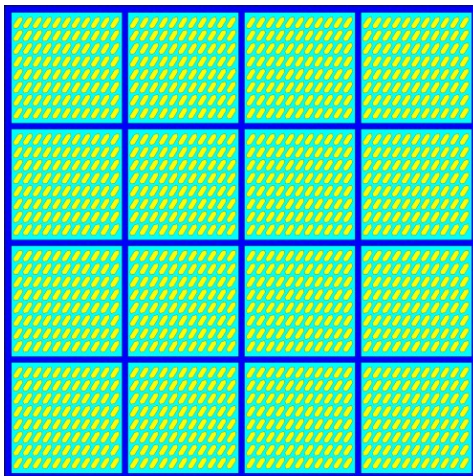


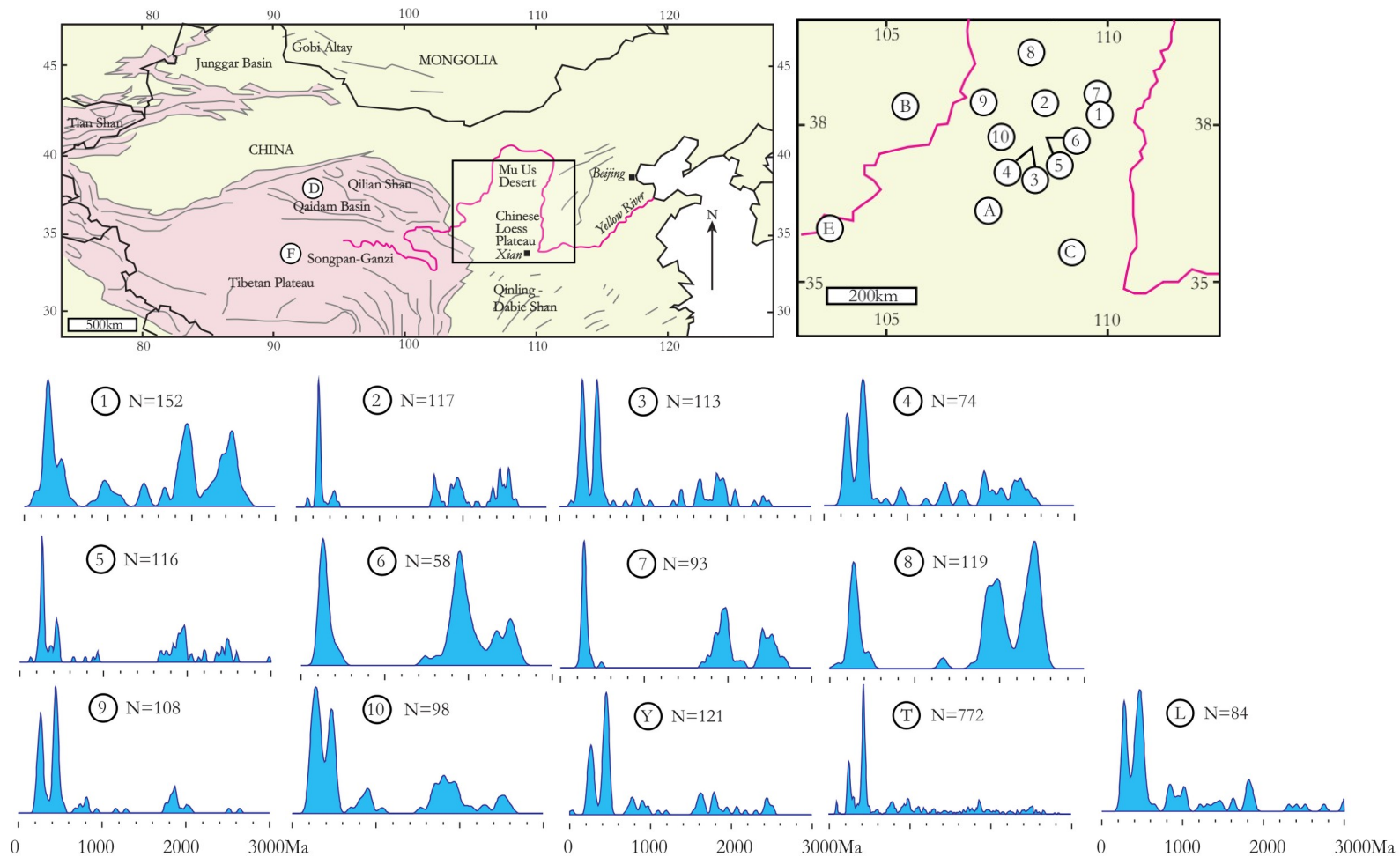
(Figure courtesy of
H. von Eynatten)



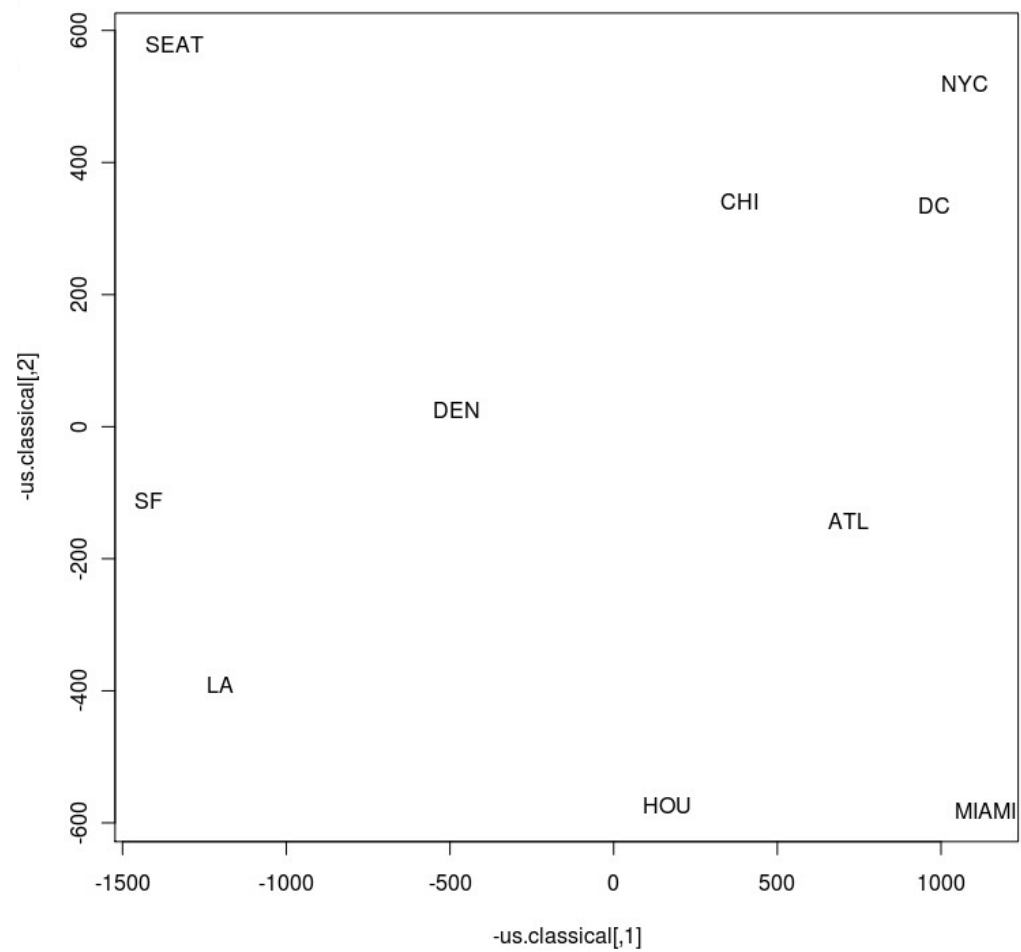




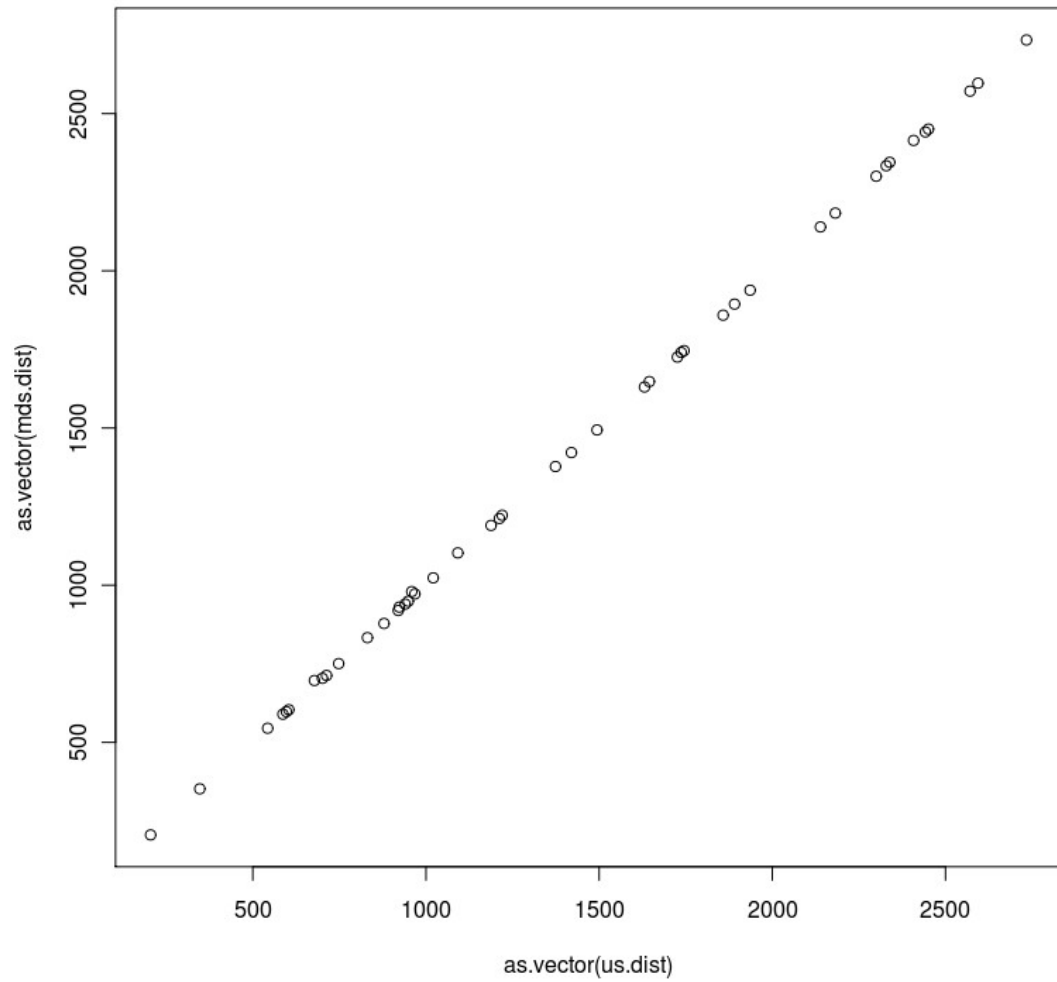


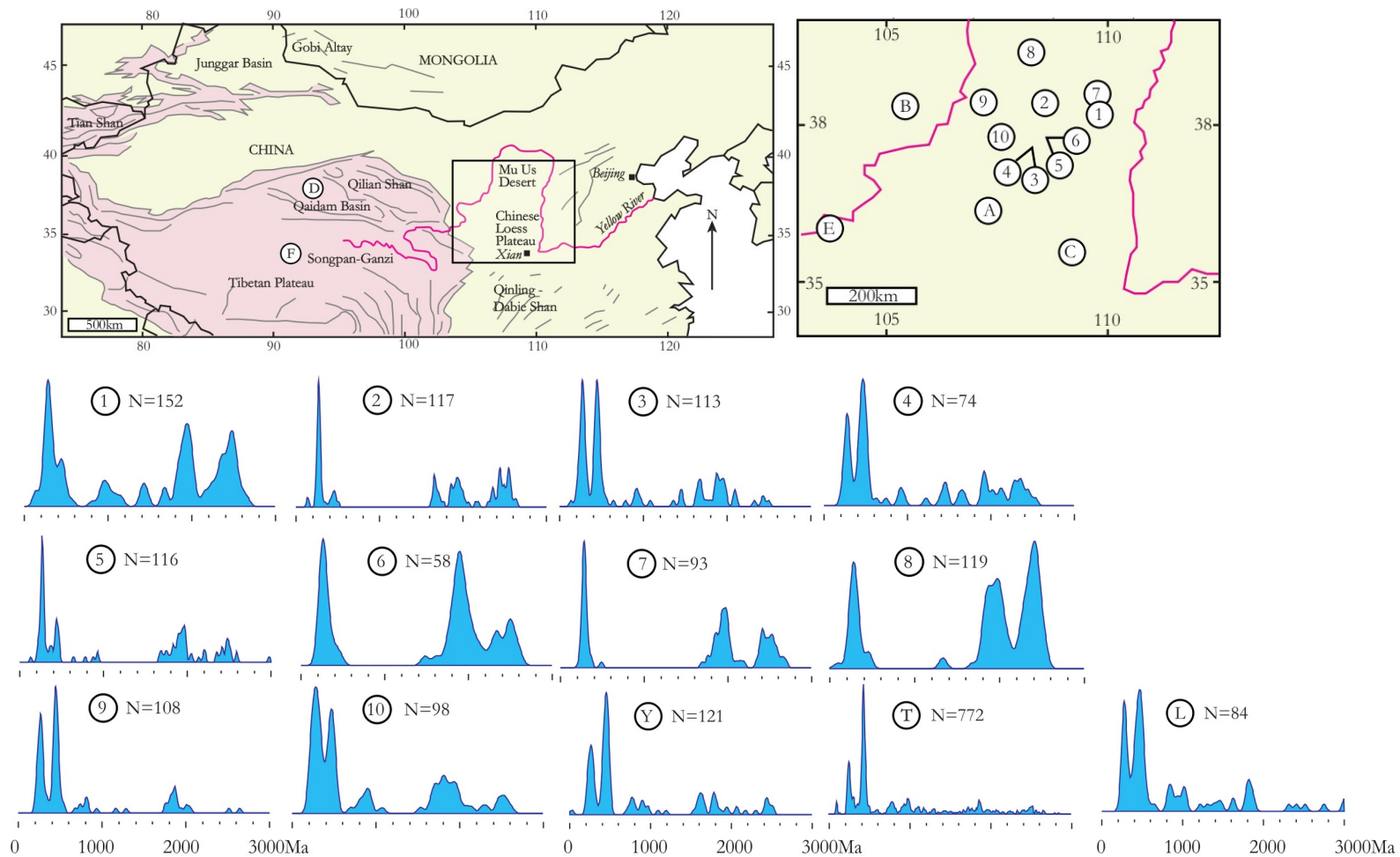


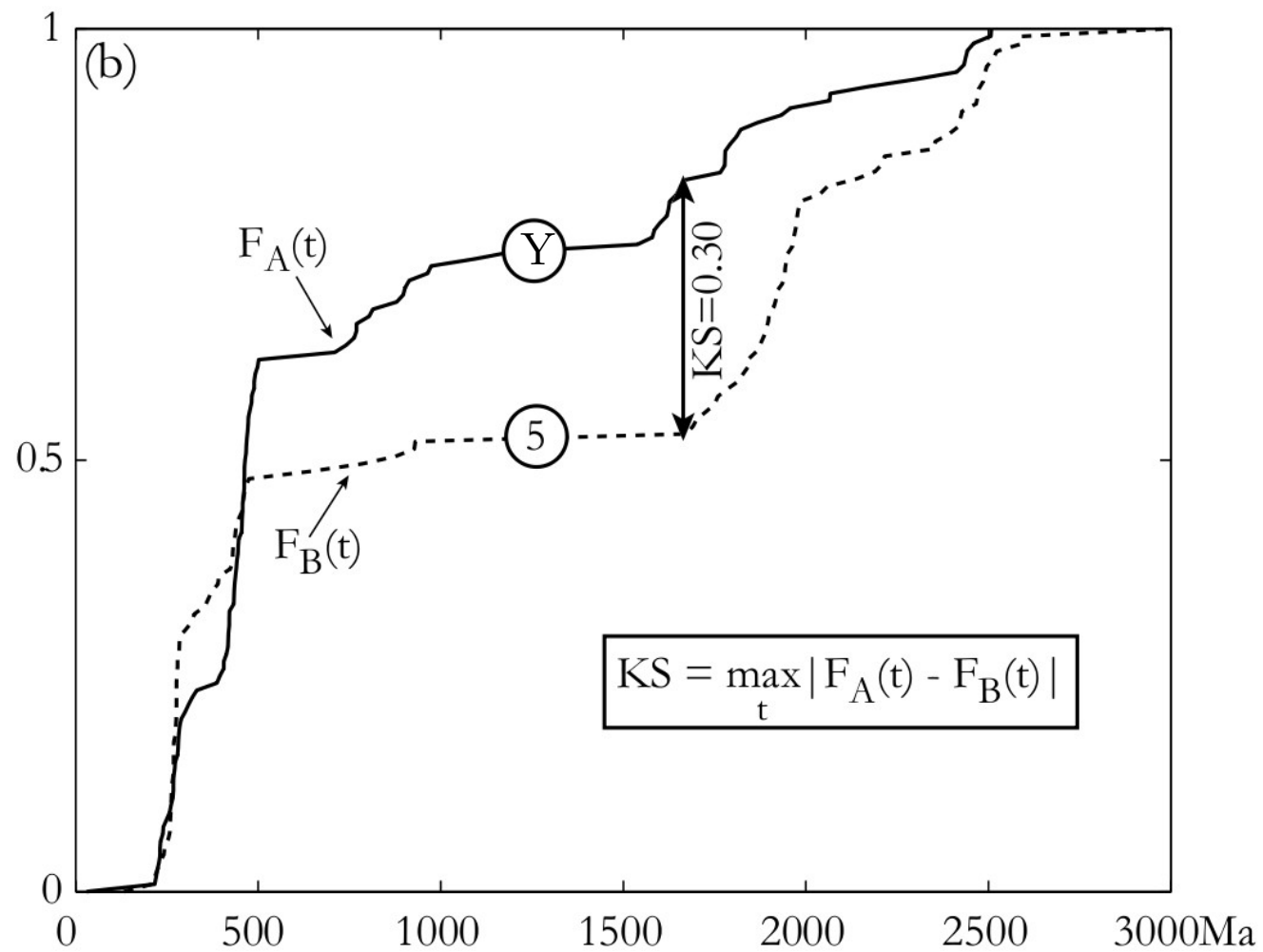
	ATL	CHI	DEN	HOU	LA	MIAMI	NYC	SF	SEAT	DC
ATL	0	587	1212	701	1936	604	748	2139	2182	543
CHI	587	0	920	940	1745	1188	713	1858	1737	597
DEN	1212	920	0	879	831	1726	1631	949	1021	1494
HOU	701	940	879	0	1374	968	1420	1645	1891	1220
LA	1936	1745	831	1374	0	2339	2451	347	959	2300
MIAMI	604	1188	1726	968	2339	0	1092	2594	2734	923
NYC	748	713	1631	1420	2451	1092	0	2571	2408	205
SF	2139	1858	949	1645	347	2594	2571	0	678	2442
SEAT	2182	1737	1021	1891	959	2734	2408	678	0	2329
DC	543	597	1494	1220	2300	923	205	2442	2329	0



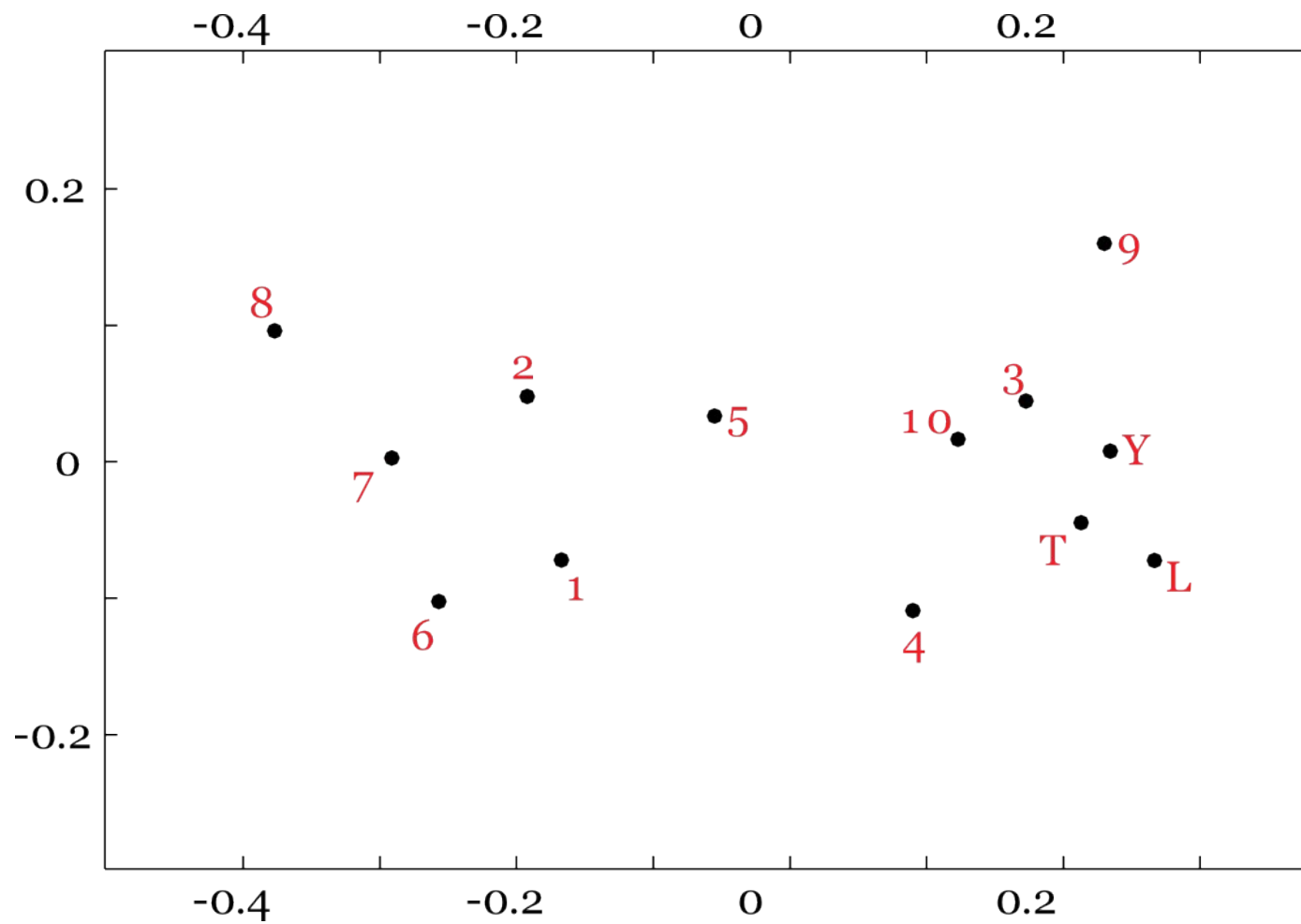
$$d_{i,j} = \sqrt{(x_i^1 - x_j^1)^2 + (x_i^2 - x_j^2)^2 + \dots + (x_i^R - x_j^R)^2}$$



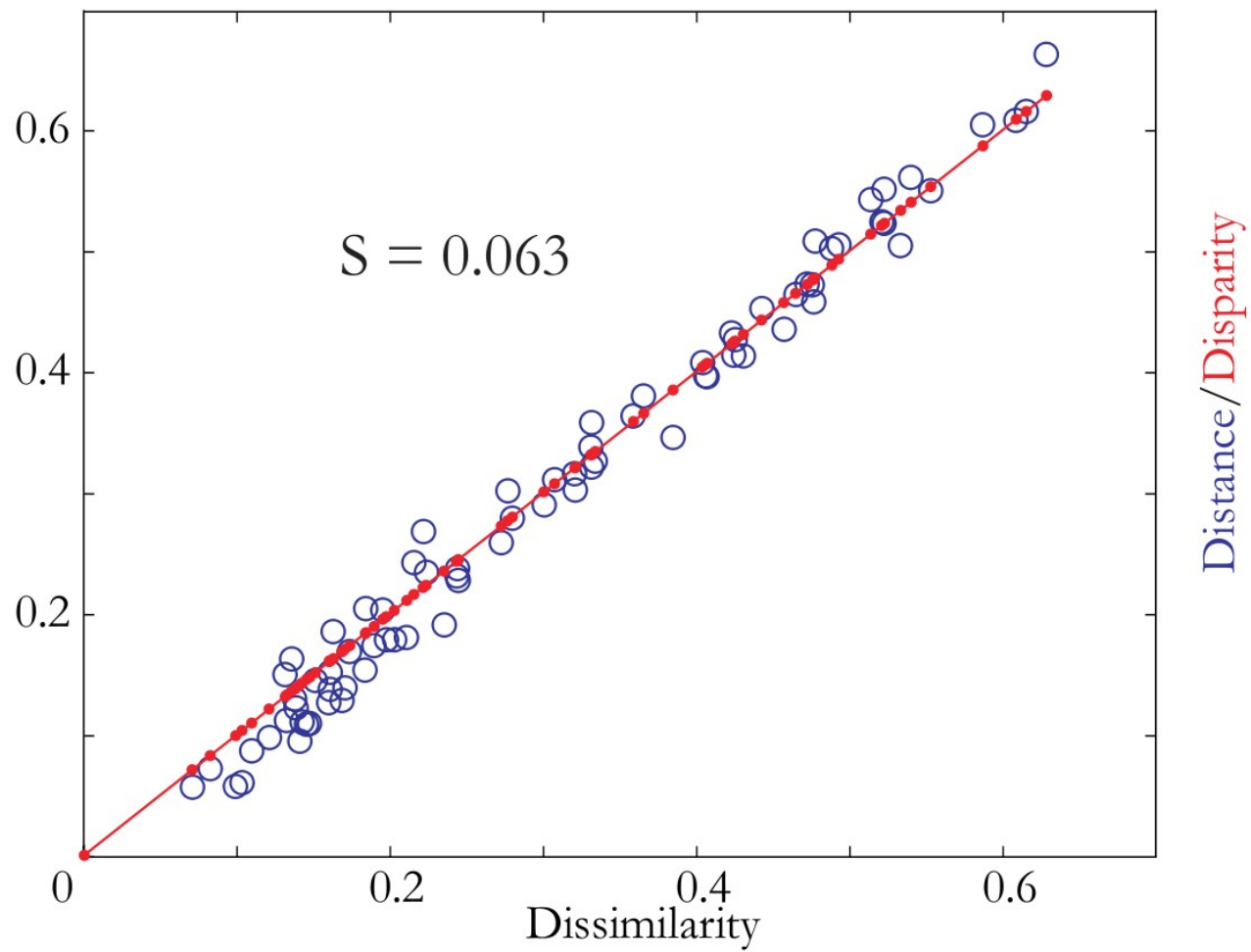


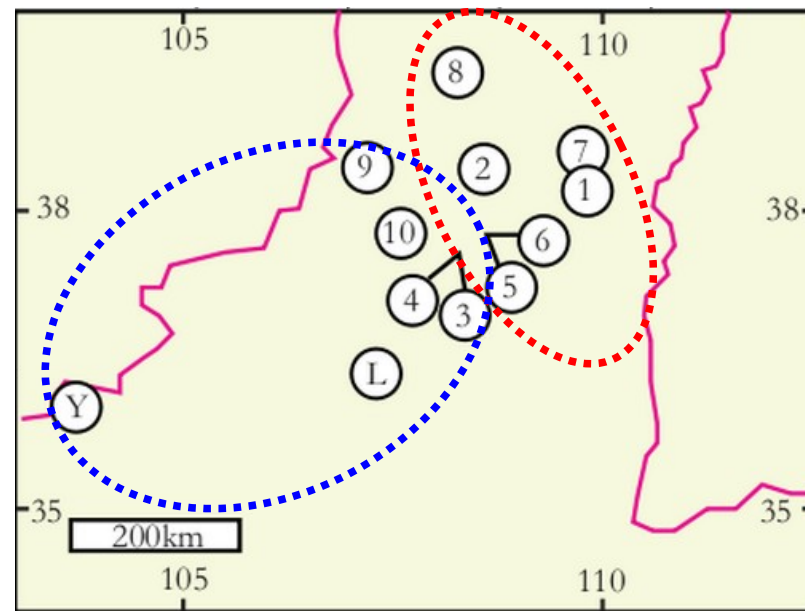
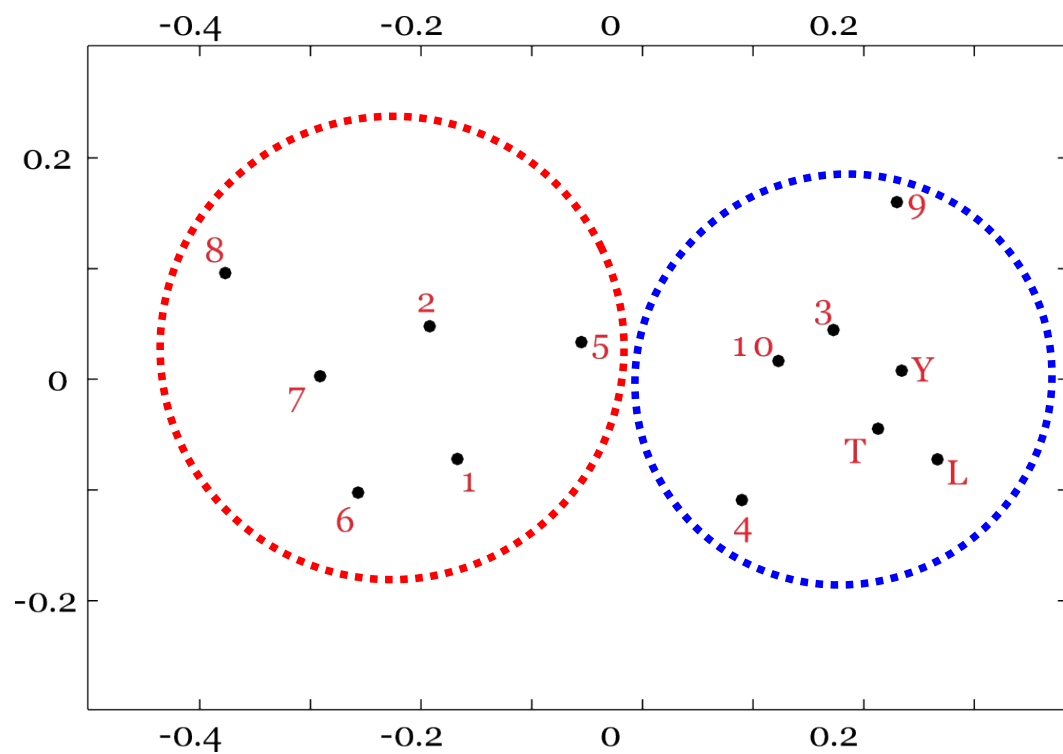


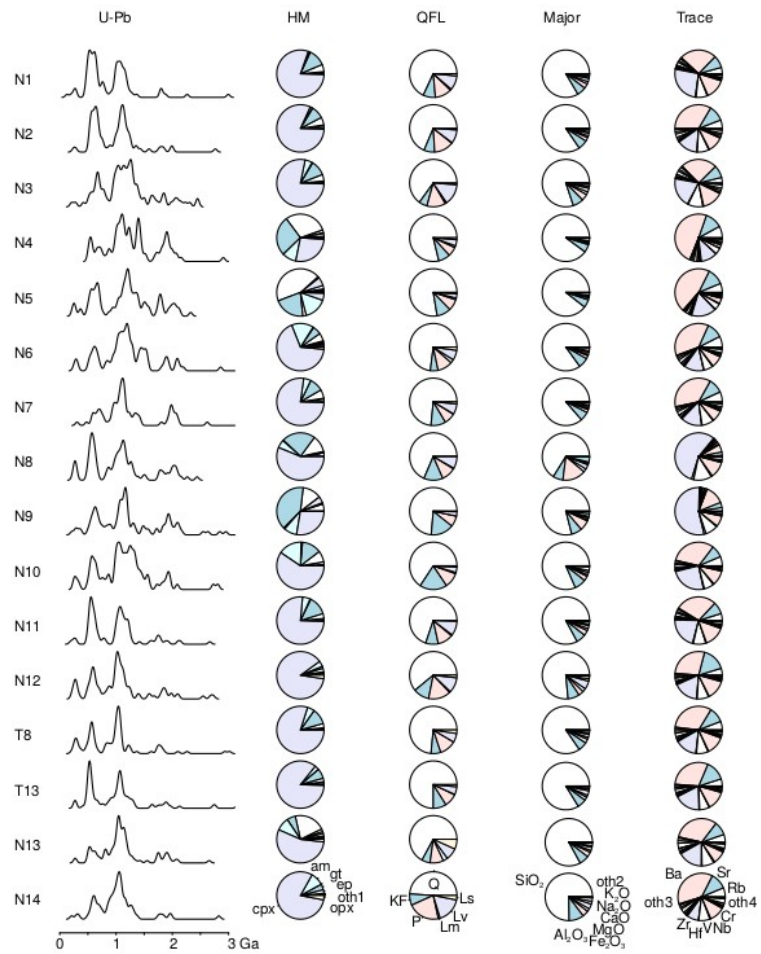
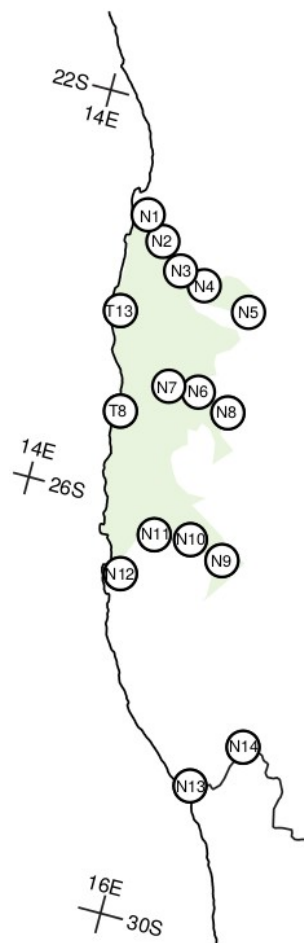
$$\delta = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & L & T & Y \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ L \\ T \\ Y \end{matrix} & \left(\begin{array}{cccccccccccccc} 0 & 14 & 33 & 27 & 18 & 14 & 15 & 22 & 48 & 32 & 42 & 37 & 40 \\ 14 & 0 & 36 & 33 & 16 & 14 & 15 & 24 & 46 & 32 & 47 & 42 & 43 \\ 33 & 36 & 0 & 19 & 24 & 44 & 47 & 55 & 17 & 10 & 13 & 12 & 8 \\ 27 & 33 & 19 & 0 & 20 & 38 & 41 & 48 & 28 & 14 & 21 & 17 & 16 \\ 18 & 16 & 24 & 20 & 0 & 22 & 24 & 33 & 31 & 20 & 33 & 28 & 30 \\ 14 & 14 & 44 & 38 & 22 & 0 & 14 & 24 & 52 & 41 & 52 & 48 & 49 \\ 15 & 15 & 47 & 41 & 24 & 14 & 0 & 16 & 51 & 43 & 54 & 49 & 52 \\ 22 & 24 & 55 & 48 & 33 & 24 & 16 & 0 & 61 & 53 & 63 & 59 & 62 \\ 48 & 46 & 17 & 28 & 31 & 52 & 51 & 61 & 0 & 20 & 22 & 18 & 16 \\ 32 & 32 & 10 & 14 & 20 & 41 & 43 & 53 & 20 & 0 & 17 & 15 & 13 \\ 42 & 47 & 13 & 21 & 33 & 52 & 54 & \boxed{63} & 22 & 17 & 0 & 10 & 11 \\ 37 & 42 & 12 & 17 & 28 & 48 & 49 & 59 & 18 & 15 & 10 & 0 & \boxed{7} \\ 40 & 43 & 8 & 16 & 30 & 49 & 52 & 62 & 16 & 13 & 11 & 7 & 0 \end{array} \right) \end{matrix}$$

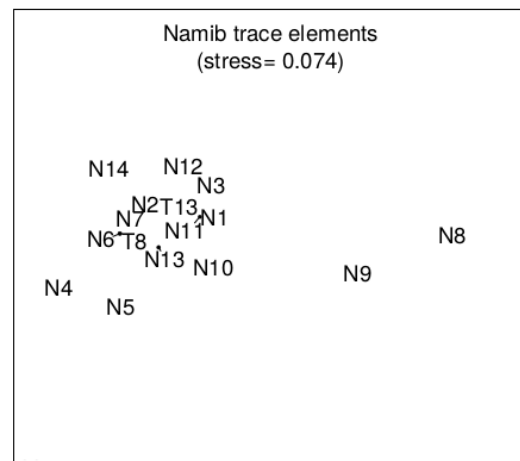
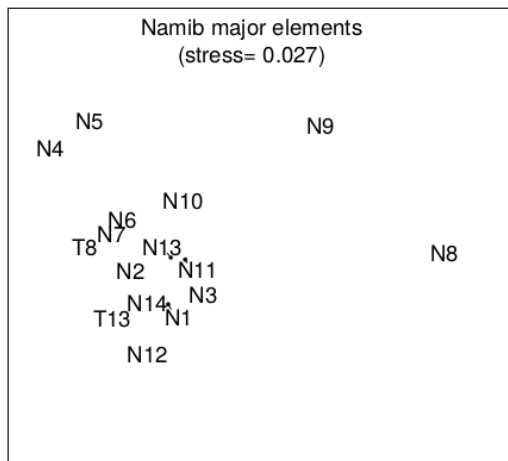
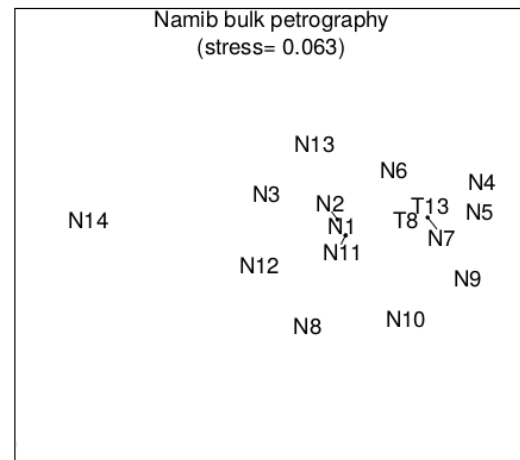
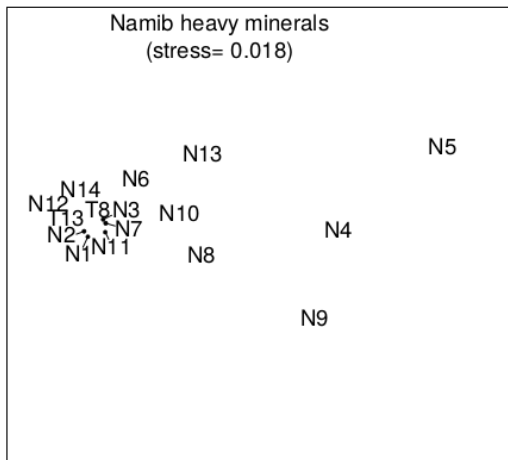
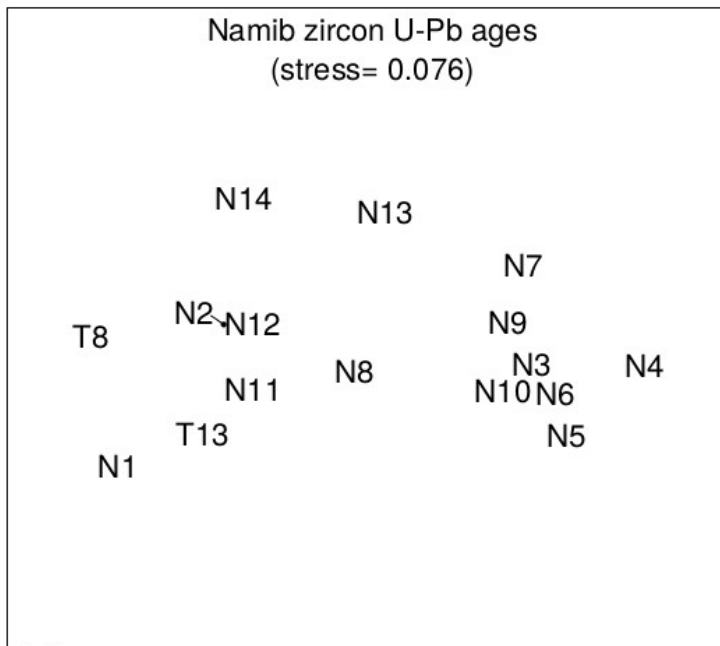
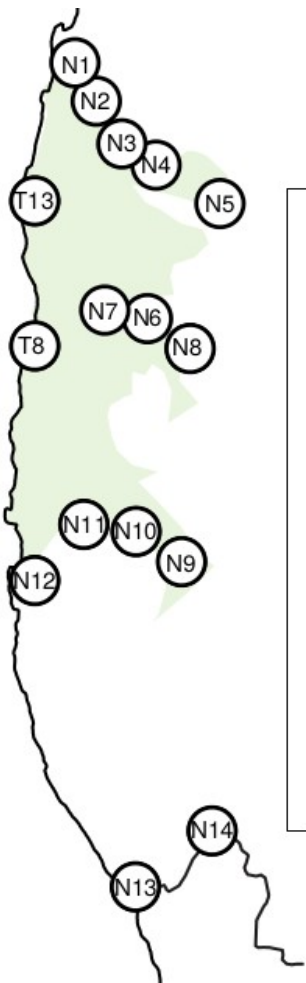


$$d = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & L & T & Y \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ L \\ T \\ Y \end{matrix} & \left(\begin{array}{ccccccccccccccc} 0 & 12 & 35 & 26 & 18 & 11 & 18 & 23 & 48 & 32 & 42 & 38 & 41 \\ 12 & 0 & 36 & 30 & 14 & 7 & 9 & 18 & 44 & 33 & 45 & 40 & 43 \\ 35 & 36 & 0 & 10 & 23 & 42 & 45 & 54 & 17 & 3 & 9 & 5 & 7 \\ 26 & 30 & 10 & 0 & 18 & 35 & 39 & 47 & 27 & 8 & 16 & 12 & 15 \\ 18 & 14 & 23 & 18 & 0 & 21 & 22 & 31 & 31 & 20 & 32 & 27 & 29 \\ 11 & 7 & 42 & 35 & 21 & 0 & 7 & 13 & 51 & 39 & 50 & 46 & 48 \\ 18 & 9 & 45 & 39 & 22 & 7 & 0 & 9 & 52 & 42 & 54 & 49 & 51 \\ 23 & 18 & 54 & 47 & 31 & 13 & 9 & 0 & 61 & 50 & 62 & 58 & 60 \\ 48 & 44 & 17 & 27 & 31 & 51 & 52 & 61 & 0 & 19 & 21 & 20 & 18 \\ 32 & 33 & 3 & 8 & 20 & 39 & 42 & 50 & 19 & 0 & 12 & 8 & 10 \\ 42 & 45 & 9 & 16 & 32 & 50 & 54 & \boxed{62} & 21 & 12 & 0 & 5 & 4 \\ 38 & 40 & 5 & 12 & 27 & 46 & 49 & 58 & 20 & 8 & 5 & 0 & \boxed{3} \\ 41 & 43 & 7 & 15 & 29 & 48 & 51 & 60 & 18 & 10 & 4 & 3 & 0 \end{array} \right) \end{matrix}$$

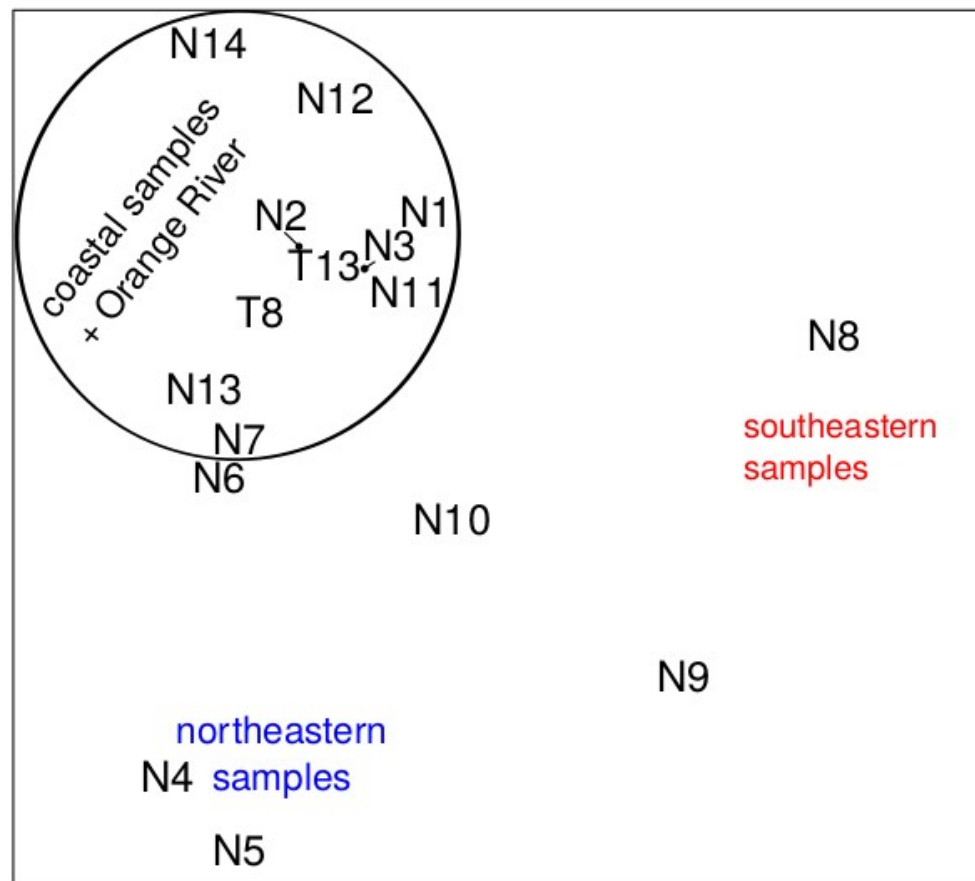
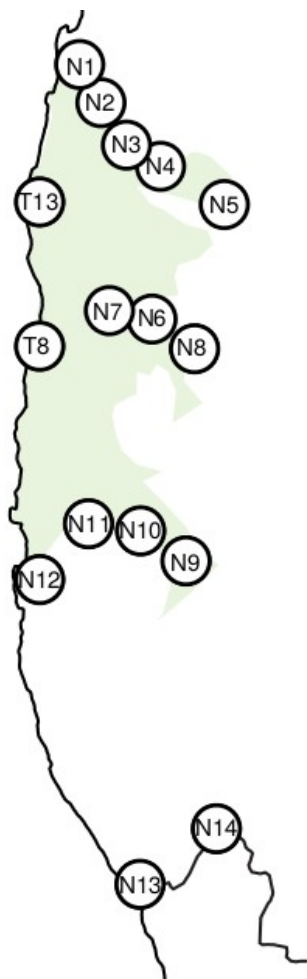


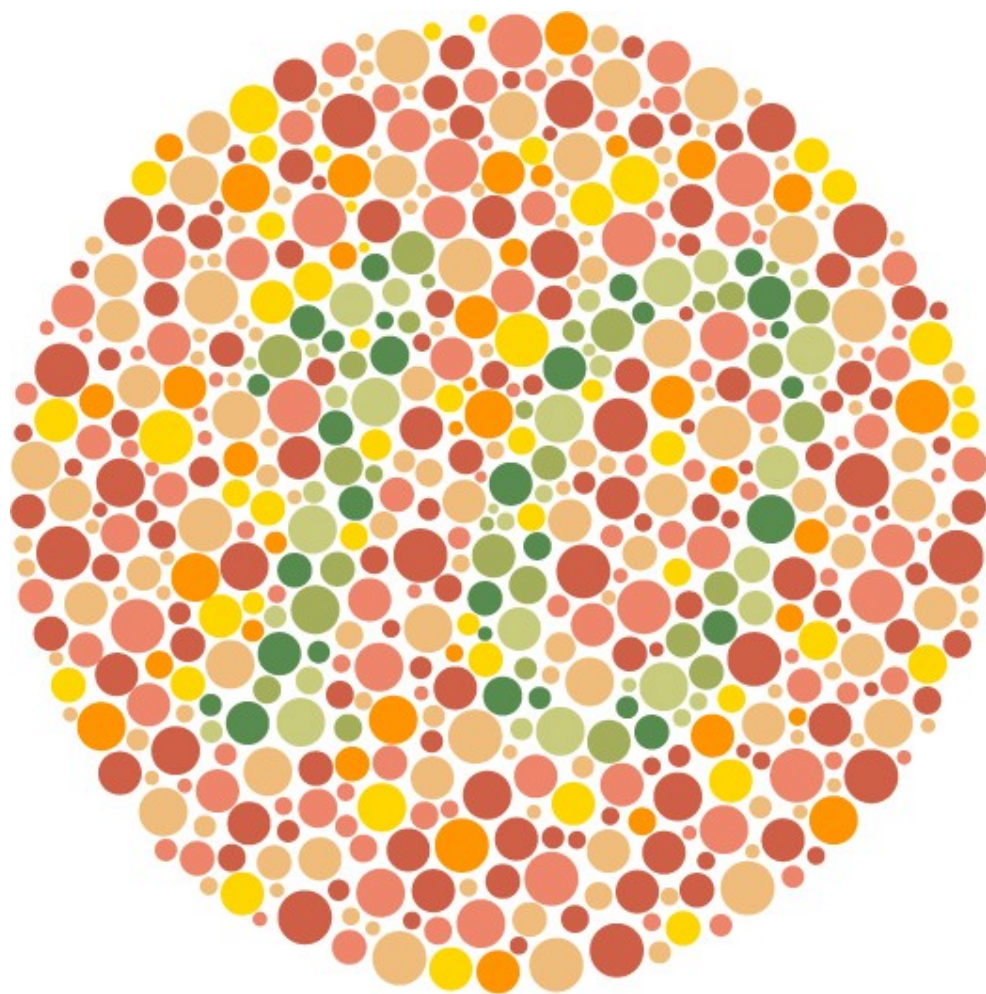












Each of these 16 9x9 distance matrices can be plotted as an MDS map

\$N1

	Pur	R	ed	Y	el	G	y1	G	y2	Green	Blue	B	lP	Pur1
Red	6.8													
Yel	12.5	5.4												
Gy1	13.8	8.3	5.2											
Gy2	14.2	10.4	7.2	3.7										
Green	12.5	11.6	9.5	5.9	4.2									
Blue	11.0	13.8	11.3	10.1	6.9	4.3								
BlP	8.6	14.3	13.5	11.1	10.2	6.8	4.8							
Pur1	5.5	11.8	14.6	12.3	12.1	9.9	7.4	4.5						
Pur2	3.5	8.9	14.1	12.5	11.2	10.7	8.7	6.1	3.6					

Gy2 9.6 9.3 R Pur R ed Y el G y1 G y2 Green Blue B lP Pur1
 Green 10.8 9.9 Red 6.8
 Blue 9.7 11.7 1 Yel 12.5 5.4
 BlP 8.5 11.6 1 Gy1 13.8 8.3 5.2
 Pur1 4.9 10.3 1 Gy2 14.2 10.4 7.2 3.7
 Pur2 3.5 8.0 1 Green 12.5 11.6
 Blue 11.0 13.8
 BlP 8.6 14.3
 Pur1 5.5 11.8
 Pur2 3.5 8.9

4

.N8

	Pur	R	ed	Y	el	G	y1	G	y2
Red	7.5								
Yel	9.1	4.4							
Gy1	10.2	7.9	5.7						
Gy2	12.1	10.4	8.3	3.9					
Green	12.5	11.2	10.2	6.5	4				
Blue	9.7	12.6	11.3	8.7	7				
BlP	9.8	11.4	12.2	10.3	9				
Pur1	8.3	11.3	11.9	10.7	11				
Pur2	6.7	10.4	10.7	12.6	11				

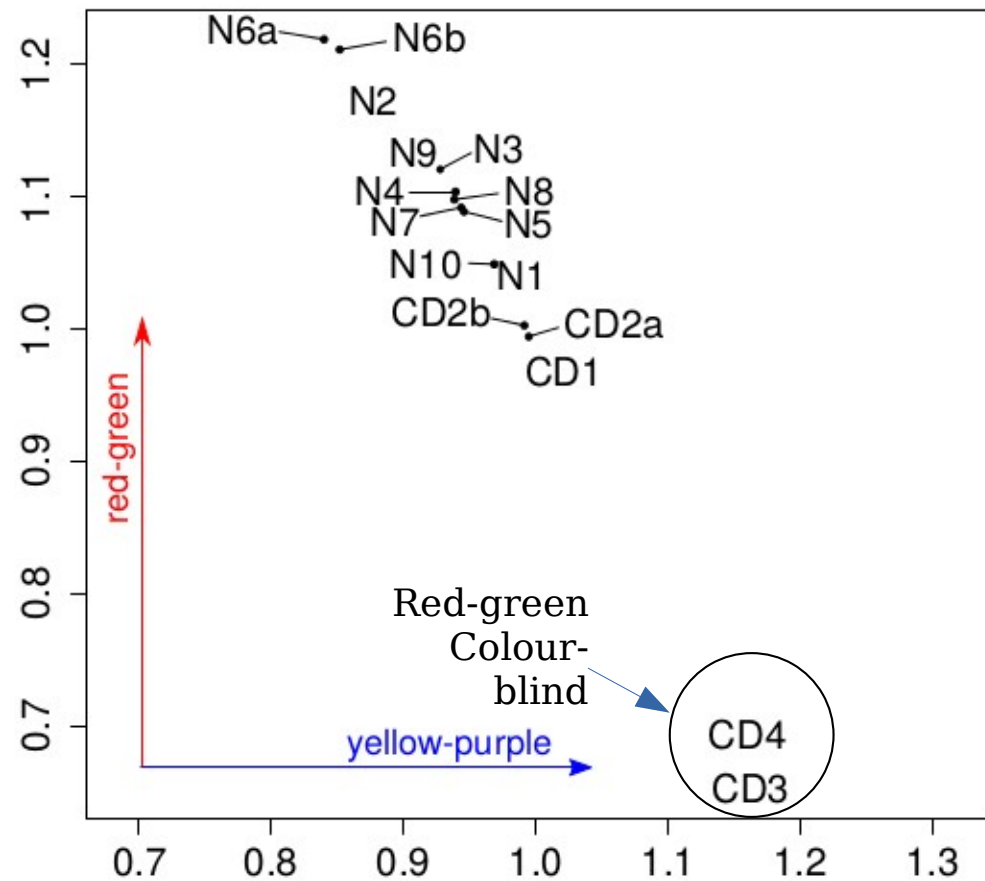
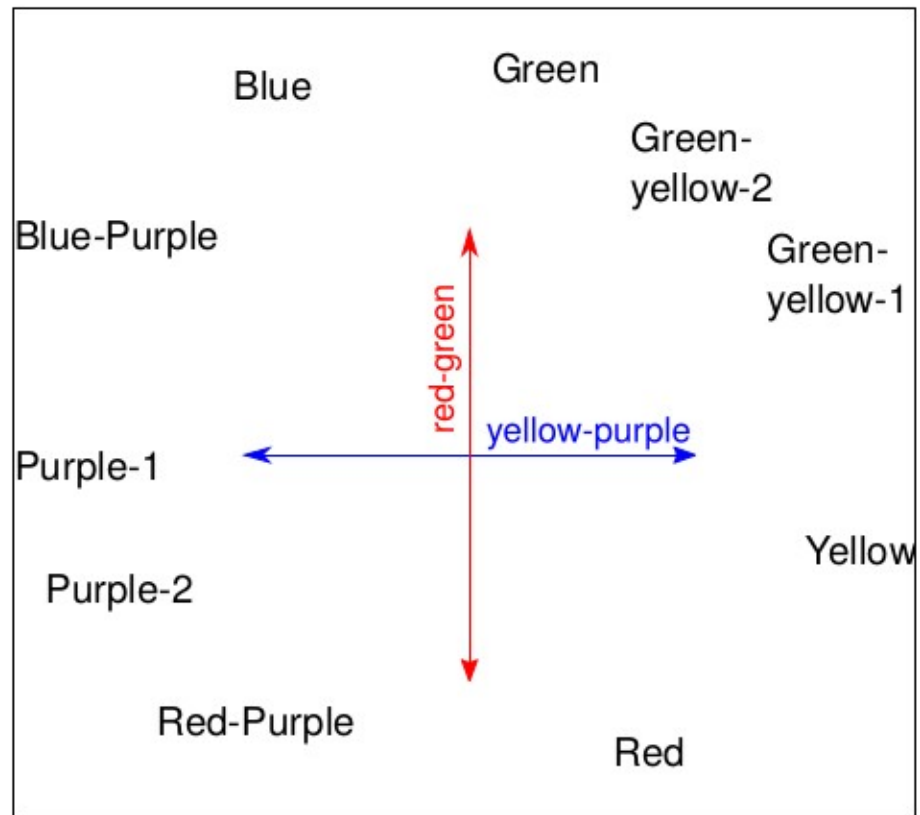
\$CD2a

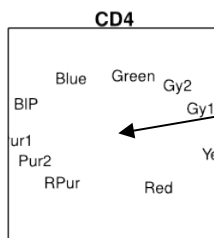
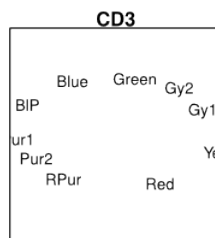
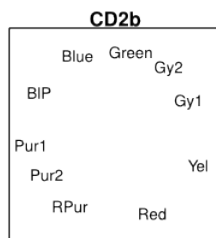
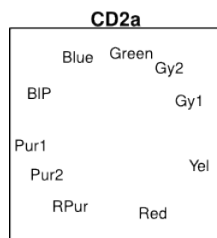
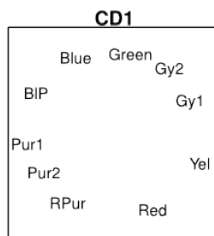
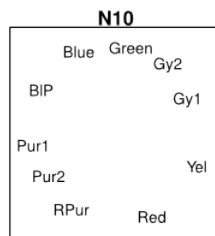
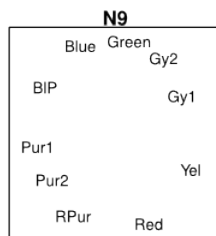
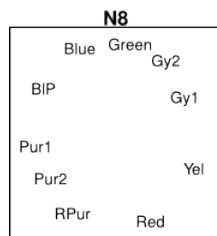
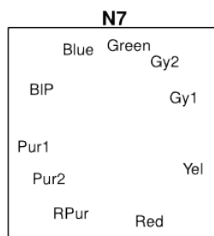
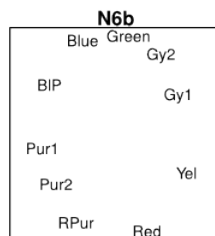
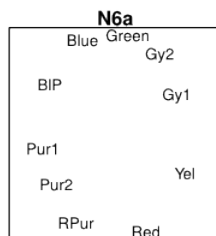
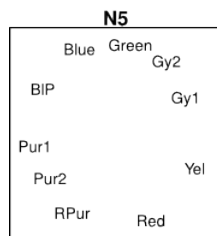
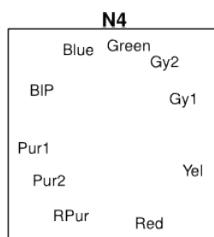
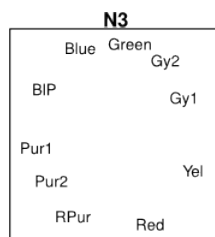
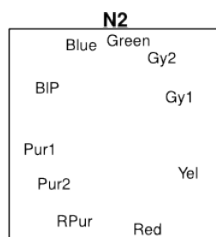
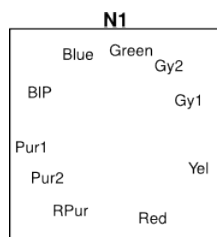
	P	R	ed	Y	el	G	y1	G	y2	Green	Blue	B	lP	Pur1
Red	1													
Yel	1	4.4												
Gy1	1	7.9	5.7											
Gy2	1	10.4	8.3	3.9										
Green	1	11.2	10.2	6.5	4									
Blue	1	12.6	11.3	8.7	7									
BlP	1	11.4	12.2	10.3	9									
Pur1	1	11.3	11.9	10.7	11									
Pur2	1	10.4	10.7	12.6	11									

\$N5

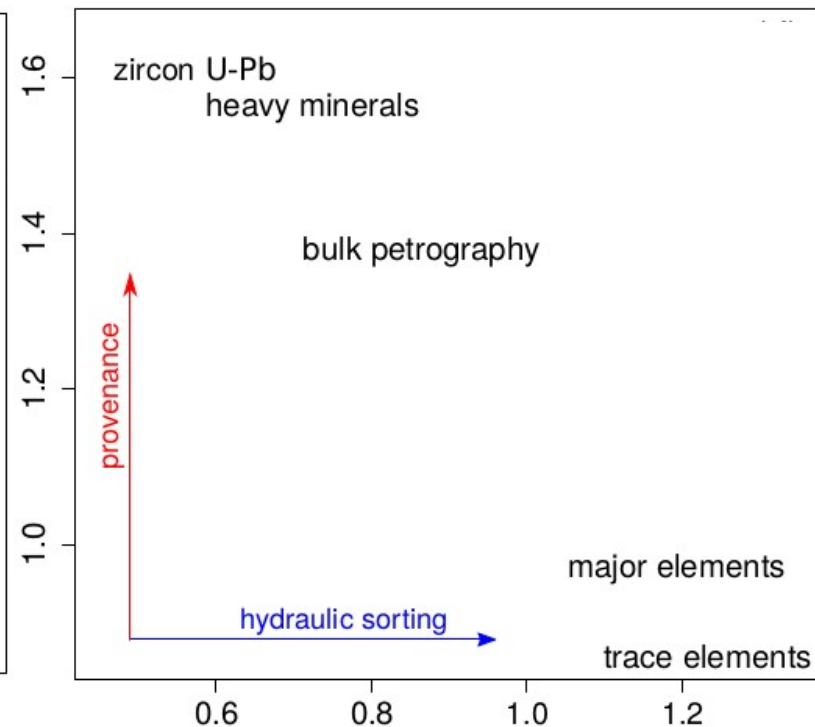
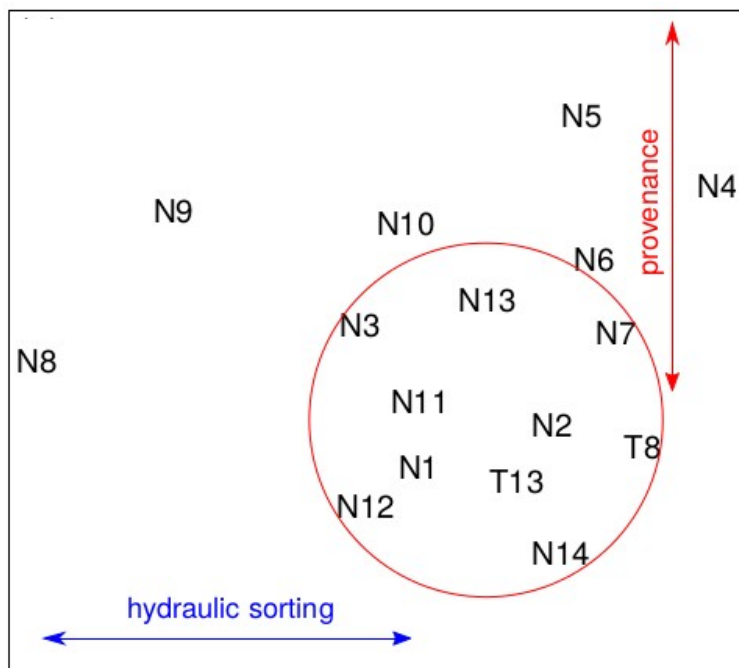
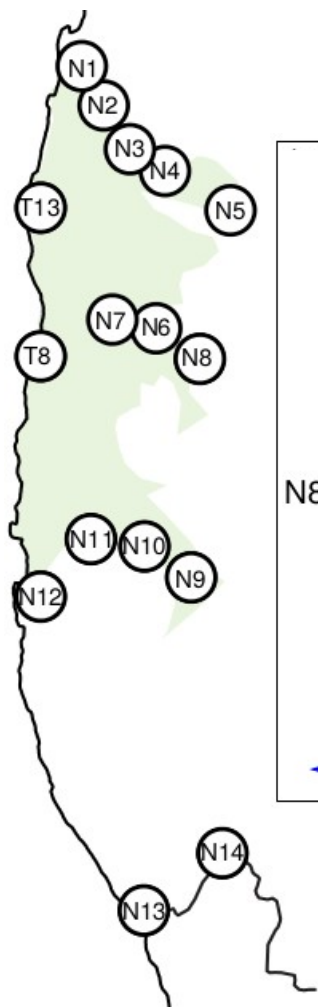
	Pur	R	ed	Y	el	G	y1	G	y2	Green	Blue	B	lP	Pur1
Red	6.6													
Yel	10.5	5.5												
Gy1	10.2	9.6	7.2											
Gy2	9.6	9.3	8.3	4.7										
Green	10.8	9.9	9.3	6.2	3.3									
Blue	9.7	11.7	11.3	8.9	6.3	4.2								
BlP	8.5	11.6	11.9	10.3	9.1	8.9	6.6							
Pur1	4.9	10.3	11.8	11.6	11.1	9.4	8.9	5.8						
Pur2	3.5	8.0	11.5	10.2	10.4	10.6	9.2	7.3	2.9					

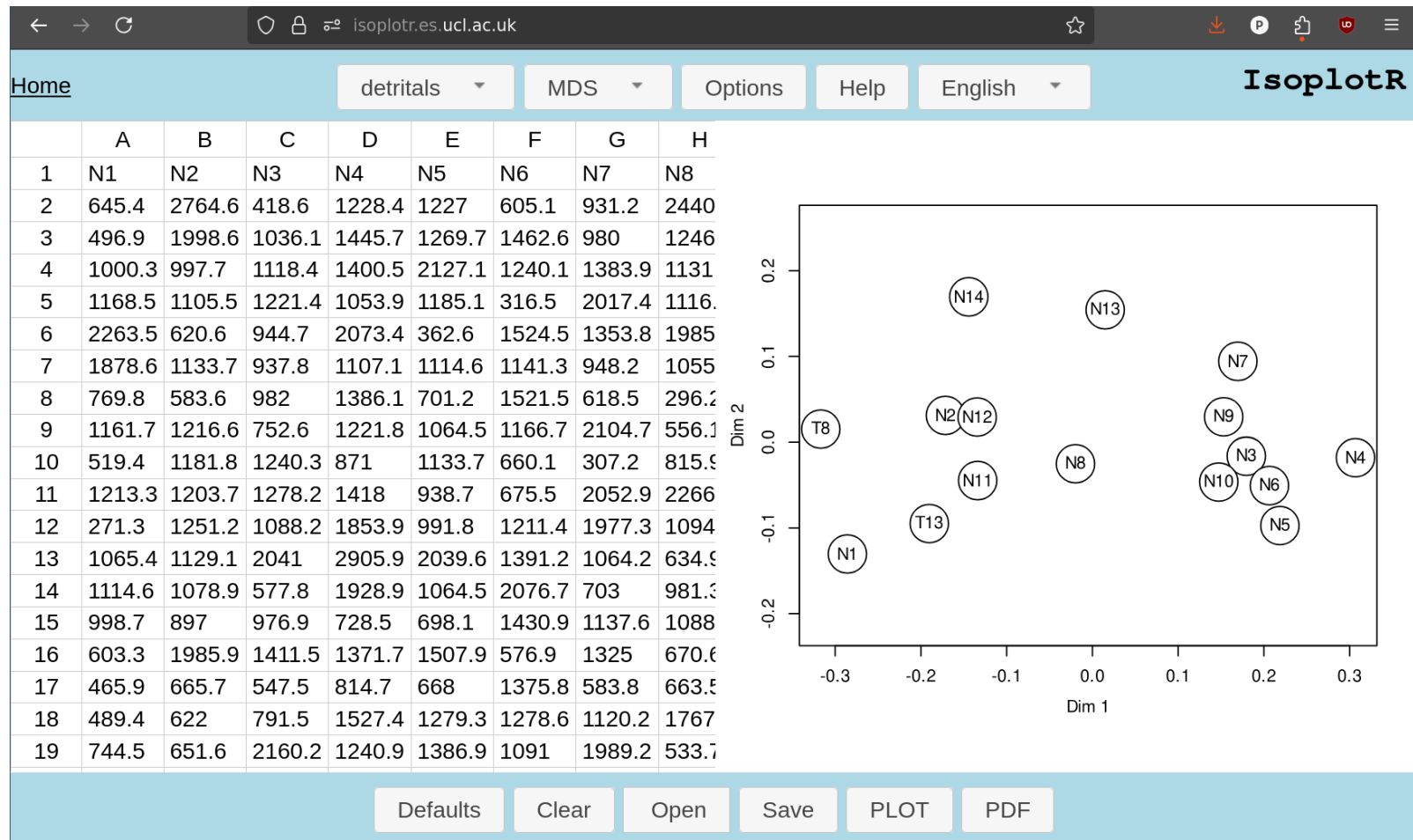
4.6					
7.8	6.3				
9.9	9.6	4.8			
11.2	10.6	6.8	4.6		
11.6	11.6	9.1	7.4	5.2	





Red-green
colour blind

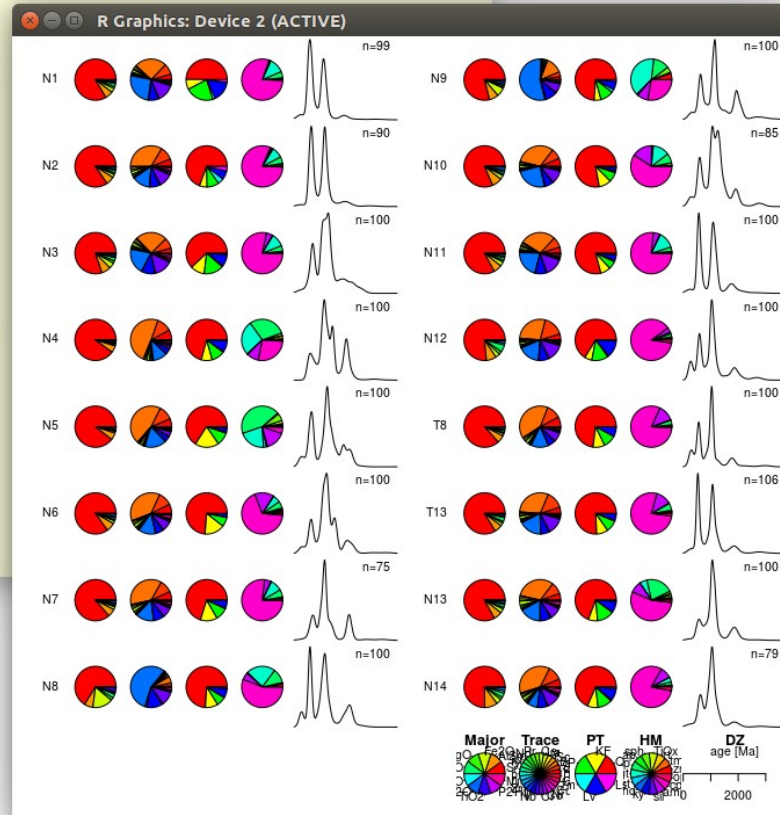


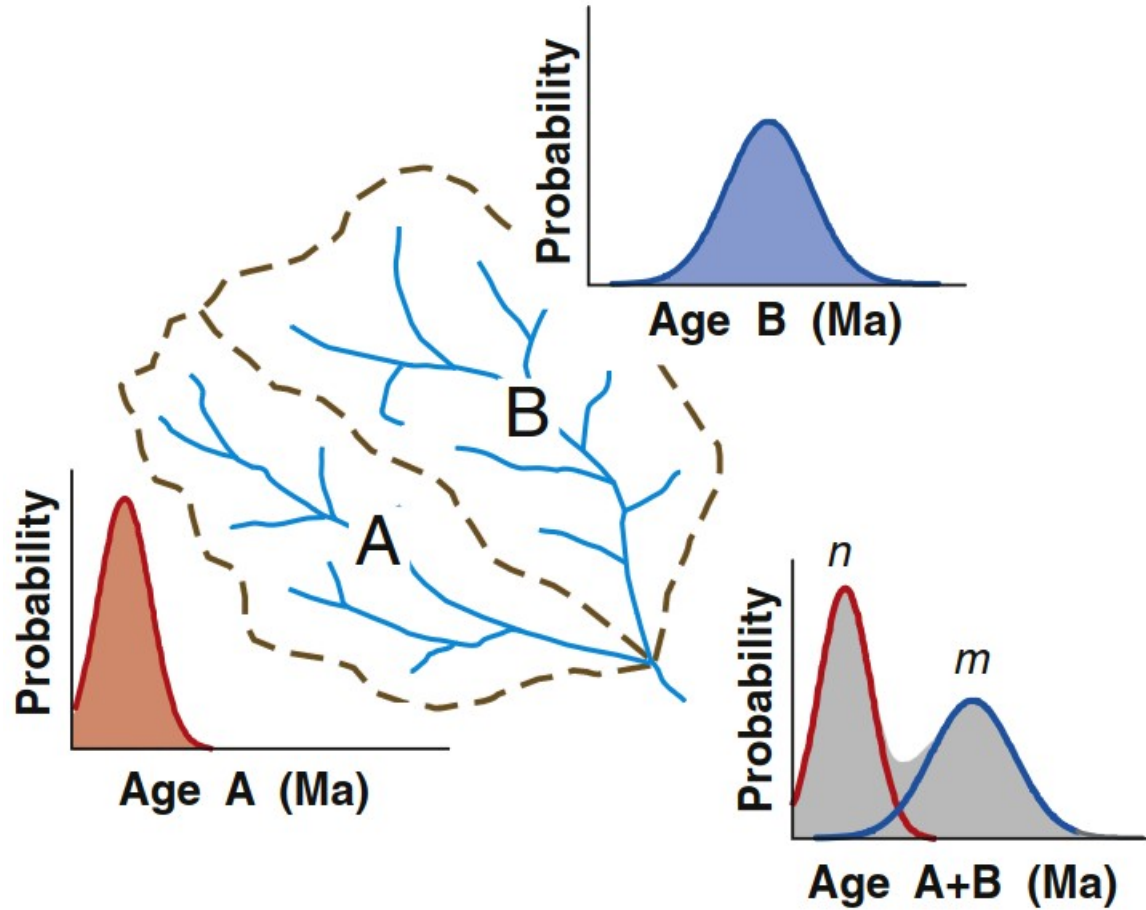


```

Enter file name: DZ.csv
Options:
1 - Subset samples
2 - Combine samples
3 - Load analytical uncertainties
c - Continue
c
Options:
1 - Set minimum age
2 - Set maximum age
3 - Turn off adaptive density estimation
4 - Plot on a log scale
5 - Set bandwidth
6 - Use the same bandwidth for all samples
7 - Normalise area under the KDEs
c - Continue
c
Number of columns? 2

```





Hydraulic sorting and mineral fertility bias in detrital geochronology

Marco G. Malusà, Alberto Resentini *, Eduardo Garzanti

ZIRCON FERTILITY MAP OF THE ALPS

